SEMINAR

# EEG-Based Guilty Knowledge Testing

## Methods, Evidence, Limitations, and Future Directions

*From Psychophysiological Recognition to Brain-Based Concealed-Information Detection*

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Public edition v1.7 · October 2026

# The Fundamental Question

*Why start here: every method in this seminar is an attempt to answer this one question — and only this one.*

*"Can the brain reveal whether a person recognizes information they are trying to conceal?"*

**The scientific version**

Can physiological, behavioral, and particularly neural responses reveal recognition of information that a person may be attempting to conceal?

**Not the question**

"Can EEG detect lies?" — the test probes recognition, not deception.

CONCEPTUAL ANCHOR

# Recognition ≠ Deception ≠ Guilt

*Why first: every later claim depends on keeping these three levels apart.*

## Recognition

Information appears familiar or cognitively significant

What the CIT is designed to probe

≠

## Deception

An intentional act of communicating something believed to be false

An act — not a single physiological signal

≠

## Guilt

Responsibility for an act — a legal and investigative conclusion

Never an EEG measurement

EEG-based CIT investigates concealed recognition. A differential response to crime-relevant information can also arise from legitimate familiarity, media exposure, previous questioning, investigator leakage or incidental exposure.

Sources: [1], [11], [14], [53]

SEMINAR ROADMAP

# Six Questions, Six Sections

*Why a roadmap: each section answers one question, and each builds on the assumptions of the one before.*

**I** — **What is EEG-based GKT, and why is it needed?**

Traditional measures · Lykken's logic · why EEG

**II** — **How does it work?**

EEG · preprocessing · stimuli · ERP · P300 · decisions

**III** — **What does a real experiment look like?**

Mock theft · six images · files · first look at the data

**IV** — **What can computation add?**

Features · ML · deep learning · validation · results on real data

**V** — **What prevents real-world use?**

Recognition ≠ guilt · memory · countermeasures · ethics

**VI** — **What is needed next?**

Better experiments · better computation · a translation ladder

**Each layer inherits the assumptions of the one before: a classifier cannot repair a badly chosen probe**

# How to Read This Deck

*Why first: the deck mixes literature, simulations and real data, and every slide says which one it shows.*

### Learning objectives

- Explain why the CIT probes recognition, not deception
- Describe how a P300 probe effect is measured and decided
- Read a real public EEG-CIT dataset and its limits
- Judge a computational result by its validation design

### Prerequisites

- Basic statistics: mean, variability, resampling
- No EEG background needed; ML basics help in Section IV

### Citations

[1]–[59] published studies · [S1]–[S3] supplementary AI-architecture papers · “Own analysis · data [37]”: newly computed from the public dataset

### Source legend

| | |
|---|---|
| Schematic | illustration of an idea; no data |
| Real data | own analysis of the Dryad files |
| Reported [n] | stated in a cited paper |
| Verified | checked directly in the released files |
| Assumed | not documented; inferred and labelled |

### Notation

µV microvolt · ms millisecond · Hz samples per second
L09 / H09 = participant 9 in the lying / honest folder
d* group-level effect size · AUC area under the ROC curve

# From Traditional Psychophysiological Methods to EEG-Based GKT

Understanding the foundations of concealed-information testing and the motivation for EEG-based approaches.

I · What & why
II · Methods
III · Experiment & data
IV · Computation
V · Limitations
VI · Future

# Traditional Psychophysiological Detection

*Why start here: the CIT inherited its measures from this tradition — and its central inferential problem.*

### Electrodermal activity

GSR / SCR / skin conductance

### Respiration

Breathing pattern, e.g. respiration line length

### Cardiovascular

Heart rate; blood pressure where recorded

### Behavioral

Reaction time and response patterns

**MEASURED**

A physiological change in response to a stimulus or question

→

**INFERRED**

Deception or knowledge — a conclusion that depends on the test logic, not on the signal alone

Sources: [1], [7], [16], [34]

# Electrodermal Activity: GSR, SCR and EDA

*Why first: electrodermal recording is the measure the GKT was built on, and still the most studied CIT channel.*

**EDA**
Electrodermal activity: the umbrella term

**GSR**
Galvanic skin response: Lykken's historical term

**SCR**
Skin conductance response: phasic, stimulus-evoked

**SCL**
Skin conductance level: slow tonic baseline

## How it works

**Two electrodes on fingers or palm**

↓

**Small constant voltage across the skin**

↓

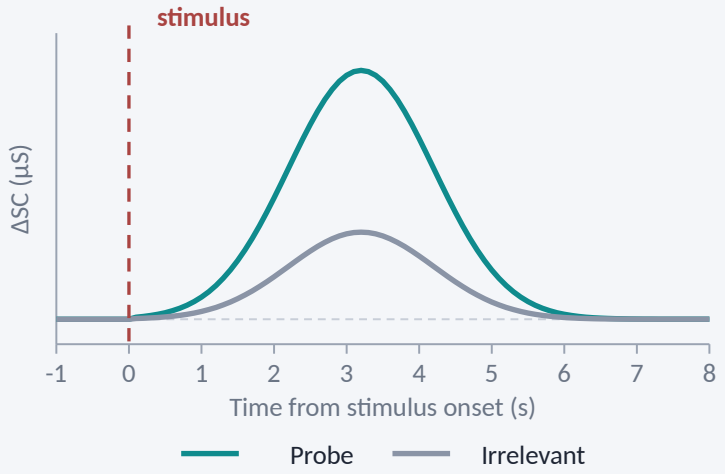
**Sympathetic activation fills eccrine sweat glands**

↓

**Conductance rises → SCR a few seconds later**

*General physiological background*

## SCR to probe vs irrelevant




*Schematic — timing approximate; amplitude not to scale*

## How it supports the GKT

- Recognized probe → larger SCR than irrelevants [1]
- Behaves most like an orienting response [39]
- Scored per item, e.g. amplitude or area [55]

## Limits

Slow responses need long intervals; habituates with repetition [10]; some people barely respond [38]; reduced by a mental countermeasure [9].

**Group-level synthesis: mean d* = 1.55 for SCR in laboratory CIT studies [34] — not an individual accuracy**

Sources: [1], [9], [10], [34], [38], [39], [55]

# Respiration: RLL and Breathing Suppression

*Why separately: respiration responds to the probe in the opposite direction from SCR — and for a different reason.*

**RLL**
Respiration line length: total length of the breathing trace in a window

**Pneumograph**
Elastic belt sensor around chest or abdomen

**Suppression**
Shallower, slower breathing → shorter RLL

## How it works

**Elastic belts around chest and abdomen**

↓

**Belt stretch tracks breathing movement**

↓

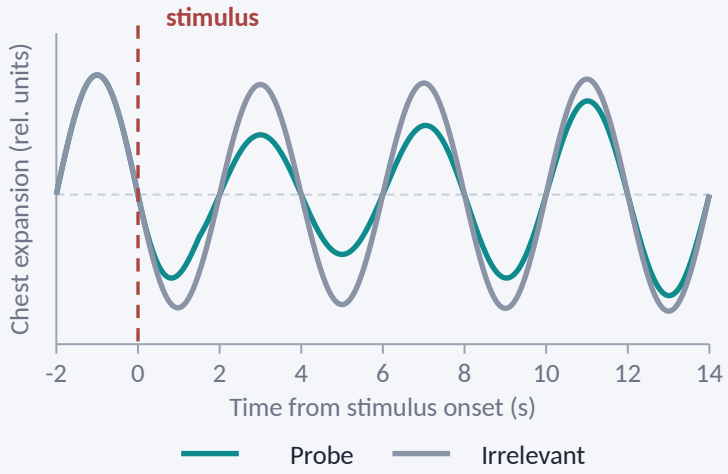
**Trace length summed over a post-stimulus window**

↓

**Shorter line = suppressed breathing**

*General physiological background*

**Breathing to probe vs irrelevant**




*Schematic — timing approximate; amplitude not to scale*

### How it supports the GKT

- Recognized probe → breathing suppression (shorter RLL)
- More consistent with arousal inhibition than orienting [39], [41]
- Objective scoring compared with human evaluation [7]

### Limits

Sensitive to speech and movement; depends on motivation to conceal; lab–field differences examined [40].

**Group-level synthesis: mean d* = 1.11 for RLL [34]; in one study a mental countermeasure affected EDA but not RLL [9]**

Sources: [7], [9], [34], [39], [40], [41]

# Cardiovascular Measures: HR, ECG and BP

*Why include it: heart rate adds a third response pattern, and its direction surprises many audiences.*

**HR**
Heart rate: beats per minute

**ECG**
Electrocardiogram: electrical record of each heartbeat

**IBI**
Inter-beat interval: time between successive beats

**BP**
Blood pressure: recorded in polygraph practice

## How it works

**ECG electrodes on chest or limbs**

↓

**R-peak detected in each heartbeat**

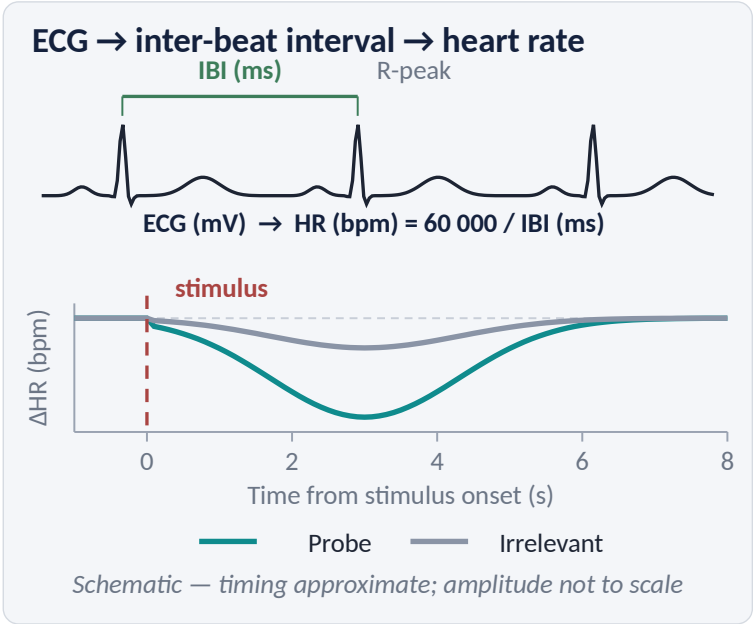
↓

**Inter-beat intervals → heart rate over time**

↓

**Probe: heart rate tends to slow**

*General physiological background*




*Schematic — timing approximate; amplitude not to scale*

## How it supports the GKT

- Recognized probe → heart-rate deceleration, not acceleration
- Grouped with respiration: arousal inhibition [39], [41]
- Orienting vs defensive responses distinguished [17]

## Limits

Smallest pooled autonomic effect [34]; reactivity differs between individuals [19]. BP is not a focus of the reviewed literature.

**Group-level synthesis: mean d* = 0.89 for HR in laboratory CIT studies [34] — not an individual accuracy**

Sources: [17], [19], [34], [39], [41]

# Behavioral Measures: RT and the RT-CIT

*Why include behavior: reaction times detect concealed information without physiological sensors, so they are cheap and scalable.*

**RT**
Reaction time: ms from stimulus onset to key press

**RT-CIT**
Reaction-time Concealed Information Test

**aIAT**
Autobiographical Implicit Association Test

## How it works

**Stimulus appears on screen**

↓

**Participant presses a key, e.g. ‘no’ to every item**

↓

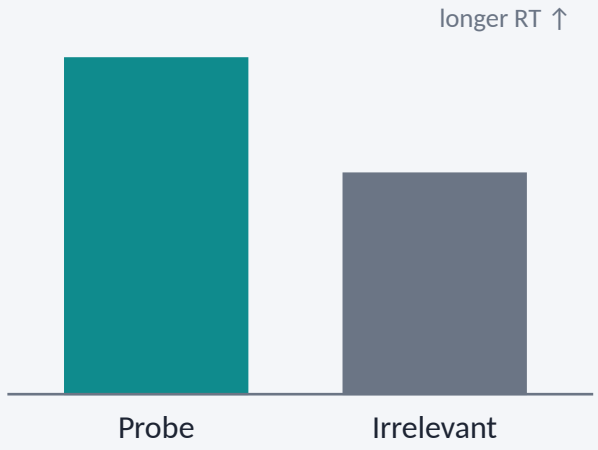
**Software logs the response time per trial**

↓

**Probe: responses tend to be slower**

*General physiological background*

## Mean RT: probe vs irrelevant




*Schematic — timing approximate; amplitude not to scale*

## How it supports the GKT

- Denying a recognized item creates response conflict → slower RT [12], [26]
- Validity compared with the polygraph [26]
- Combined with P300: CTP [24]; with the aIAT [30]

## Limits

Responses are under voluntary control, so strategies can alter them; resistance must be tested per protocol.

**Behavioral evidence is cited qualitatively; the meta-analytic d* values cover SCR, RLL, HR and P300 only [34]**

Sources: [12], [24], [26], [30]

# Lykken (1959): The Guilty Knowledge Test

*Why it matters: the GKT redefined the problem — from finding a lying signature to testing differential recognition.*

**Before: search for a lying signature**

- Is there a physiological pattern unique to lying?
- Inference depends on reactions being specific to deception

→

**Lykken's shift: test differential recognition**

- Does this person respond differently to the correct detail?
- Inference depends on equivalence of the alternatives

## Why several items strengthen the logic

1 **Single item**

An uninformed person can single out the correct alternative by chance

2 **Several independent items**

Consistent selection of correct alternatives by chance becomes progressively less likely

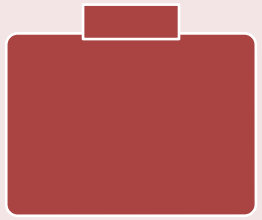
3 **Equivalent alternatives**

Alternatives must look equally plausible to an uninformed examinee

Sources: [1], [16]

# The Red Toolbox

*Why one example: a single scenario keeps every later method comparable.*



**Red toolbox**

PROBE



**Blue suitcase**

IRRELEVANT



**Black backpack**

IRRELEVANT



**Green storage box**

IRRELEVANT

## What makes it a usable probe

- Nonpublic detail of the investigation
- Alternatives equally plausible to an uninformed person

## What can break it

- Media reports, questioning or leaked case files
- Alternatives that differ in salience from the probe

Sources: [11], [44], [51]

# Why Does the Probe Elicit a Different Response?

*Why ask: if the mechanism is unknown, the meaning of a positive result is unknown.*

### Orienting

Attention captured by a significant item

### Recognition

Memory match to a known detail

### Stimulus significance

Personal or task relevance of the item

### Inhibition

Effort to suppress arousal while concealing

### Conflict

Tension between knowledge and required response

### Attention

Allocation of processing resources

### Response fractionation

- Skin conductance: more consistent with orienting
- Respiration and heart rate: more consistent with arousal inhibition
- Pupil vs fixations/blinks show a similar split [54]

**Different measures can reflect different mechanisms**

Sources: [17], [39], [41], [45], [50], [53], [54]

# Why EEG for Concealed-Information Research?

*Why ask: EEG's value lies in timing — matching brain activity to each individual stimulus.*

1. **Non-invasive**
   Scalp electrodes; no injection or radiation
2. **High temporal resolution**
   Activity tracked at the millisecond scale
3. **Event / stimulus locking**
   Responses aligned to each probe or irrelevant
4. **Cognitive processing**
   Sensitive to attention, memory and significance
5. **ERP extraction**
   Averaging or single-trial analysis isolates components

**Trade-off**

Limited spatial precision: scalp EEG does not localize cortical sources directly.

**Next**

One ERP component became particularly influential: the P300.

Sources: [3], [47]

# What Makes a CIT EEG-Based?

*Why define it: the title promises EEG-based GKT, so its ingredients should be explicit before the methods.*

## The five ingredients

CIT paradigm: probe · target · irrelevants → Stimulus-locked EEG: every item marked in the recording → Neural marker: P300 (also N200 and feedback-related ERPs) → Within-person comparison: probe vs irrelevants → Decision rule: bootstrap test or classifier

### What it measures

- A recognition-related cortical response: the probe is processed as significant
- Not a deception signal, and never guilt

### A simple test of the definition

- Remove the EEG: it is an autonomic or behavioural CIT
- Remove the probe–irrelevant comparison: it is no longer a CIT

**EEG-based GKT = CIT logic + stimulus-locked EEG + a within-person probe–irrelevant decision**

Sources: [3], [24], [33], [47], [48], [56]

# Methods: How Does EEG-Based GKT Work?

EEG and ERP foundations · the P300 · experimental design · acquisition and preprocessing · quantification and decision rules.

I · What & why
**II · Methods**
III · Experiment & data
IV · Computation
V · Limitations
VI · Future

# From Scalp to Decision: The Method in Eight Stages

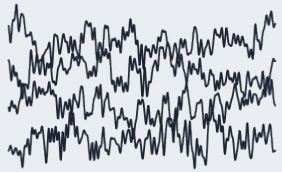
*Why a map first: every stage is a methodological choice, and an error at any stage propagates to the conclusion.*

**1 Present stimuli**



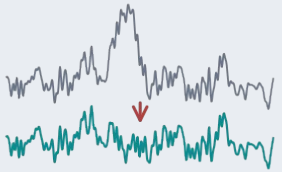

Target · probe · irrelevants

**2 Record EEG**



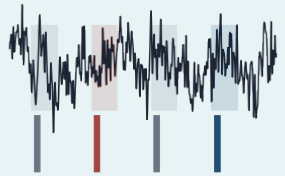
Markers written as items appear

**3 Clean**



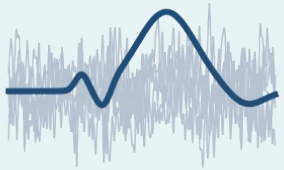
Filter · remove artifacts

**4 Epoch**



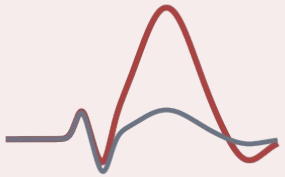
Window around each marker

**5 Average**



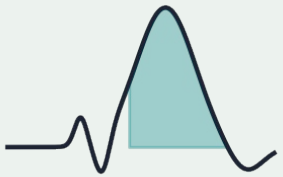
ERP emerges from noise

**6 P300 response**



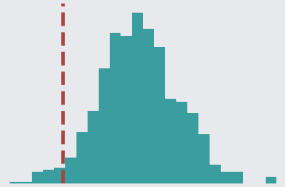
Larger to rare, meaningful items

**7 Quantify**



Amplitude · latency · area

**8 Decide & interpret**



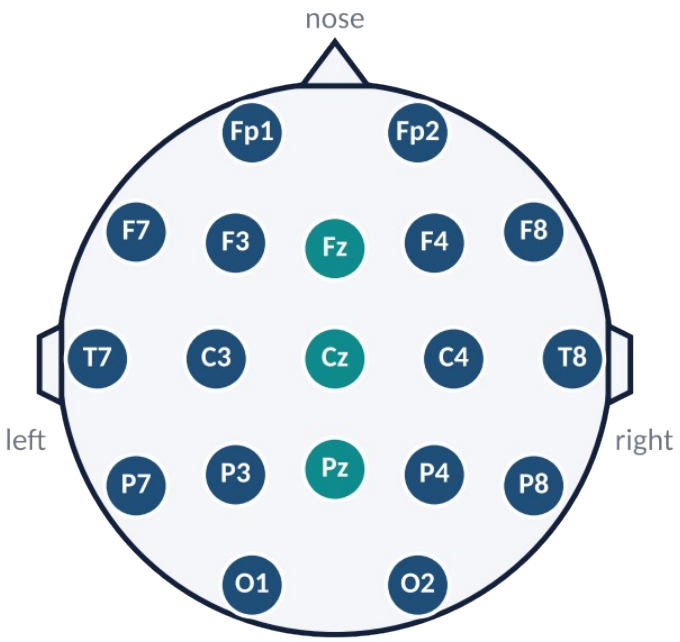
Within-person test → recognition?

**Two routes after stage 4:** averaging → statistical CIT tradition (this section) · single trials → machine learning (Section IV)

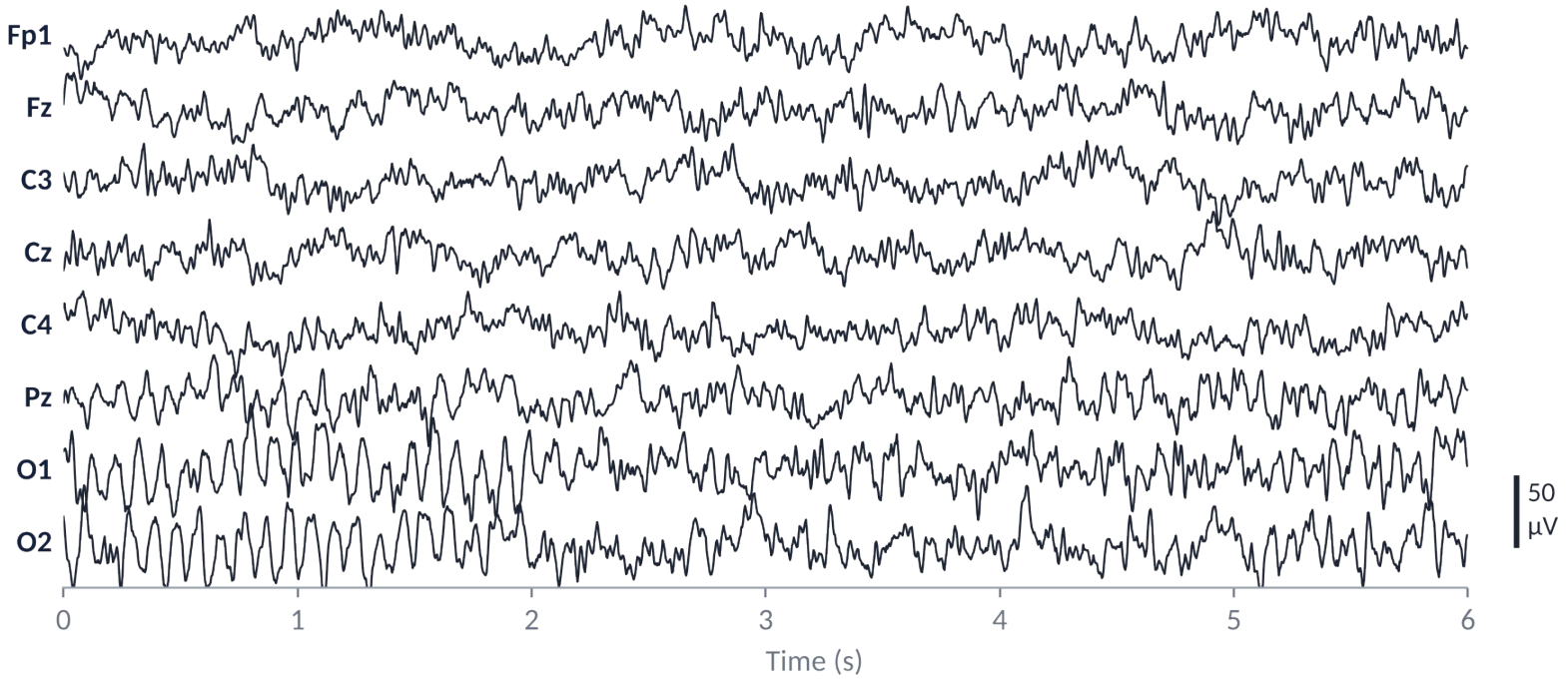
Sources: [21], [37], [47]

# What Is EEG?

*Why only the essentials: the CIT needs three properties of EEG — scalp voltage, continuity and millisecond timing.*




International 10–20 system (subset)
teal = midline sites (z)




*Simulated signals for illustration — typical scalp magnitudes, not recorded data. Posterior channels show ~10 Hz alpha.*

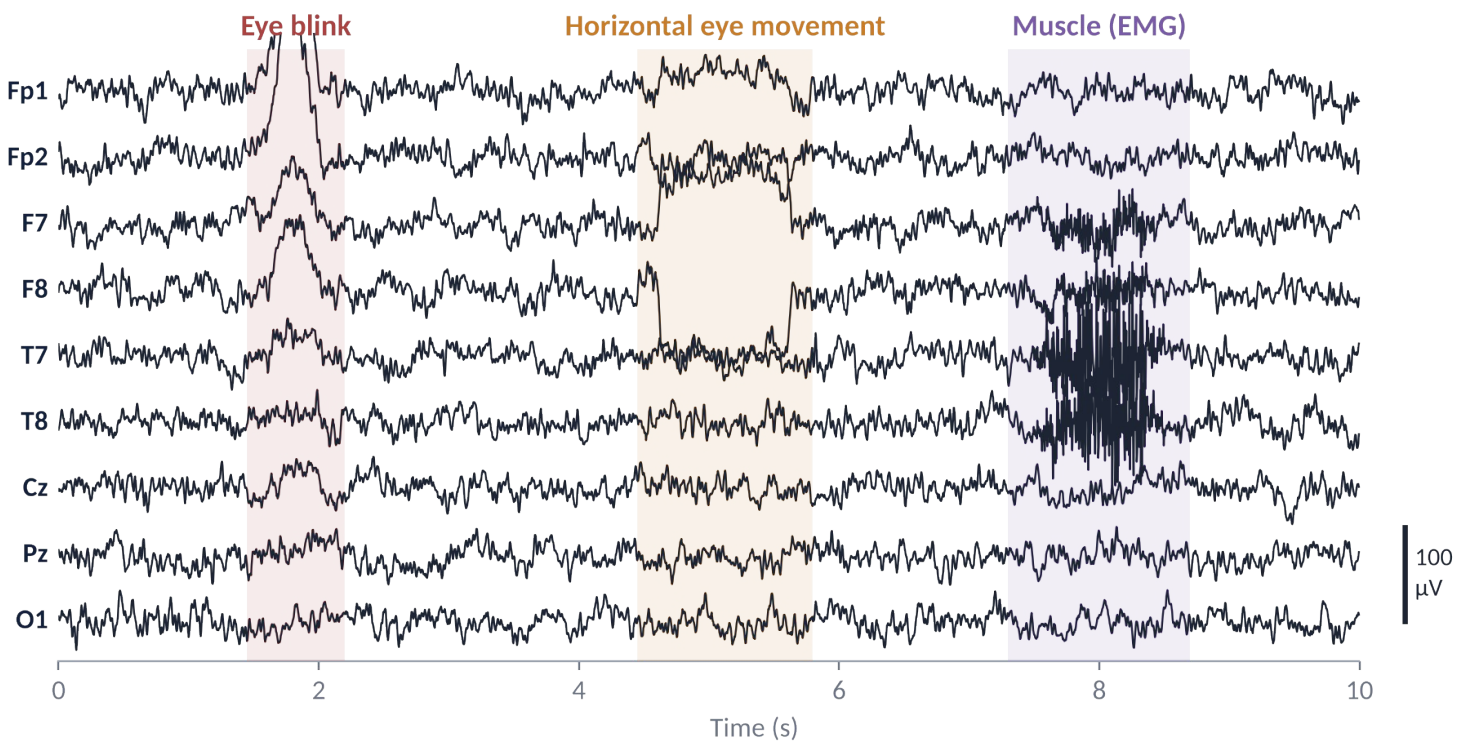
**Scalp voltage:** µV-scale differences from synchronized cortical activity

**Continuous:** Background activity is larger than any single stimulus response

**Millisecond timing:** Sampled hundreds of times per second

# EEG Artifacts: Blinks, Eye Movements, Muscle

*Why now: artifacts are often far larger than the P300, so every preprocessing and modelling step must deal with them.*




*Simulated signals for illustration — typical scalp magnitudes, not recorded data.*

### Eye blink

- Frontal: Fp1 / Fp2
- Large positive deflection, ~hundreds of ms
- Shrinks towards posterior sites

### Horizontal eye movement

- Lateral frontal: F7 / F8
- Box-shaped step
- Opposite polarity left vs right

### Muscle (EMG)

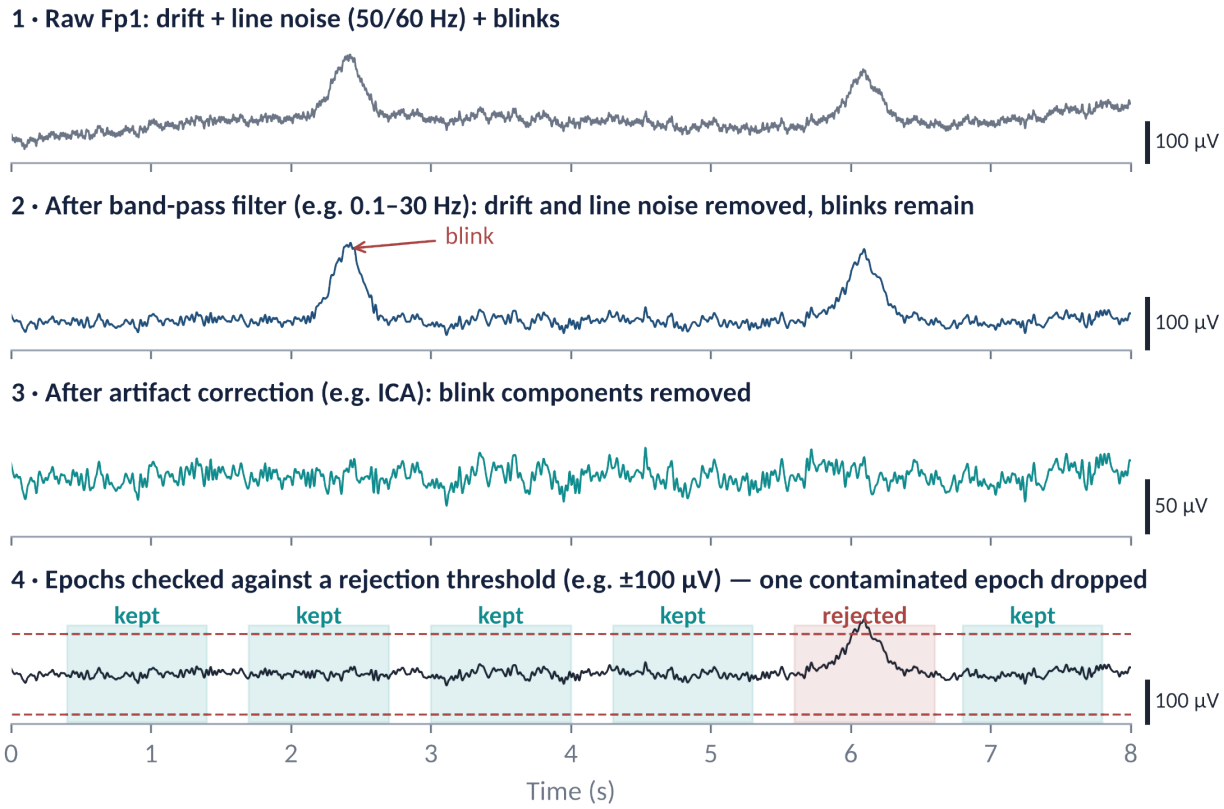
- Temporal: T7 / T8
- High-frequency bursts
- Jaw clench, swallowing, tension

**Why it matters:** artifacts can be time-locked to stimuli and mimic an ERP difference; a model can learn artifacts instead of recognition. Handled by EOG monitoring, rejection or correction (e.g. ICA [37]).

Sources: [37]

# Preprocessing in Practice: Four Cleaning Steps

*Why show the signal: each step visibly changes the data, so each must be justified and fixed before outcomes are inspected.*




*Simulated signals for illustration — typical scalp magnitudes, not recorded data. Filter band and threshold are examples.*

**Filter**

Removes drift and line noise; must preserve the slow P300

**Correct**

ICA separates and removes blink components [37]

**Reject**

Epochs beyond a threshold are dropped — report how many

**Caution**

ICA does not inherently improve accuracy; ML implications in Section IV

Sources: [21], [25], [37]

# Three Stimulus Types: Target, Probe, Irrelevant

*Why three: the target keeps attention on every item, the irrelevants give each person a baseline, and the probe carries the question.*

## TARGET



**White crate**

| | |
|---|---|
| Role | Task-relevant for everyone |
| Frequency | **≈ 20% — infrequent** |
| Response | "Target" key (e.g. YES) |
| Expected | P300 in everyone → attention check |

## PROBE



**Red toolbox**

| | |
|---|---|
| Role | Nonpublic crime detail |
| Frequency | **≈ 20% — infrequent** |
| Response | "No" key — the concealment |
| Expected | P300 may be enhanced if recognised |

## IRRELEVANTS



**Blue suitcase · black backpack · green storage box**

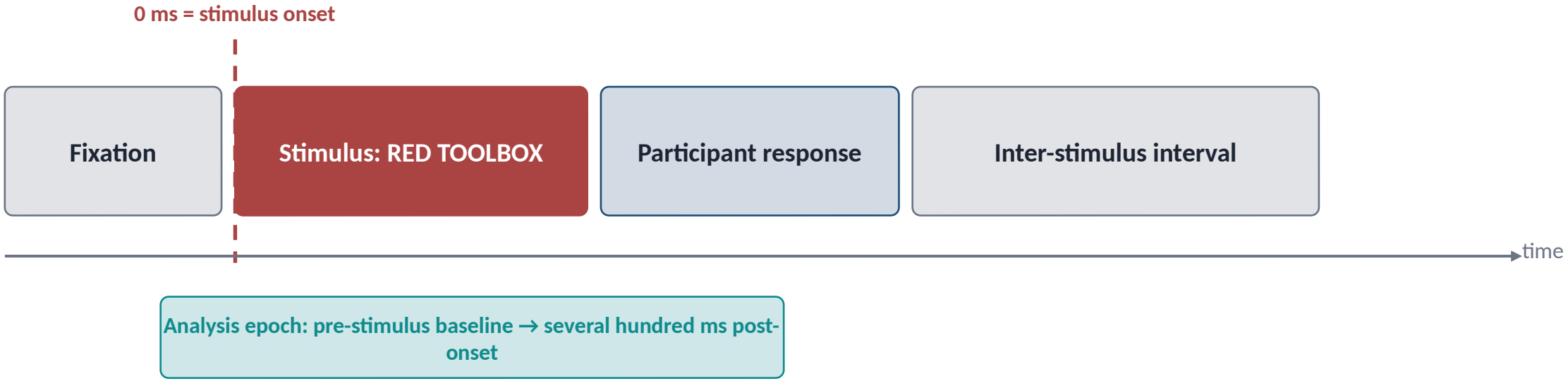
| | |
|---|---|
| Role | Plausible alternatives |
| Frequency | **≈ 20% each · ≈ 60% together** |
| Response | "No" key |
| Expected | Each person's baseline |

**Target and probe are equally infrequent;** only the irrelevants as a class are frequent. Proportions follow this five-item example; paradigms differ [24].

Sources: [3], [20], [24], [47]

# Anatomy of One Trial

*Why timing matters: every analysis is defined relative to stimulus onset at 0 ms.*




## Event synchronization

Stimulus onset is written as a marker into the EEG; timing errors blur the averaged response.

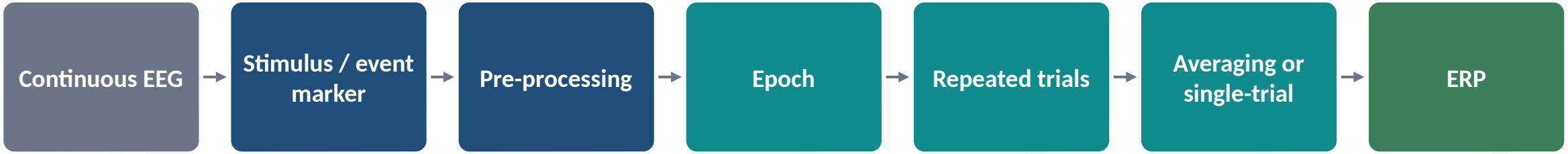
## Illustrative durations

Segment lengths are schematic; real protocols differ in timing and response requirements.

Sources: [24], [47]

# From Continuous EEG to Event-Related Potentials

*Why this pipeline: the CIT compares responses to events, so the signal must first be organized by events.*




## Averaging

- Background activity unrelated to the stimulus tends to cancel
- Consistent stimulus-related activity remains
- Requires many repetitions per stimulus type

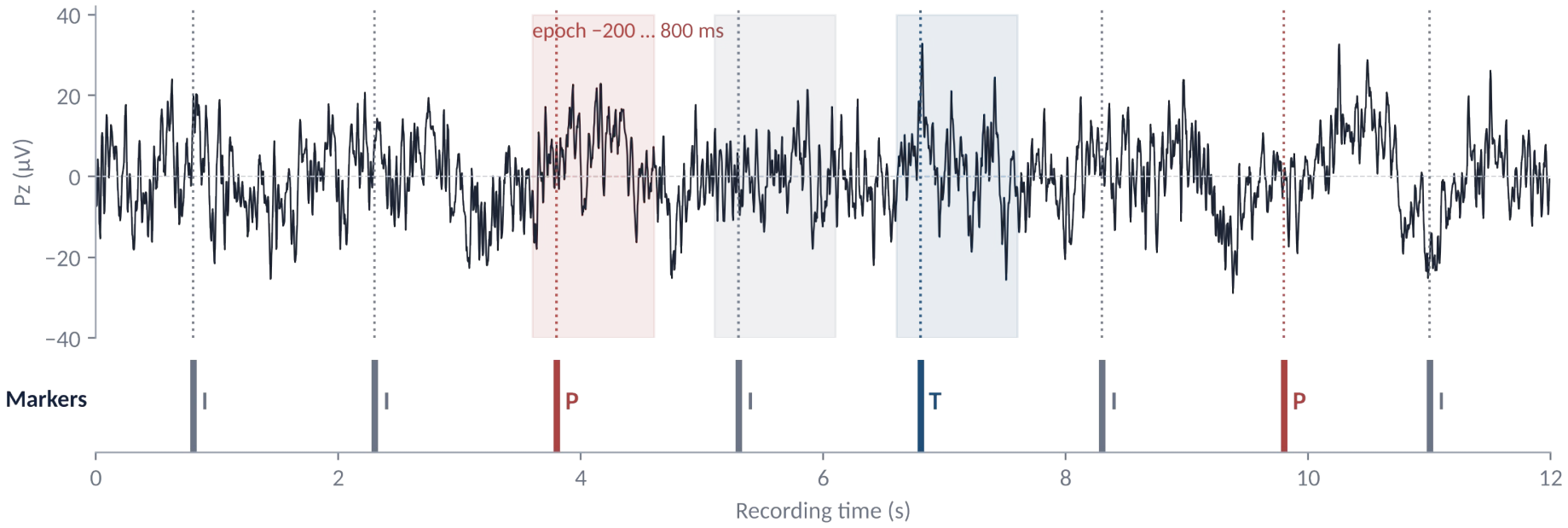
## Single-trial analysis

- Analyzes individual epochs
- Retains trial-to-trial variability
- Common input for classifiers; lower signal-to-noise ratio

Sources: [21], [37], [47]

# Event Markers: Linking Each Stimulus to the EEG

*Why markers: the EEG itself does not record when a stimulus appeared; the marker track supplies that timing.*




*Simulated signals for illustration — typical scalp magnitudes, not recorded data. Markers: I = irrelevant (gray) · P = probe (red) · T = target (blue).*

**Marker**

Event code + sample index, e.g. P at sample 950 (3.8 s × 250 Hz)

**Epoch**

Fixed window around each marker, here −200 to 800 ms

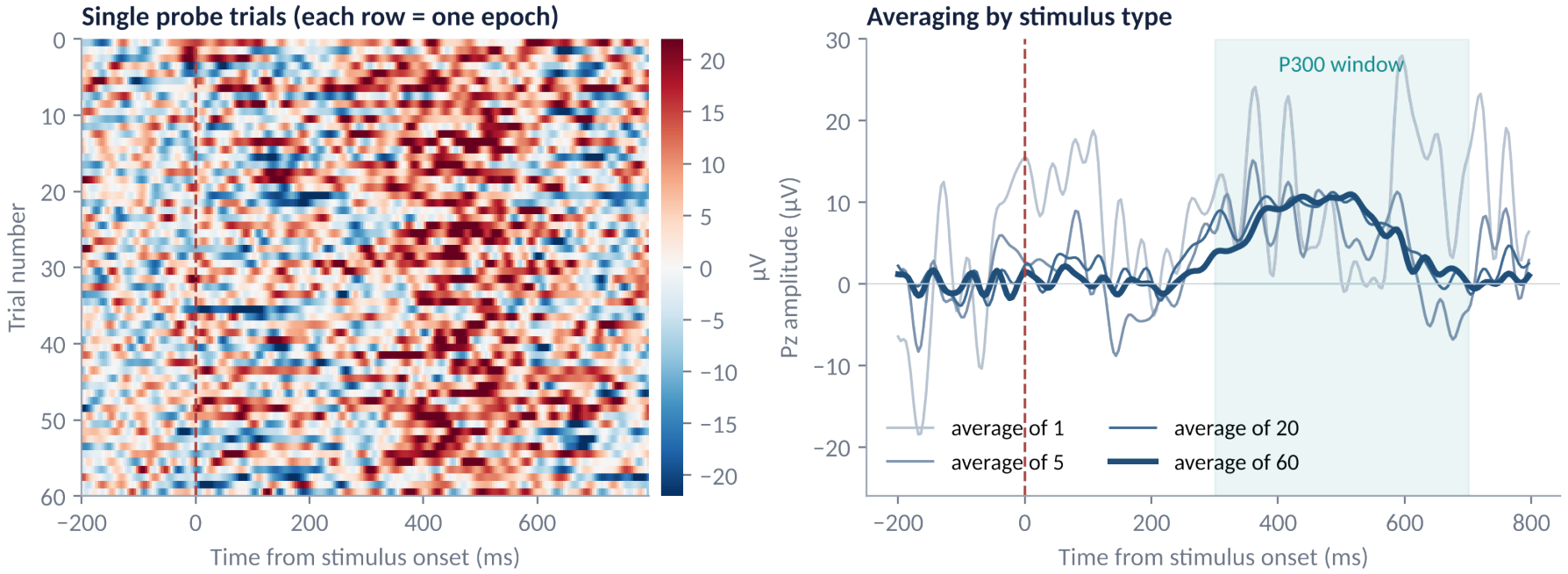
**Single trial**

P300 not visible by eye — background activity dominates

Sources: [21], [37], [47]

# Averaging: How the ERP Emerges From Noise

*Why average: the P300 is a few μV inside tens of μV of background activity; averaging epochs of one type reveals it.*




*Simulated signals for illustration — typical scalp magnitudes, not recorded data. Each epoch baseline-corrected to the pre-stimulus interval.*

**Cancels**

Activity not time-locked to the stimulus averages towards zero

**Improves with N**

Residual noise falls roughly with 1/√N

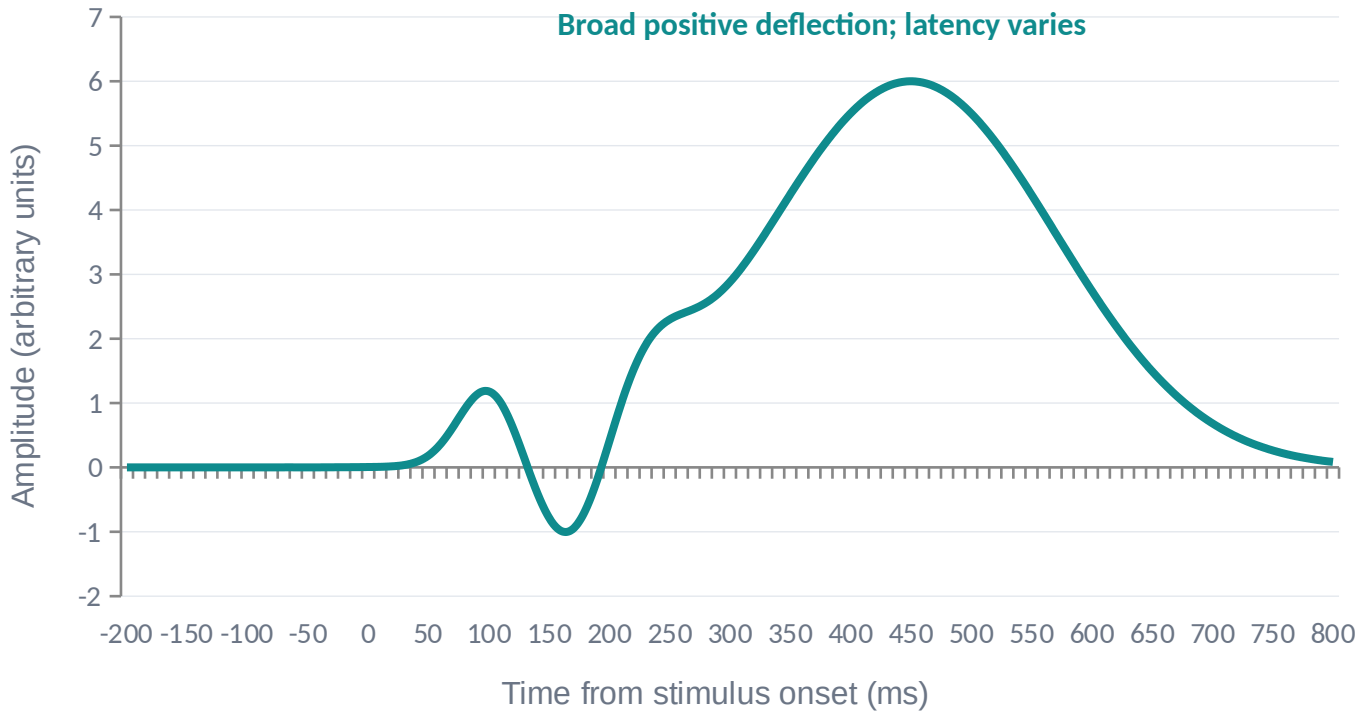
**An estimate**

An ERP comes from many trials; single-trial models face the noisy rows

Sources: [21], [37], [47]

# What Is the P300?

*Why define it carefully: popular accounts attribute meanings to the P300 that the evidence does not support.*




*Conceptual illustration — positive plotted upward; not a universal waveform*

## Associated with

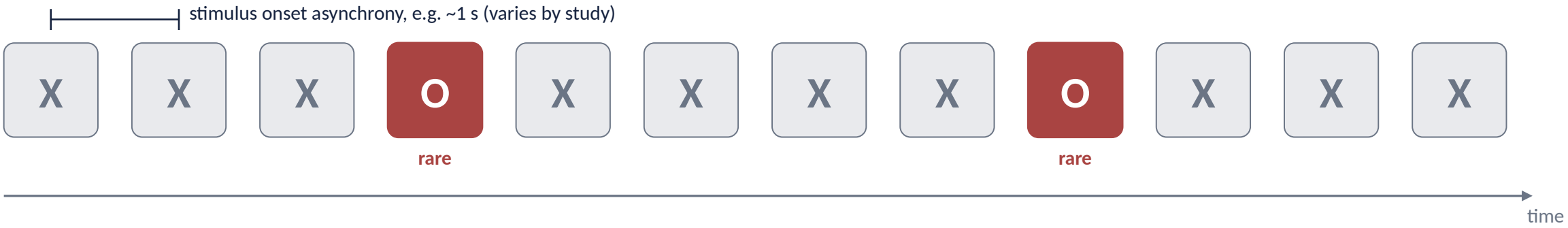
- Stimulus probability
- Task relevance and attention
- Stimulus significance
- Context updating
- Recognition and memory

**P300 IS NOT A LIE SIGNAL**

Sources: [3], [29], [47]

# The Oddball Paradigm

*Why introduce it: the P300 was characterised in oddball tasks, and the CIT inherits this structure.*




**Probabilities (example)**




**1 · Task**

Count, or press a button for, the rare stimulus

**2 · Processing**

Rare + task-relevant → context updating

**3 · Response**

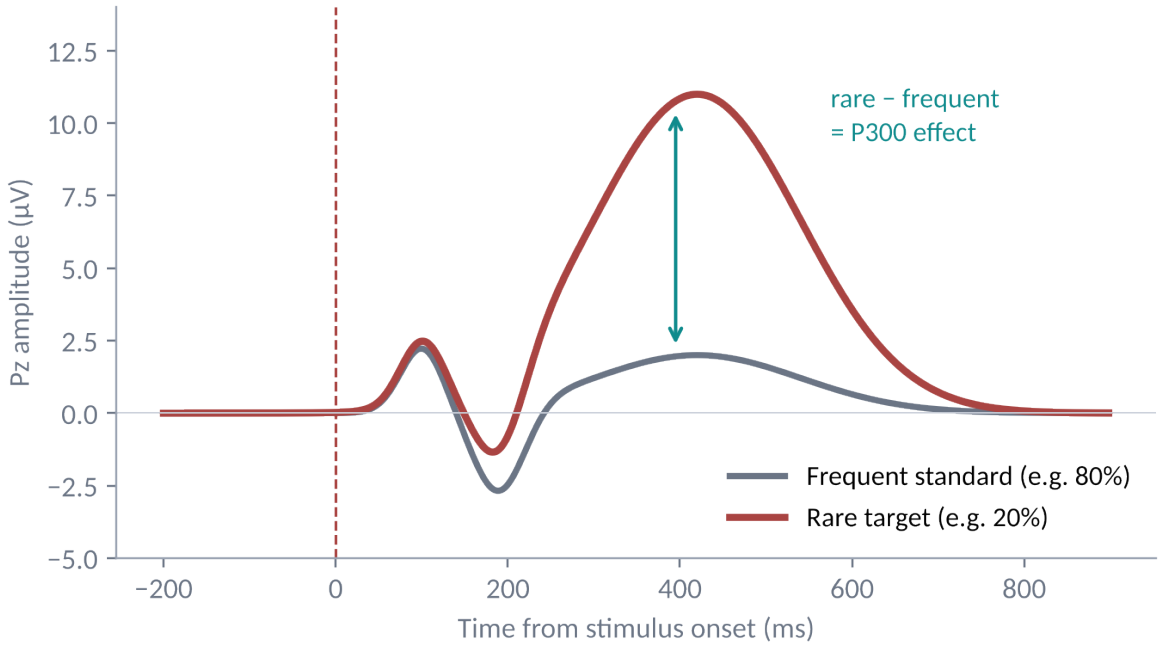
Larger P300 to the rare stimulus — shown next

**The P300 indexes rarity, task relevance and context updating — not deception [47]**

Sources: [3], [47]

# Rare Stimuli Evoke a Larger P300

*Why it matters: the CIT recreates an oddball — the probe can stand out for someone who recognises it.*




*Simulated signals for illustration — typical scalp magnitudes, not recorded data.*

## Same structure in the CIT (running example)

| Target white crate | Probe red toolbox | Irrelevants: blue suitcase · black backpack · green storage box |
|---|---|---|

each item ≈ 20% of trials · irrelevants together ≈ 60%

**Knowledgeable person**

Probe may be meaningful → can stand out from the frequent irrelevants → larger P300 likely

**No knowledge**

Probe is just another irrelevant → response resembles irrelevants

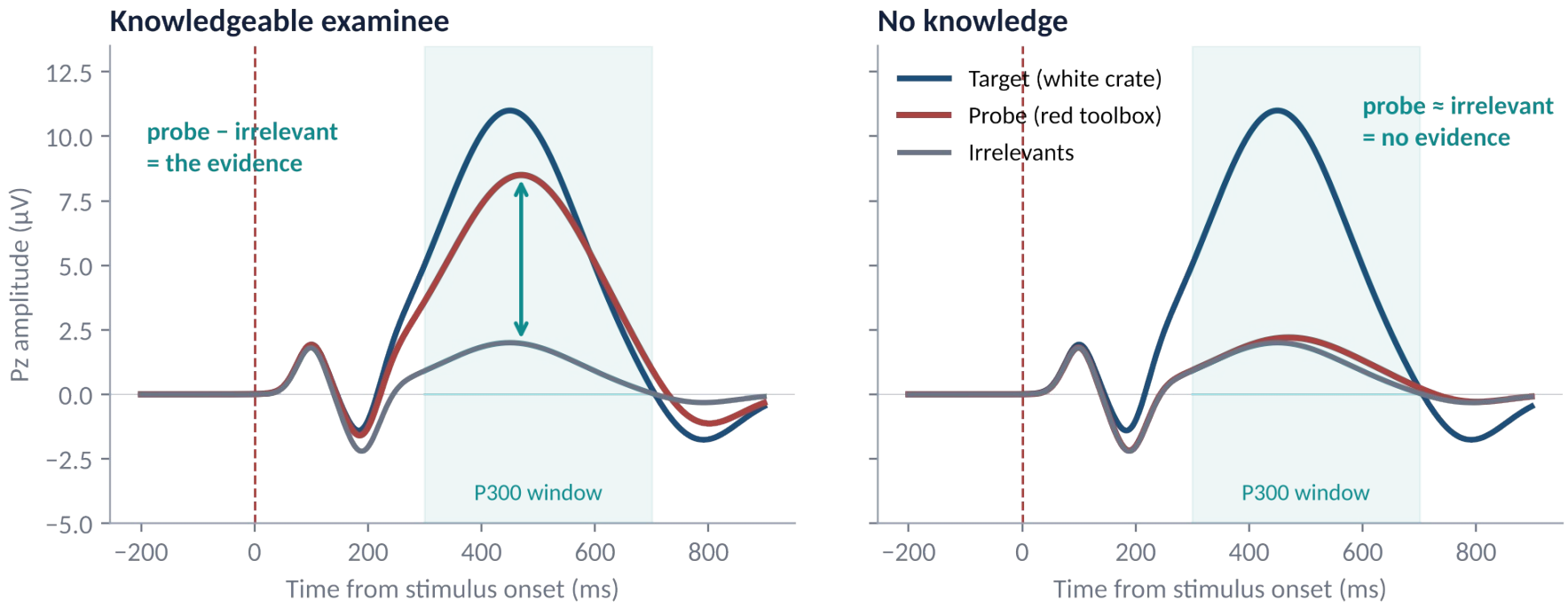
*Classic three-stimulus design; protocols such as the CTP restructure trials [24].*

**Equal frequencies, different subjective significance: recognition may enhance the probe P300 — not a deception signal [47]**

Sources: [3], [24], [47]

# Expected ERPs: Knowledgeable vs No Knowledge

*Why show this: the prediction is explicit and testable — the evidence is the probe–irrelevant difference, not the presence of a P300.*




*Simulated signals for illustration — typical scalp magnitudes, not recorded data. Not a universal waveform or diagnostic threshold.*

**Target:** Large P300 in everyone → attention check

**Probe: may be enhanced if recognised**

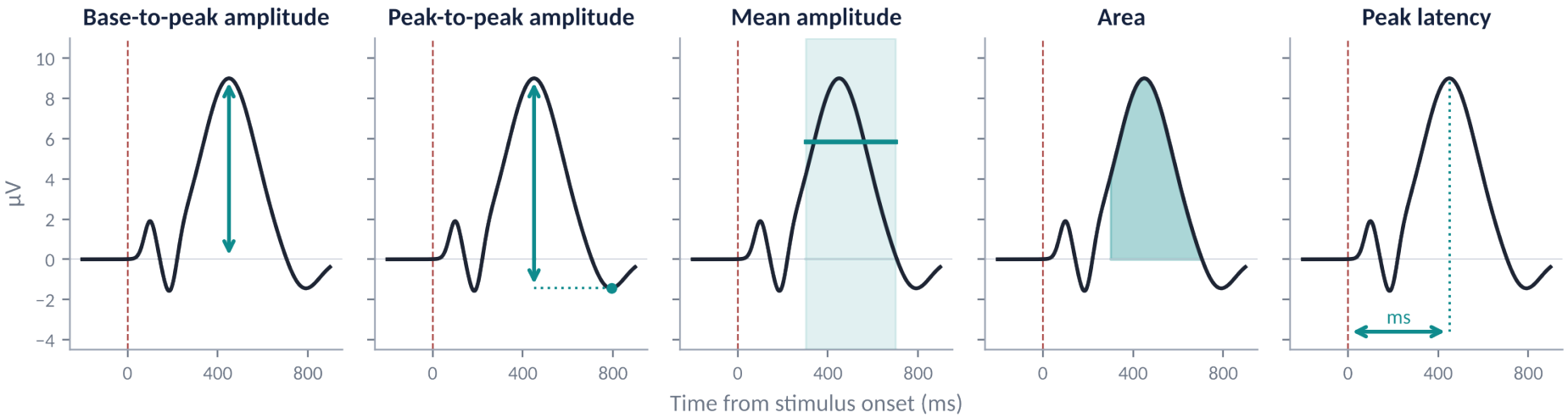
**Irrelevants:** Baseline for the comparison

**Evidence = PROBE − IRRELEVANT difference**

Sources: [3], [24], [47]

# Measuring the P300: Five Waveform Measures

*Why it matters: different measures of the same waveform can lead to different decisions about the same person.*




*Simulated waveform; window shown 300–700 ms as an example.*

**Base-to-peak**

Largest value in a window, relative to the pre-stimulus baseline

**Peak-to-peak**

Peak minus the following trough; common in P300-CIT work [24], [47]

**Mean amplitude**

Average over a fixed window; less sensitive to noise

**Area**

Integrated positivity in the window

**Latency**

When the peak occurs — timing, not size

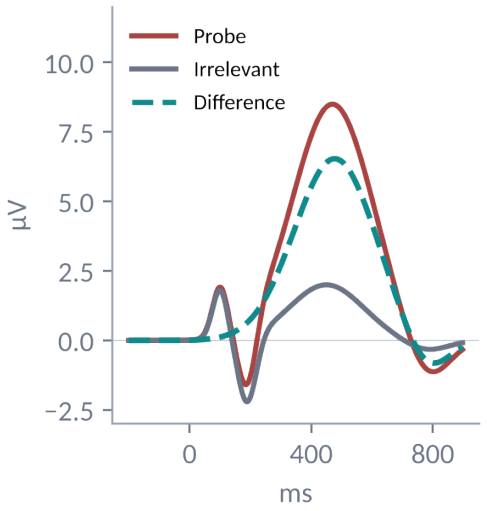
**Fix the window and measure before inspecting results — methods have been compared directly [21]**

Sources: [21], [24], [47]

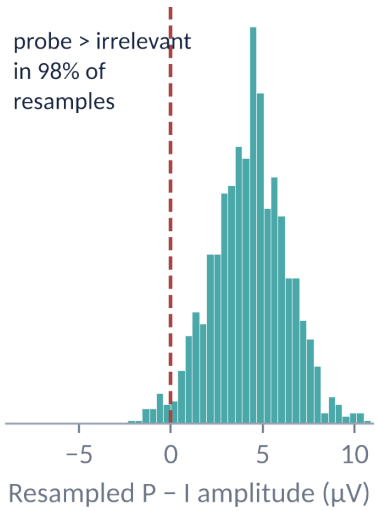
# From Waveform to Decision: Four Approaches

*Why: a decision about one person needs a within-person contrast and an estimate of how reliable it is.*

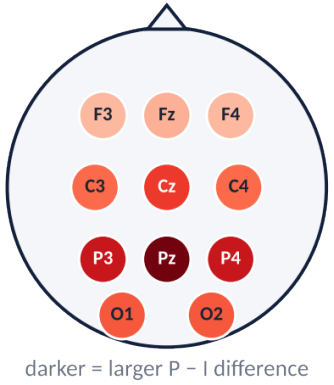
**Probe − irrelevant difference**




**Bootstrap test (one person)**

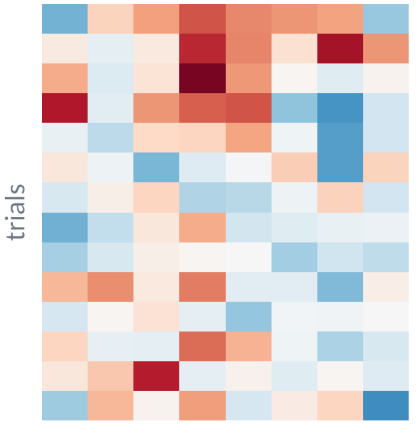



**Multichannel pattern**




darker = larger P − I difference

**Single-trial feature matrix**




features (channels × time windows)

*Simulated illustrations.*

### Difference

The within-person contrast that carries the inference

### Bootstrap

Resample trials; e.g. probe > irrelevant in ≥ 90–95% of resamples (criterion varies) [21], [24]

### Multichannel

Spatial pattern across sites, not one electrode

### Single trial

Feature rows for classifiers — Section IV [37]

Sources: [21], [24], [37], [47]

# A Worked Number Example: Probe – Irrelevant

*Why compute it once: the decision rule becomes concrete when the numbers are small enough to follow.*

### 1 One person's epochs (µV, Pz, 300–800 ms)

**Probe** 5.2 | 1.1 | 6.4

**mean = 4.23 µV**

**Irrelevant** 1.9 | 3.0 | 0.6 | 2.7

**mean = 2.05 µV**

**Difference = 4.23 − 2.05 = +2.18 µV**

### 2 How stable is +2.18 µV?

1. Draw 3 probe and 4 irrelevant values with replacement
2. Recompute probe mean − irrelevant mean
3. Repeat for every possible draw (6,912)

**94.1% of resampled differences > 0**

### 3 Decision depends on the criterion

**Criterion 90%**

94.1% ≥ 90% → consistent with recognition

**Criterion 95%**

94.1% < 95% → not decided (≠ not knowledgeable)

**Illustrative numbers, not data. Fix the criterion before seeing the result — the same epochs pass at 90% and fail at 95%.**

Sources: [21], [24]

# How One Bootstrap Draw Is Generated

*Why show one draw: the 94.1% summarises thousands of draws, and each draw is a simple calculation.*

| Step | Values (µV) | Result |
|---|---|---|
| **Original** | Probe 5.2, 1.1, 6.4 · irrelevant 1.9, 3.0, 0.6, 2.7 | +2.18 |
| **Resample probe** | 5.2, 5.2, 1.1 (with replacement) → 11.5 / 3 | 3.833… |
| **Resample irrelevant** | 1.9, 1.9, 0.6, 2.7 (separately) → 7.1 / 4 | 1.775 |
| **New difference** | 3.833… − 1.775, before rounding | +2.058… ≈ +2.06 |
| **Repeat** | all $3^3 \times 4^4 = 6{,}912$ equally likely draws | 6,502 > 0 = 94.1% |

### A draw can reverse the sign

Probe 1.1, 1.1, 1.1 vs irrelevant 3.0 × 4 → −1.90 µV

### Exact vs simulated

All draws: 94.1% exactly · random simulation: ≈ 94%; the estimate varies with the seed

### What 94.1% is not

Not a probability of guilt · not a p-value · “not decided” ≠ “not knowledgeable”

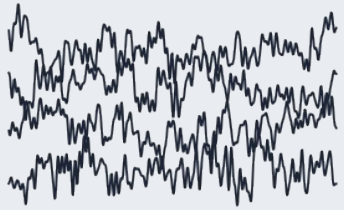
**94.1% = share of resampled differences above zero: stability of the effect, not a probability of guilt**

Sources: [21], [24] · illustrative numbers, not data

# From Measurement to Inference: A Worked Example

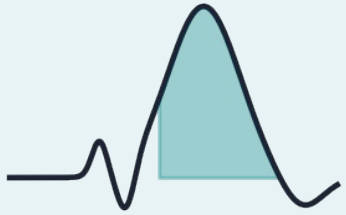
*Why separate the levels: an error at any level propagates, and only the last step is an interpretation.*

### SIGNAL



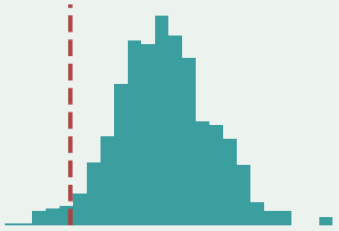
Pz voltage while the red toolbox, irrelevants and target appear

### FEATURE



Probe − irrelevant mean amplitude, 300–700 ms

### DECISION RULE



Bootstrap: is the difference reliable for this person?

### INTERPRETATION

✓ **Consistent with recognition of the red toolbox**

× not: Deception

× not: Guilt

*Recognition may be innocent [11], [14]*

**A strong decision rule cannot rescue an invalid interpretation**

Sources: [11], [14], [21], [47]

# Experimental Design and Dataset

One public P300-CIT study: the mock theft, the participants, the six images and the data files

# One Experiment, One Dataset: Why and How

*Why this section: methods become concrete only in a real experiment, and the same data feed the machine-learning section.*

**Why this dataset**

- Public: downloadable from Dryad
- Genuine three-stimulus P300-CIT: probe · target · irrelevants [37]
- Two groups: knowledgeable and control
- Already used for a machine-learning pipeline [37] — the bridge to Section IV

SECTION ROADMAP


1 The study at a glance

2 Participants, groups and roles

3 The mock theft and the six images

4 Stimulus presentation and EEG recording

5 The files on disk: what the numbers are

6 A first look at the data


**Sources of every fact:** the paper [37] · the Dryad README · inspection of the files. Anything added for clarity is marked **Assumed.**

Sources: [37] · Dryad doi:10.5061/dryad.2qc64 (README and data files)

# The Study at a Glance

*Why first: every later slide refers back to these numbers, so they are collected in one place with their source.*

| | | |
|---|---|---|
| **Study** | Gao et al. (2014), PLOS ONE [37] | paper |
| **Data** | Dryad (2015): one 568 MB RAR archive + README | Dryad |
| **Participants** | Abstract: 34 · Methods section and release: 30 (unresolved) | paper · files |
| **Groups** | 15 in folder lying (knowledgeable) · 15 in folder honest (control) | files |
| **Paradigm** | Mock theft → three-stimulus P300-CIT with 6 object photographs | paper |
| **Presentation** | 0.5 s per image · ISI 1.6 s · each object 30× per session | README |
| **Sessions** | 5 per participant → max 150 probe · 150 target · 600 irrelevant | README |
| **Recording** | 14 channels (12 EEG + 2 EOG) · 500 Hz · 0.1–30 Hz · right-earlobe reference | paper · files |
| **Released form** | Cut, baseline-corrected epochs: −300 to +1298 ms = 800 samples | files (verified) |

*Right-hand tags show where each fact comes from.*

Sources: [37] · Dryad doi:10.5061/dryad.2qc64 (README and data files)

# Key Terms: Groups, Roles and Stimulus Types

*Why define them: the paper, the folders and this seminar use different words for the same things.*

| | PAPER | FOLDER | TERM IN THIS SEMINAR | MEANING IN THIS STUDY |
|---|---|---|---|---|
| Group | *guilty* | *lying* | **Knowledgeable (experimental) group** | Committed the mock theft; knows the stolen object; conceals it |
| Group | *innocent* | *honest* | **Control group** | Same procedure, no theft: never saw the probe; nothing to conceal |
| Stimulus | *probe* | *probe* | **Probe** | Crime-relevant object: stolen (knowledgeable) · never seen (control) |
| Stimulus | *target* | *target* | **Target** | Object seen in the safe; known to everyone; keeps attention on the task |
| Stimulus | *irrelevant* | *irrelevant* | **Irrelevant (×4)** | Objects never linked to the event; the comparison baseline |

**Why a control group?**

It goes through everything except learning the probe, so it shows the probe response without knowledge.

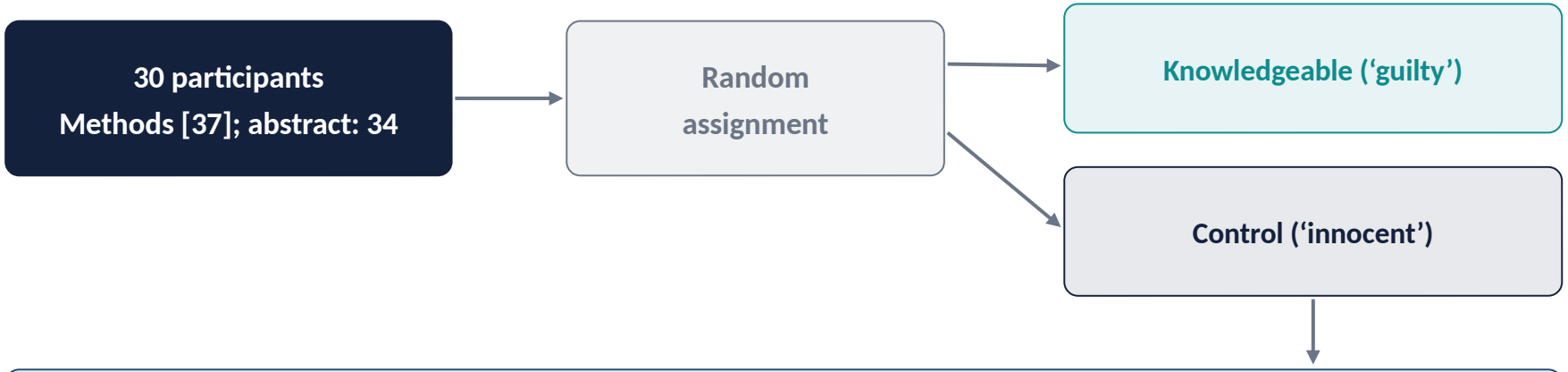
**Labels ≠ measurement**

'Guilty' and 'lying' name the assigned role. The EEG can at most show recognition of the probe.

Sources: [37] · Dryad doi:10.5061/dryad.2qc64 (README and data files)

# Participants: Who Is in Which Folder?

*Why ask: the folder names suggest one person might appear twice; the design rules that out.*




**Public release: 30 participants — 15 in lying/ · 15 in honest/ (the abstract reports 34; the difference is unresolved)**

EACH PARTICIPANT IN THE RELEASE

| 5 | 3 | 15 | 450 |
|---|---|---|---|
| sessions | files per session | files per person | files in total |

**Same person twice?**

- No: groups are between subjects
- A thief cannot later be unaware of the theft
- Numbering restarts per folder: lying/subject9 ≠ honest/subject9
- Seminar codes: L09 and H09

Sources: [37] · Dryad doi:10.5061/dryad.2qc64 (README and data files)

# The Mock Theft, Step by Step

*Why start here: a probe effect can mean recognition only if the two groups had different exposure to the probe.*

**Knowledgeable (experimental-guilty)**

Operator places 2 random objects in a safe → Participant opens the safe and 'steals' one → Writes down style and colour of the safe's objects

Resulting stimulus roles

- **Probe** — the stolen object
- **Target** — the other object seen
- **Irrelevants** — 4 objects never seen

**Control (innocent)**

Operator places 1 random object in a safe → Participant only watches the object → Writes down style and colour of the safe's object

Resulting stimulus roles

- **Probe** — drawn at random from the 5 unseen — never seen
- **Target** — the watched object
- **Irrelevants** — remaining 4 objects

**EEG test:** all six objects appear repeatedly in random order; knowledgeable participants conceal the stolen object. **Same probe role, different history:** seen and stolen vs never seen.

Sources: [37] · procedure as depicted in Fig. 2 of [37]

# What Each Participant Did During the Test

*Why it matters: the instruction decides what a probe response can mean — recognition with concealment, or no recognition at all.*

**Knowledgeable participant**

- Knows two objects: the stolen one (probe) and the other one in the safe (target)
- Instructed to conceal the theft: treats the probe as unfamiliar
- Probe = recognized and concealed

**Control participant**

- Knows one object: the watched one (target)
- Nothing to conceal: responds truthfully
- Probe = just another unfamiliar object

**Reported in [37]: YES for the known target, NO for every other object, so a knowledgeable participant's NO to the probe is the concealing response. The raw event table shows the two response codes.**

**Prediction:** knowledgeable → probe behaves like the target (larger P300 than irrelevants) · control → probe behaves like an irrelevant

Sources: [37] (Experimental Protocol; Fig. 2) · event table of the raw .cnt file (L12, session 1)

# The Six Stimulus Images

*Why look: a probe should stand out only because it is recognized, not because it looks different from the alternatives.*



Object 1



Object 2



**Object 3 · pendant**



Object 4



Object 5



Object 6

### What the objects are

- Five bead or pearl bracelets
- One metal chain pendant: visually distinctive

### How roles were assigned

- Drawn at random for every participant
- Spreads, but does not remove, item effects [44]

### What the files do not say

- Which object was probe, target or irrelevant for each person

Sources: [37], [44] · Images: Gao et al. (2014), PLOS ONE 9(11): e109700, Fig. 2, CC BY licence

# One Session: How the Images Were Presented

*Why it matters: rarity drives the P300, so the number and order of presentations define the comparison.*




*Order and roles in the strip are illustrative; the actual trial order is not stored in the files.*

**180**
**presentations per session**
30 probe · 30 target · 120 irrelevant

**× 5**
**sessions**
→ 150 · 150 · 600 per person

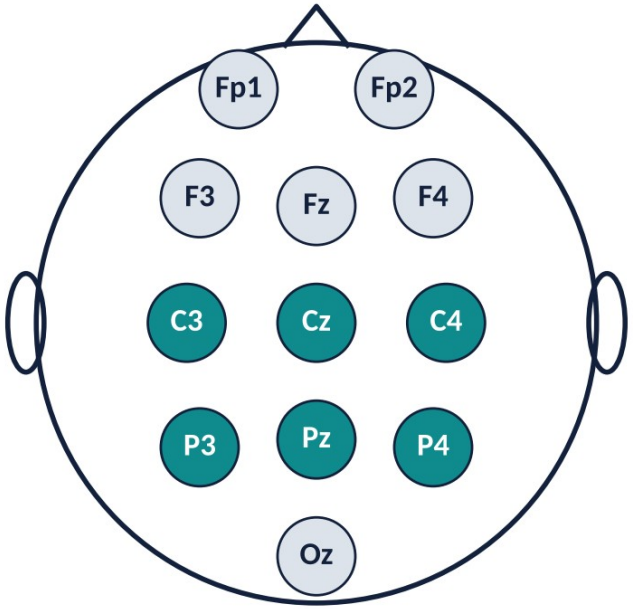
**1/6**
**probe = target rarity**
rarity alone cannot separate them

**≈ 5–6 min**
**per session (Assumed)**
ISI onset vs offset not stated

Sources: [37] · README of Dryad doi:10.5061/dryad.2qc64 (README and data files)

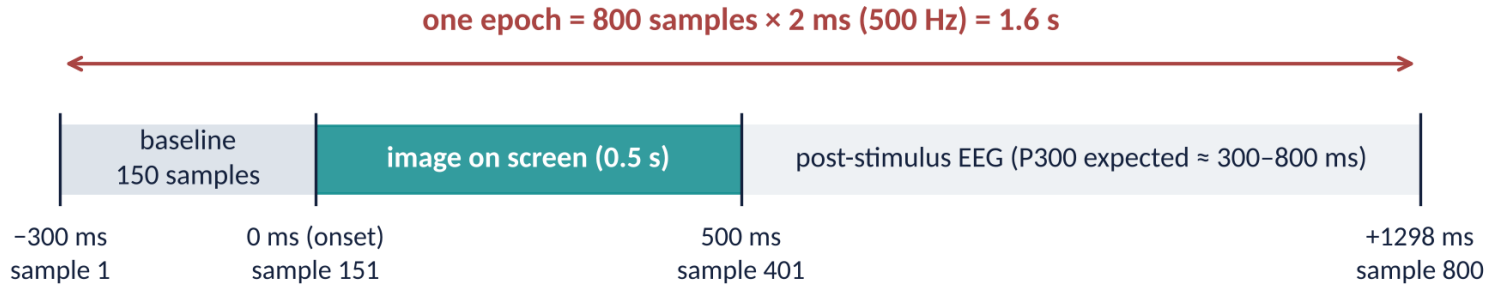
# EEG Recording and the Epoch Window

*Why both: the montage says where the signal comes from; the epoch window says which part of it was kept.*




12 EEG sites (10–20 names) + HEOG and VEOG eye channels

teal = centro-parietal sites where the P300 is usually largest




## Recording

- 14 channels: 12 EEG + HEOG, VEOG
- 500 Hz = 1 sample every 2 ms
- Right-earlobe ref., online 0.1–30 Hz [37]; not in README

## Released epochs

- Cut by the authors around each image
- Contaminated trials already removed
- 300 ms baseline already subtracted

Sources: [37] · Dryad doi:10.5061/dryad.2qc64 (README and data files) · baseline, epoch length and time zero verified against the raw .cnt file

# The Dataset on Disk: Folder Structure

*Why start with folders: the files contain only numbers, so every label — group, person, session, stimulus — lives in the folder path.*

```
All data for 14 electrodes and 30 subjects/
└─ 14 electrodes-30 subjects/
   ├─ honest/                    ← control group, 15 people
   │  ├─ subject1/ … subject15/
   │  │  ├─ session1/ … session5/
   │  │  │  ├─ irrelevant/irrelevant.txt
   │  │  │  ├─ probe/probe.txt
   │  │  │  └─ target/target.txt
   └─ lying/                     ← knowledgeable group, 15 people
      └─ subject1/ … subject15/   (same layout)
```

**Path = label:** **lying / subject9 / session1 / probe**

### Counts

- 2 groups × 15 people × 5 sessions × 3 files = 450 text files
- Download: one .rar archive (568 MB) + README
- ≈ 2.4 GB of text once extracted (estimate)

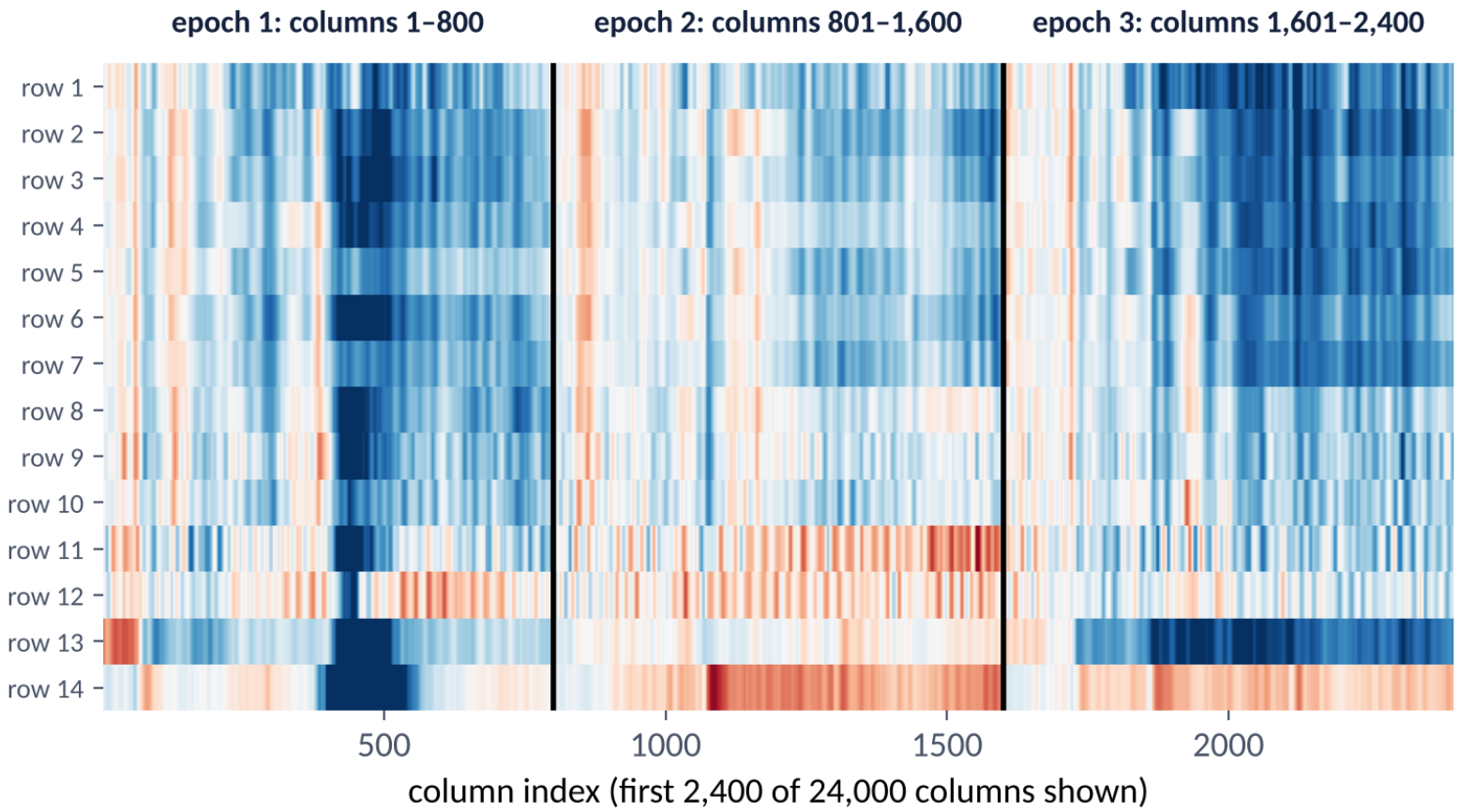
### What the README does not say

- Epoch length and time axis
- Channel names and order
- Units of the values
- Which object was the probe

Sources: README of Dryad doi:10.5061/dryad.2qc64 (README and data files)

# Inside One Text File: What the Numbers Are

*Why open a file: a row of bare numbers is useless until its rows, columns and units are known.*




```
probe.txt  (lying/subject9/session1)

rows      14   (one per channel)
columns   24,000

24,000 ÷ 800 = 30 epochs

first values in row 1:
  -4.0362  -3.7005
  -2.8612  -1.6863
  -0.5114   0.3279
   1.5028   2.8456
  …
```

Sources: Own inspection of Dryad doi:10.5061/dryad.2qc64 (README and data files) · channel order and µV unit verified against the raw .cnt header

# Decoding the File: Rows, Columns, Values, Labels

*Why decode each part: every later step depends on reading the matrix correctly, and two parts are not self-describing.*

## Rows = 14 channels

Verified*

- Exactly 14 rows in every file
- No channel names in the text files
- Order from the raw .cnt header: Oz Pz P4 C4 P3 Cz C3 Fz F4 F3 Fp2 Fp1, HEOG, VEOG

## Columns = time samples

Verified

- Epochs placed side by side, 800 columns each
- probe.txt: 24,000 = 30 epochs (one per presentation)
- irrelevant.txt: 95,200 = 119 epochs (1 of 120 rejected)

## Values = voltage

Verified*

- Unit not stated in the README; raw .cnt step = 0.168 µV
- Text values equal the raw recording in µV

## No labels inside

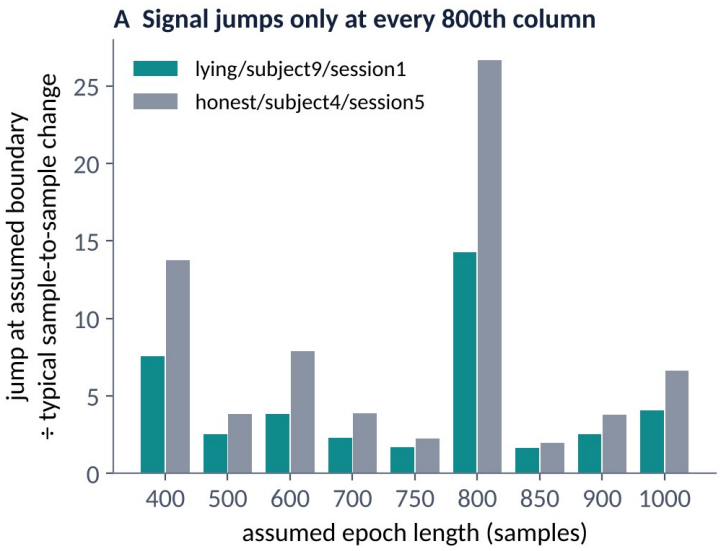
Verified

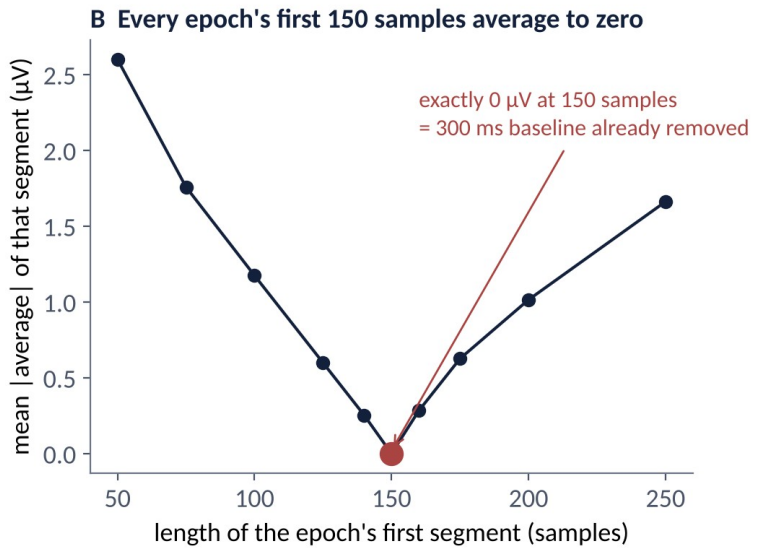
- No header, no time axis, no trial labels
- Group, person, session and stimulus come only from the folder path

Sources: Own inspection of Dryad doi:10.5061/dryad.2qc64 (README and data files) · *verified against the one raw .cnt file (L12, session 1); same export assumed for all files

# How the 800-Sample Epoch Length Was Established

*Why verify: the README never states the epoch length, and a wrong length would scramble every epoch.*




**Epoch = 800 samples: −300 to +1298 ms, onset at sample 151**

*Files inspected: lying/subject9/session1 and honest/subject4/session5 (all three stimulus files of each). The full set of usable files is cut with the same 800-sample rule; time zero is checked on the next slide.*

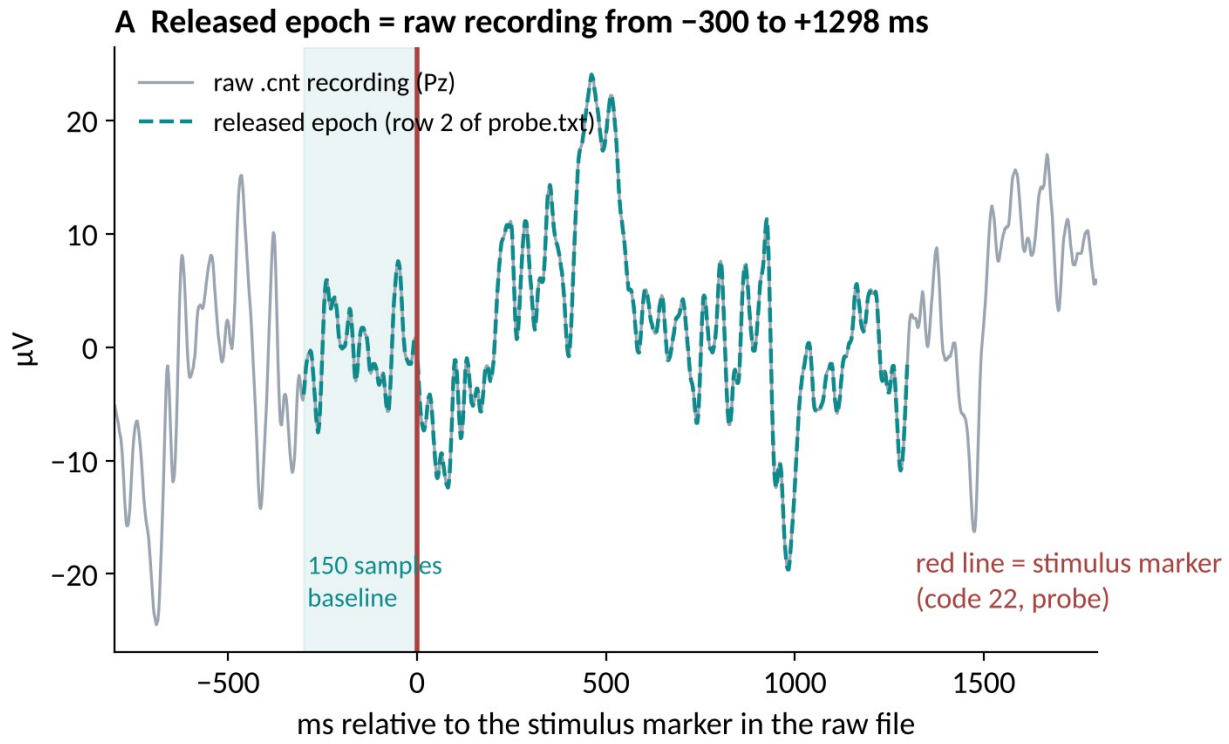
## Three checks agree

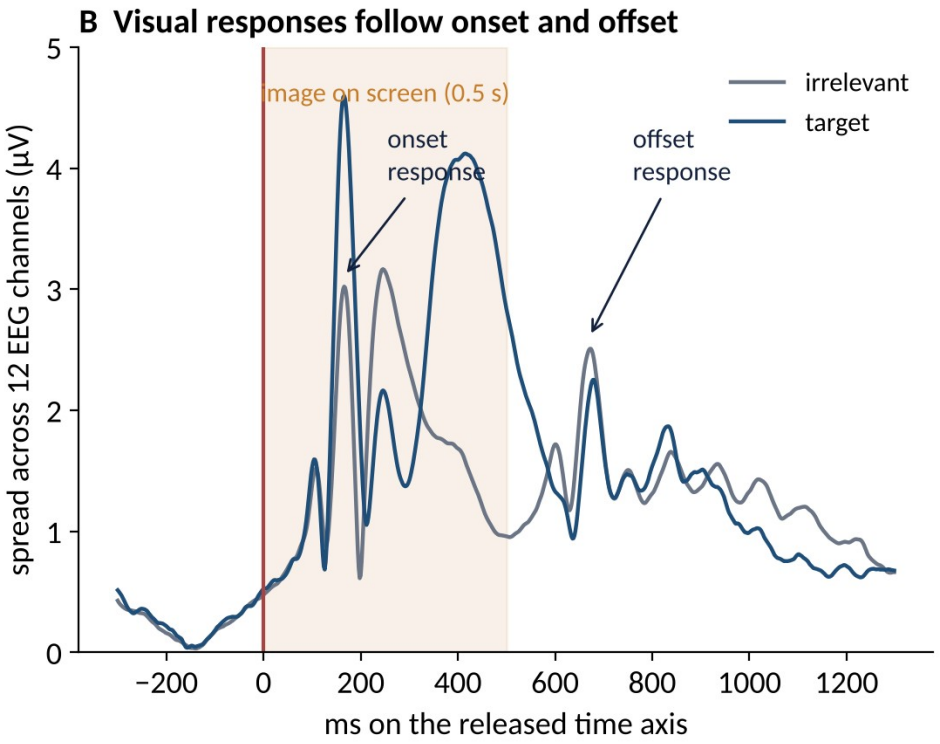
- Column counts are multiples of 800 (24,000 · 95,200 · 21,600)
- Jumps occur at every 800th column (A)
- First 150 samples average exactly 0 (B) → onset at sample 151
- 800 × 2 ms = 1.6 s: matches the README interval numerically (corroboration only)

Sources: Own inspection of Dryad doi:10.5061/dryad.2qc64 (README and data files)

# Time Zero: Article Description vs Released Files

*Why check: every latency and window in this deck depends on where 0 ms sits in the released epochs.*




**Article [37]:** −500 to +1100 ms

**Released files (used here):** −300 to +1298 ms, marker at sample 151

Sources: [37] · raw file lying/subject12/session4/xiaowushuang1.cnt (= session 1) · B: all 30 participants, epochs > ±100 µV removed

# From Text Files to a Labelled Dataset

*Why spell it out: this conversion is the first step of every analysis in Section IV, and it creates two different kinds of label.*

**1 Read file**
14 × (n·800) matrix

**2 Cut**
n = columns ÷ 800

**3 Reshape**
n × 14 × 800 epochs

**4 Label**
group · subject · session · stimulus

```
from pathlib import Path
import numpy as np

f = Path('lying/subject9/session1/probe/probe.txt')
X = np.loadtxt(f)              # 14 × 24000
n = X.shape[1] // 800          # 30 epochs
ep = X.reshape(14, n, 800)
ep = ep.transpose(1, 0, 2)     # 30 × 14 × 800
group, subj, sess, stim = f.parts[-5:-1]
# 'lying', 'subject9', 'session1', 'probe'
```

**Result**
25,740 epochs, each 14 × 800, with four labels

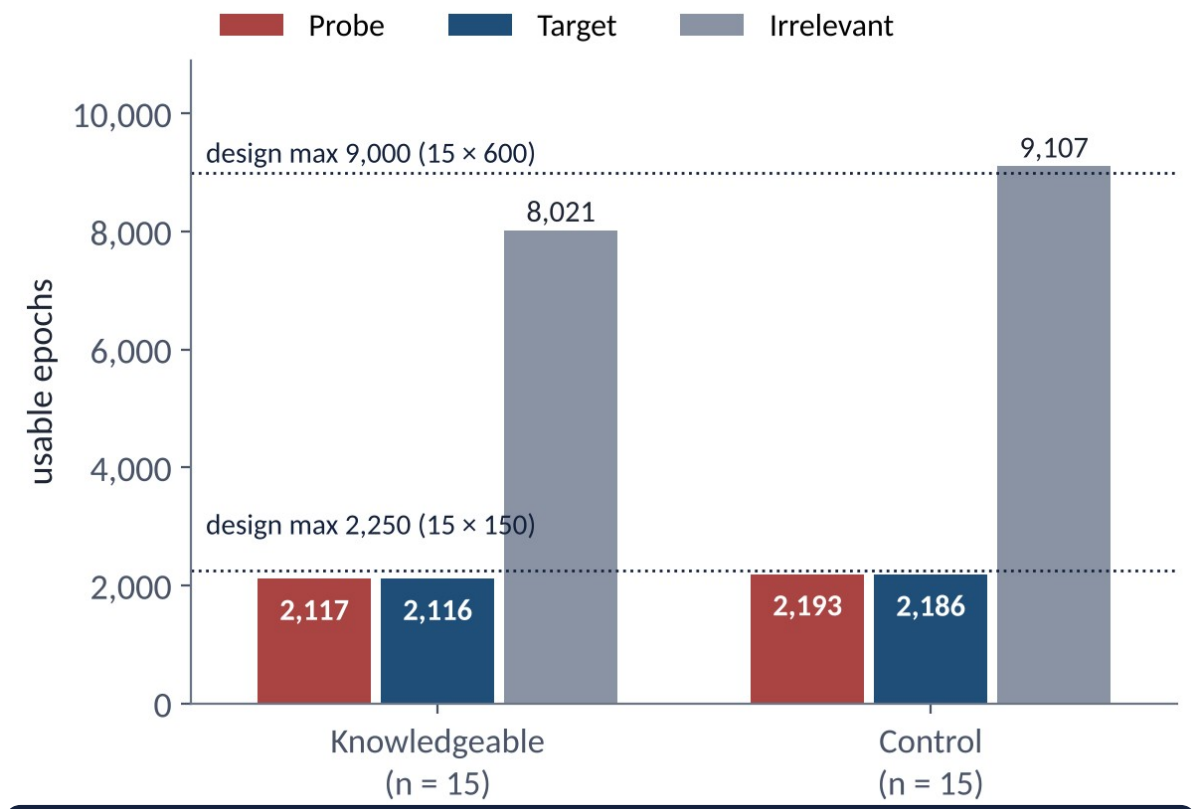
**Verified***
Channel order and µV unit from the raw .cnt (previous slides)

**Two kinds of label:** **stimulus type** = trial label (what was on screen) · **group** = person label (what the participant knew). **Trial accuracy ≠ person accuracy.**

Sources: Own processing of Dryad doi:10.5061/dryad.2qc64 (README and data files)

# Released vs Usable Data: Trial Counts

*Why count: missing or unequal trials can create or hide a probe–irrelevant difference.*




**25,740 epochs — but only 30 independent people**

## Released state

- Epochs had already passed the authors' trial rejection [37]
- 9 defective files excluded first (empty, not epoched, duplicated, conflicting labels, mis-filed)

## Usable data

- 25,740 epochs: 12,254 knowledgeable · 13,486 control
- Probe and target counts near the design maximum

## Flags to keep in mind

- 3 control files exceed 600 irrelevants (H09, H12, H15); origin unresolved
- 16.7% of released epochs still exceed ±75 µV on ≥ 1 EEG channel

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own file audit (no classification) · file manifest with MD5 checksums supplied separately

# Analysis Rules, Fixed Before Looking at Results

*Why state them: a result is interpretable only if site, window and measure were not chosen after seeing the data.*

**1 Site**

Pz — classic P300 site; all 12 channels on a later slide

**2 Window**

300–800 ms, fixed in advance; not taken from [37]

**3 Measure**

Per participant: mean probe − mean irrelevant amplitude (µV)

**4 Uncertainty**

95% bootstrap interval over that participant's trials

**Deliberately not done**

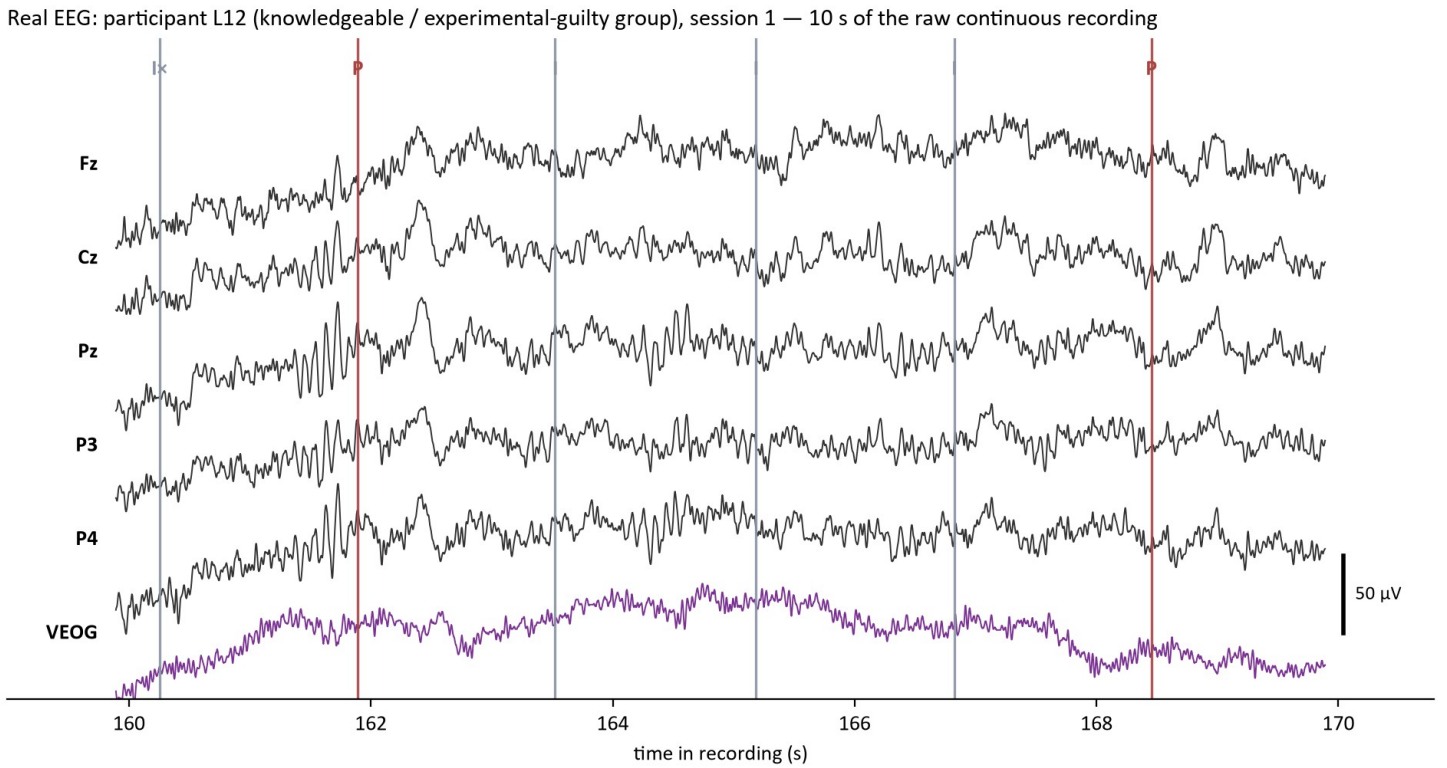
No threshold · no classifier · no accuracy · peak latencies reported only as exploratory, never reused as windows

**Descriptive analysis: it shows the pattern in one dataset; it does not test a detection method**

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification)

# Real Data: Raw Multichannel EEG

*Why show it: every average and every model starts from this raw signal — 10 s from L12, session 1, with stimulus markers.*

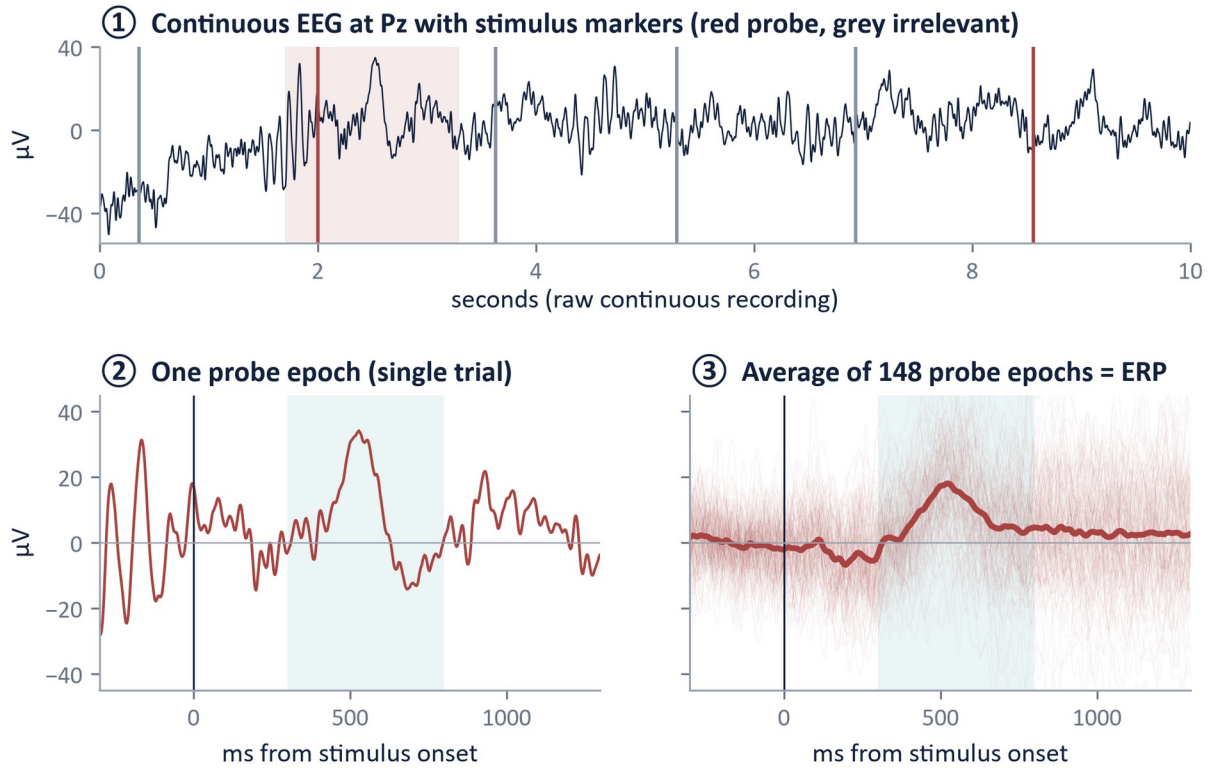



Raw Neuroscan recording as digitized (500 Hz; online 0.1–30 Hz band-pass and right-earlobe reference per Gao et al.); no offline filtering. Each trace has its segment mean removed for display only.
Vertical lines = stimulus markers: T target, P probe, I irrelevant; × = trial later rejected in the released epochs. Segment chosen by a fixed rule (starts 2 s before the first probe after the recording midpoint), not for cleanliness. VEOG = vertical eye channel.

Sources: [37] · raw file in the Dryad release: lying/subject12/session4/xiaowushuang1.cnt (= session 1; matches the released epochs sample for sample) · markers from its event table

# Real Data: From Continuous EEG to an ERP

*Why show it: a recognition-related ERP is hard to see by eye in single noisy trials; averaging makes it clearer.*




## How to read the three panels

- ① Continuous Pz recording; red and grey lines mark probe and irrelevant onsets
- ② One probe epoch cut from −300 to +1298 ms: no P300 visible by eye
- ③ Average of this participant's 148 probe epochs: a late positivity appears in the shaded window

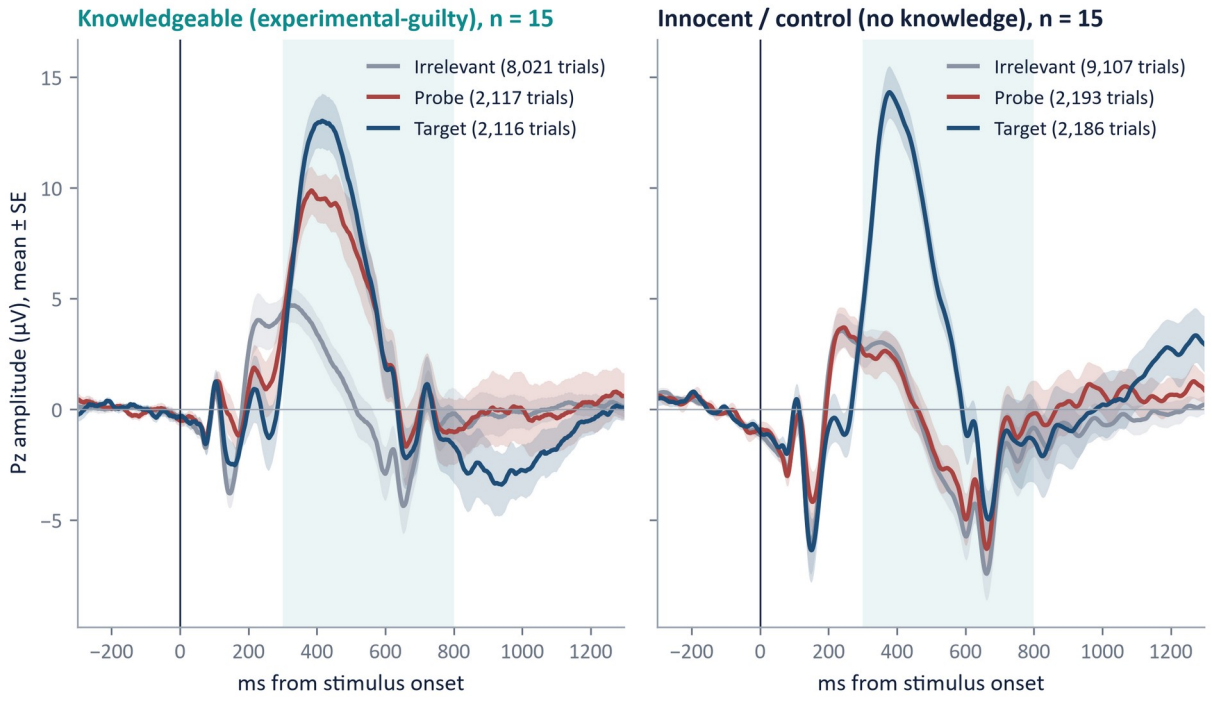
## Point

For this participant and illustration, averaging makes the late positivity clearer; single-trial models can still learn from individual epochs

Sources: [37] · Dryad doi:10.5061/dryad.2qc64 · panel ① from the raw .cnt file in the release · own descriptive analysis (no classification)

# Real ERPs: Do Probe and Irrelevant Differ?

*Why test it: the conceptual ERPs made a prediction that real data can confirm or contradict.*




### Knowledgeable (experimental-guilty)

- Probe and target both exceed irrelevants
- Probe − irrelevant +3.9 µV (300–800 ms)
- Largest mean difference at Pz among recorded channels

### Innocent / control

- Probe resembles irrelevants (+0.5 µV)
- Target P300 intact

### Caution

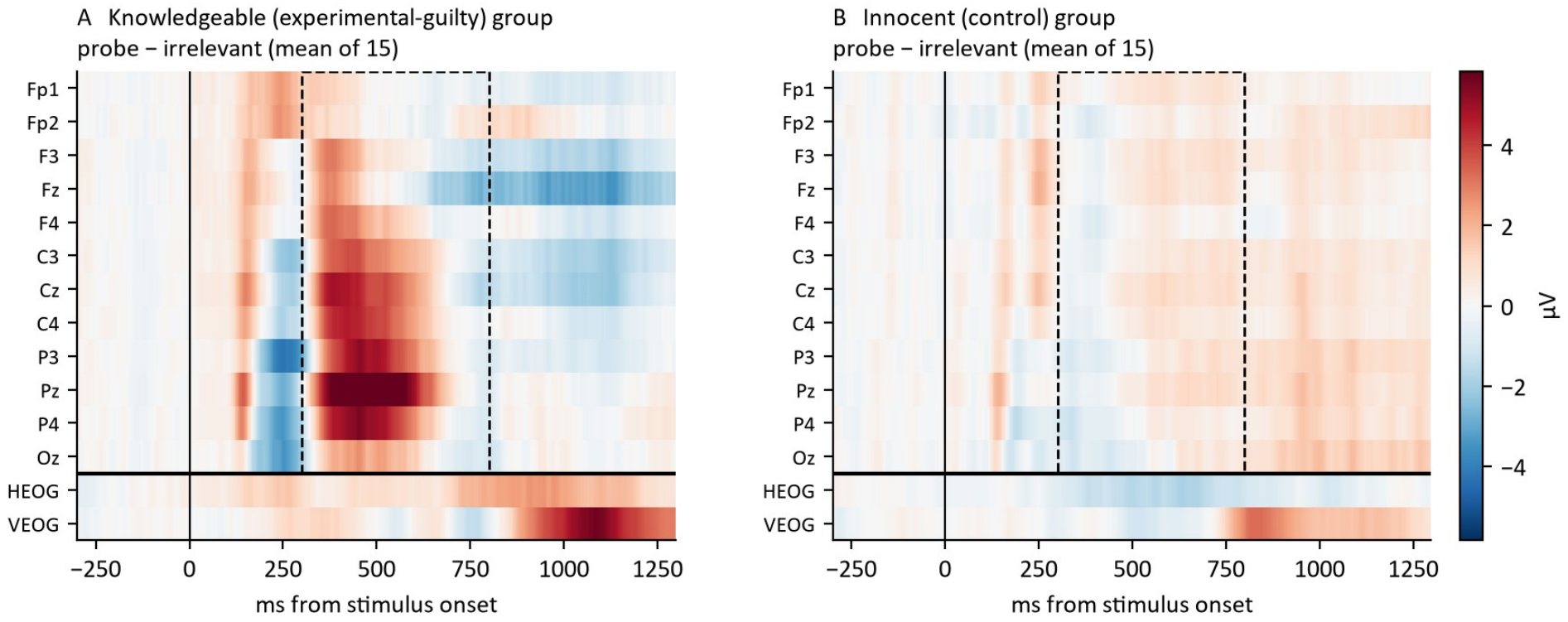
- Window fixed in advance, not from [37]; one laboratory study

**Conceptual prediction → observed pattern: the information is the conditional probe–irrelevant contrast, not the mere presence of a P300**

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification) · predeclared: Pz, 300–800 ms; peaks exploratory

# Real Data: Probe – Irrelevant by Channel and Time

*Why look at every channel and time point first: the map shows where and when the probe differs before any summary is computed.*

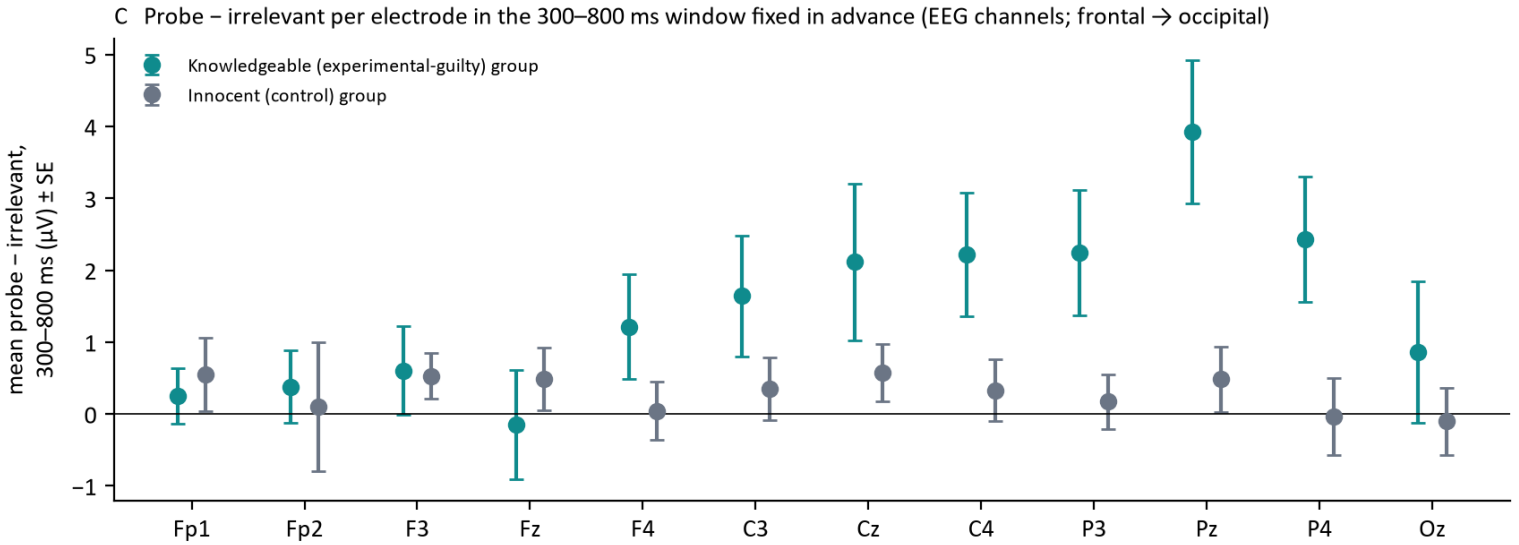


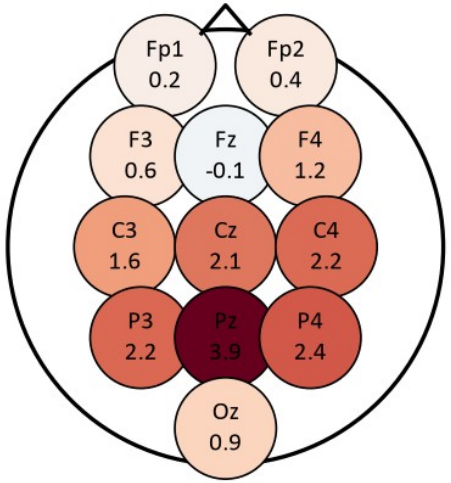

*Both groups share one colour scale · dashed box: 300–800 ms window fixed in advance · HEOG / VEOG below the black line are eye channels · rows front to back*

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification) · predeclared: Pz, 300–800 ms; peaks exploratory

# Real Data: Where on the Scalp Is the Difference?

*Why summarise per electrode: one number per site shows whether the effect has a P300-like scalp distribution.*




*Schematic 10–20 positions · electrode values only*

**Knowledgeable**

Pz 3.9 µV; P4, P3, C4, Cz 2.1–2.4 µV

**Controls and frontal sites**

Controls ≤ 0.6 µV everywhere; frontal ≈ 0

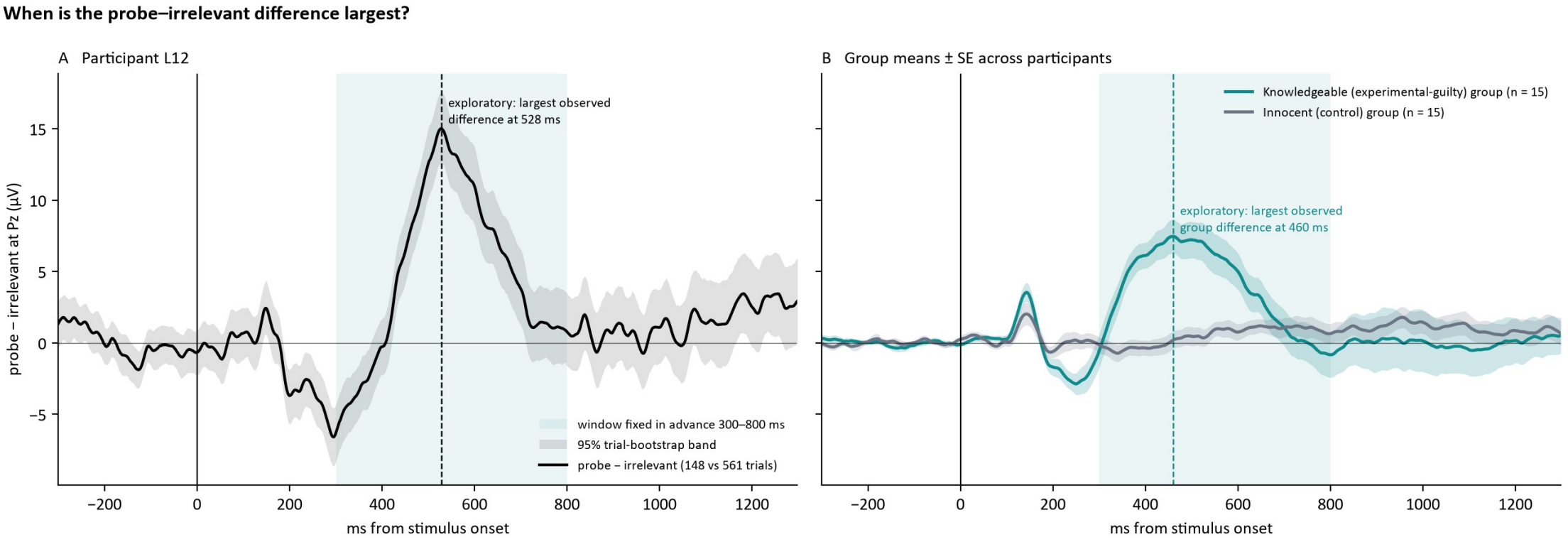
**Eye channels**

Small in the window; late VEOG effect > 900 ms

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification) · predeclared: Pz, 300–800 ms; peaks exploratory

# Real Data: When Is the Probe–Irrelevant Difference Largest?

*Why separate them: the window fixed in advance (shaded) is not the exploratory peak (dashed).*

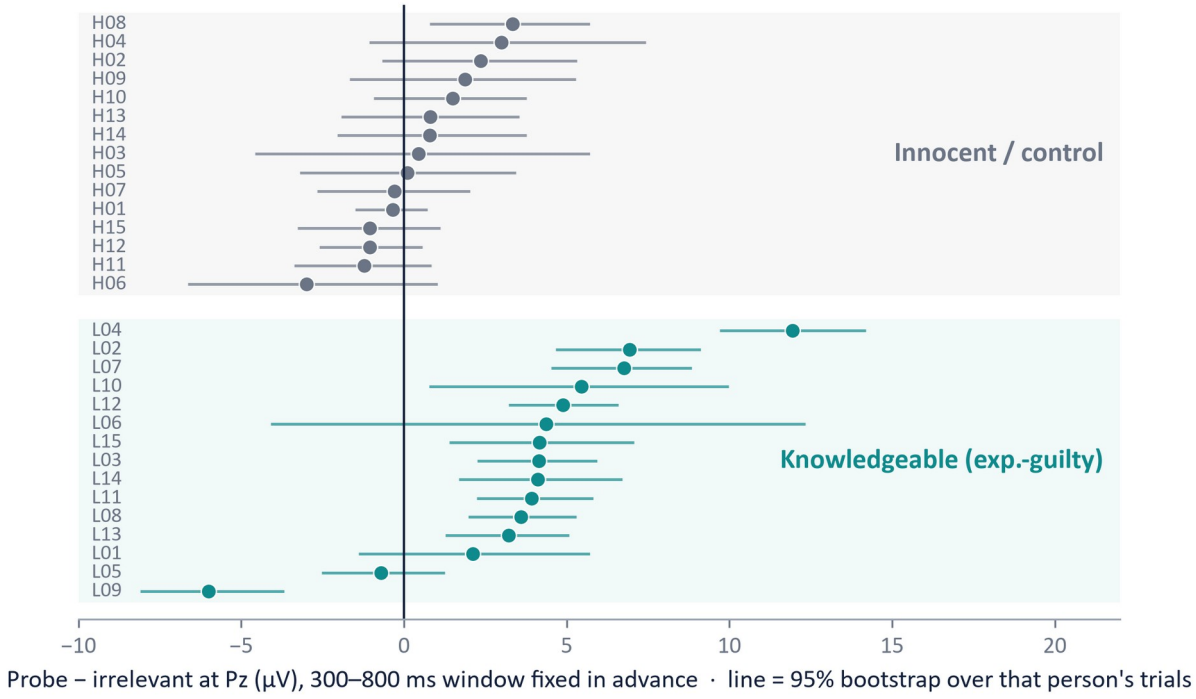



ORIGINAL STUDY WINDOW: none — Gao et al. (2014) computed features over the whole post-stimulus interval (0 to +1.1 s) and specified no P300 amplitude window.
WINDOW FIXED IN ADVANCE (shaded, 300–800 ms): a conventional late-positivity range set before this inspection, not tuned. EXPLORATORY (dashed): time of the largest observed difference — descriptive only, not a window for later analyses.

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification)

# Real Data: Group Effect vs Individual Variability

*Why individuals: a group effect does not mean that every person shows it.*




### How to read

- One row per participant; dot = probe − irrelevant at Pz, 300–800 ms
- Line = 95% bootstrap interval; right of 0 = probe larger

### What it shows

- Knowledgeable: 13/15 positive, median +4.1 µV (−6.0 to +11.9); L09 reversed
- Control: 9/15 positive, median +0.4 µV (−3.0 to +3.3) — ranges overlap

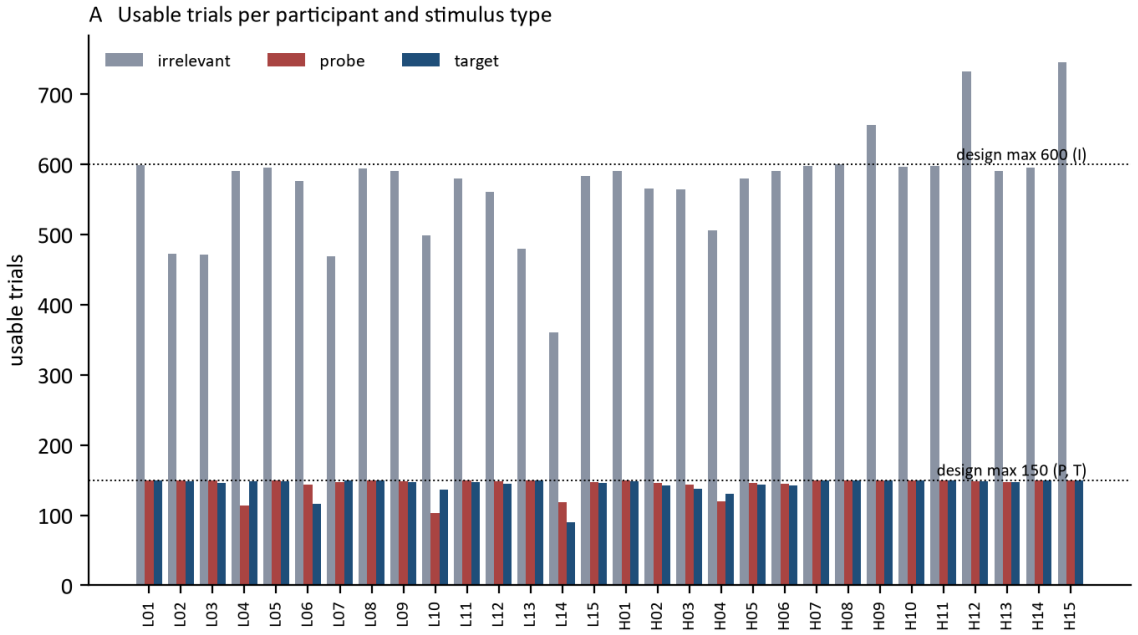
**Descriptive only: no threshold, no accuracy; N = 30**

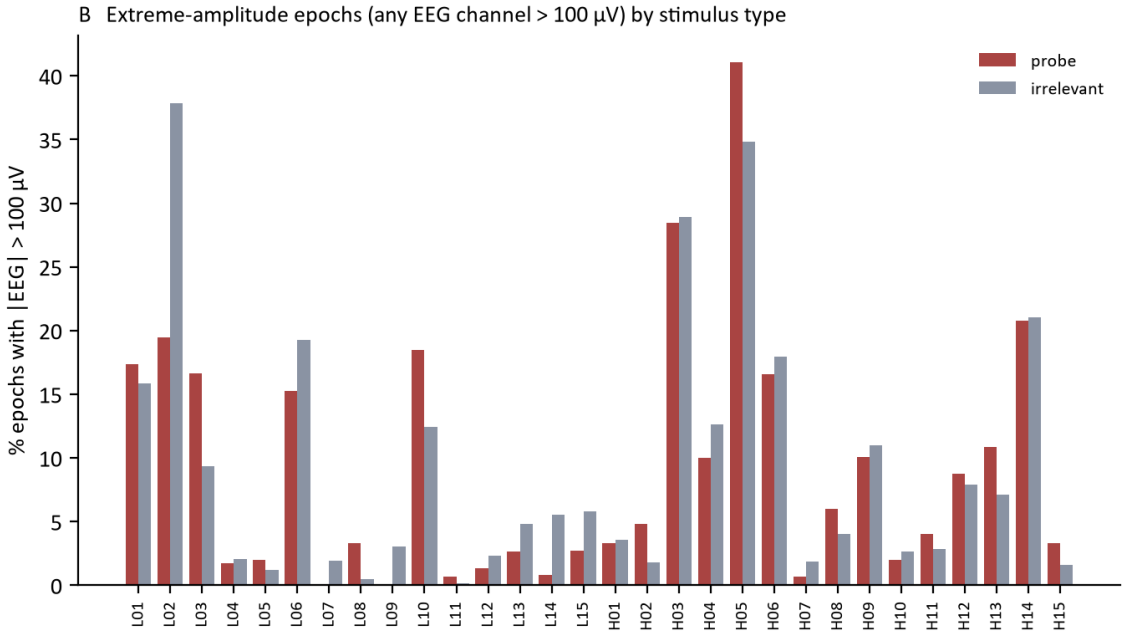
**Computational question: can multichannel, temporal and single-trial information improve person-level discrimination beyond one mean Pz amplitude?**

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification) · predeclared: Pz, 300–800 ms; peaks exploratory

# Data Quality: Are Trials Equal Across Stimulus Types?

*Why check: if probe trials were fewer or noisier than irrelevant trials, the difference could be an artifact.*




### How to read A

Bars = usable trials per participant and stimulus type; dotted lines = design maxima

### How to read B

Per participant: % of probe and irrelevant epochs with any EEG channel > 100 µV

### Takeaway

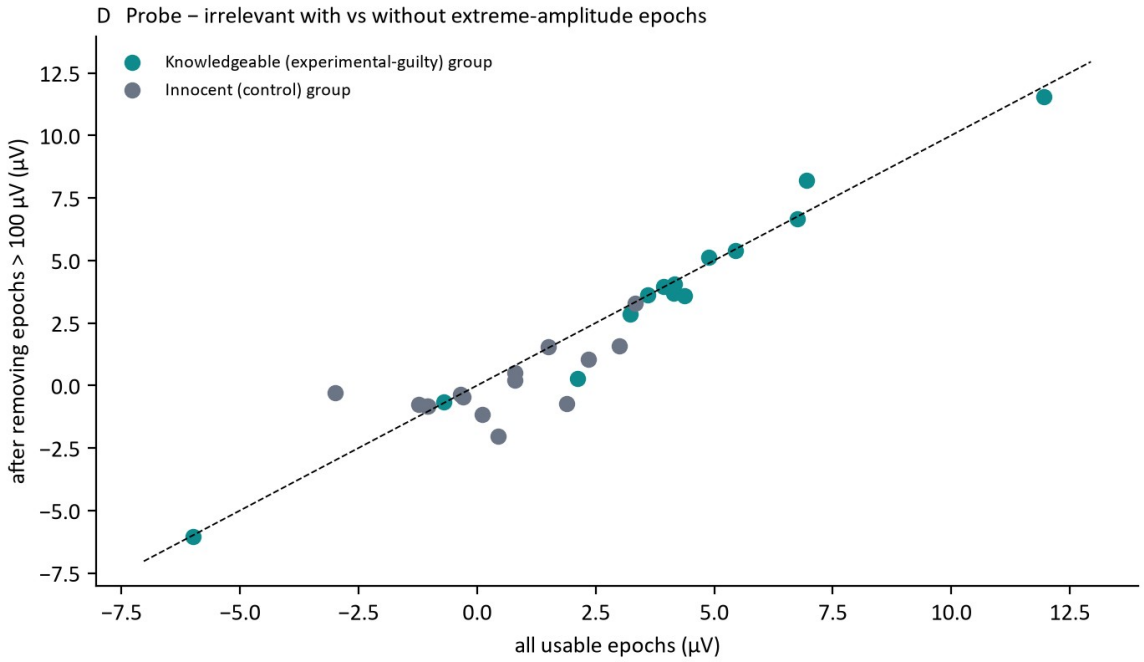
Artifact rates vary strongly across people; probe and irrelevant rates look broadly similar here — descriptive, not a test of equivalence

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification)

# Data Quality (2): Does Noise Explain the Exceptions?

*Why check: if the exceptions were simply noisy recordings, cleaning would remove them — the real-data test of that idea.*




### How to read C

Each dot = one participant: artifact rate (x) vs probe – irrelevant effect (y)

### How to read D

Effect after removing epochs > 100 µV vs original; diagonal = unchanged

### Takeaway

ρ = 0.30 / 0.03; mean 3.9 → 3.7 µV; 13/15 still positive; L09 is clean

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification)

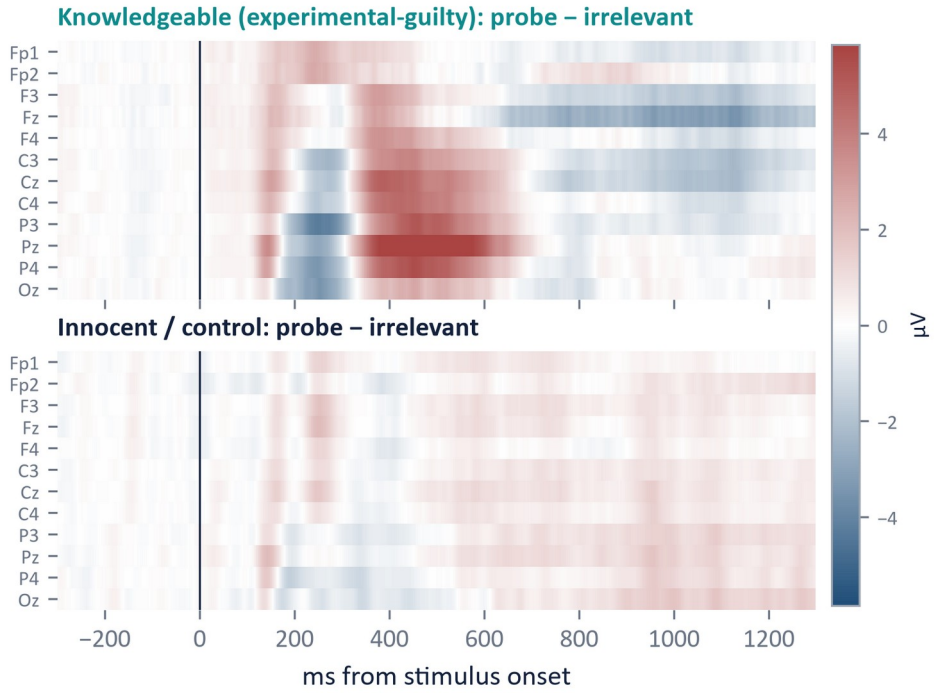
# Computational Methods: From ERP Statistics to AI

Data structure · features and classical ML · deep learning · training and validation · person-level decisions · results on the Gao data and what they mean.

I · What & why | II · Methods | III · Experiment & data | **IV · Computation** | V · Limitations | VI · Future

# What Does the Real Dataset Turn Into Computationally?

*The AI question: can models learn recognition-related EEG patterns that generalize to unseen individuals and conditions?*




| OBSERVATION IN THE REAL DATA | | COMPUTATIONAL PROBLEM |
|---|---|---|
| Single trials are noisy | → | Signal enhancement · robust representations |
| Peak timing varies (exploratory: median 454 ms, range 316–670 ms) | → | Temporal features or learned temporal filters |
| Effect spans centro-parietal channels | → | Multichannel spatial modelling, not one Pz value |
| Participants differ greatly (−6 to +12 µV) | → | Subject-independent generalization |
| Groups overlap at the edges | → | Probabilistic decisions with calibrated uncertainty |

**Goal: a representation that generalizes to an unseen person — without learning participant identity, artifacts or experimental shortcuts**

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 · own descriptive analysis (no classification) · peak timing exploratory

# What Does an EEG Dataset Actually Contain?

*Why show this: a model sees tensors plus metadata, and CIT data carry two different kinds of label.*

## From participant to epoch

Participant (30: 15 knowledgeable, 15 control)

↓

5 sessions per participant

↓

Recording: 14 channels (12 EEG + 2 EOG), 500 Hz — released as epochs

↓

Stimulus events: probe · target · irrelevant

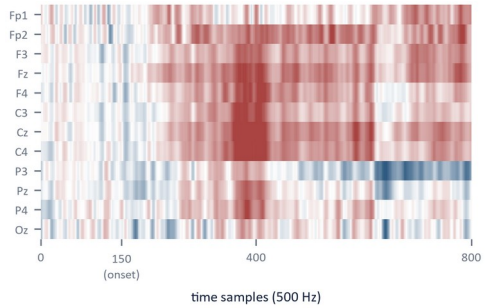
↓

Epoch: 12 EEG channels × 800 samples (−300 to +1298 ms)

↓

Dataset: 25,740 usable epochs

## One real epoch: a 12 × 800 matrix




## The dataset as a tensor

- Epochs × channels × time
- 25,740 × 12 × 800
- ≈ 247 million values
- …from only 30 participants

## Each epoch carries metadata

| Subject | Group | Stimulus | Trial | EEG matrix |
|---|---|---|---|---|
| L12 | knowledgeable | probe | 17 | 12 × 800 |
| H03 | control | irrelevant | 204 | 12 × 800 |
| L04 | knowledgeable | target | 88 | 12 × 800 |

## Two labels

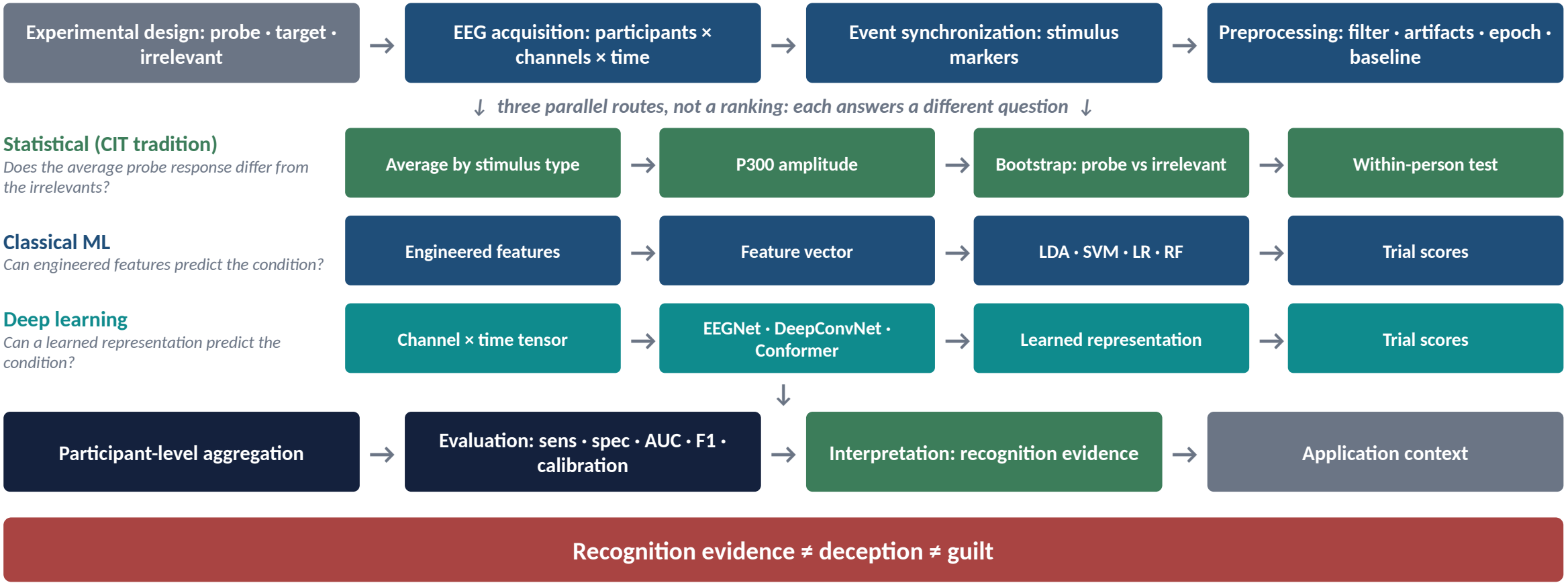
- Trial: stimulus type
- Person: knowledgeable / control

**25,740 epochs, but only 30 independent people: model complexity must match the number of people**

Sources: [37] · Data: Dryad doi:10.5061/dryad.2qc64 (Gao et al. [37]) · own descriptive analysis · example rows illustrative

# The Complete Computational EEG-CIT Pipeline

*Why one view: every later slide is one stage; the person-level question is whether accumulated evidence supports recognition.*




Sources: [3], [21], [37], [47] · architectures: [S1]–[S3] · conceptual map; four of these models are evaluated on the Gao data at the end of Section IV

# Preprocessing for ML: Why Does the Model Care?

*Why ask: each preprocessing step changes what a model can learn — including shortcuts.*

| STEP | WHAT IT DOES | WHY THE MODEL CARES |
|---|---|---|
| **Filtering** | Controls frequency content | **Removes drift and line noise a model could exploit** |
| **Bad-channel handling** | Repairs or removes failed electrodes | **Prevents learning electrode failure instead of brain activity** |
| **Artifact handling** | Removes blinks, eye movements, muscle | **Reduces ocular and muscle shortcuts** |
| **Epoching** | Cuts stimulus-locked segments | **Creates the model's samples** |
| **Baseline correction** | Removes the pre-stimulus offset | **Expresses stimulus-related change, not the DC level** |
| **Normalization / standardization** | Rescales channels or features | **Stabilizes training; parameters must come from training data** |

**Split participants first → fit pooled transformations (normalization, group ICA, feature selection) on training people only → per-person artifact correction is allowed if fixed in advance and blind to labels**

Sources: [21], [25], [37] · general ML methodology

# Three Ways to Feed EEG into a Model

*Why show this: the same epoch can enter a model as a feature vector, a matrix, or a time-frequency tensor.*

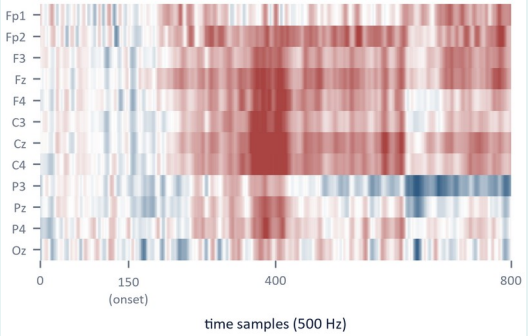
## A Handcrafted feature vector




- P300 amplitude, latency, area, band power, wavelet energy
- $x = [x_1 \ldots x_p]$: p engineered features

→ **LDA / SVM**

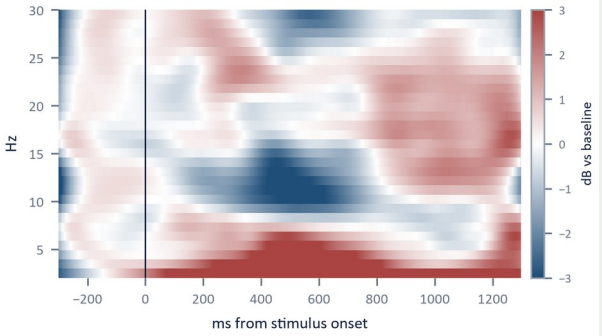
## B Raw / preprocessed epoch




- $X \in \mathbb{R}^{(C \times T)}$, C = 12 channels, T = 800 samples
- Network learns its own features

→ **CNN / EEGNet**

## C Time-frequency representation




- Wavelet or STFT per channel
- Channel × frequency × time tensor

→ **CNN or other network**

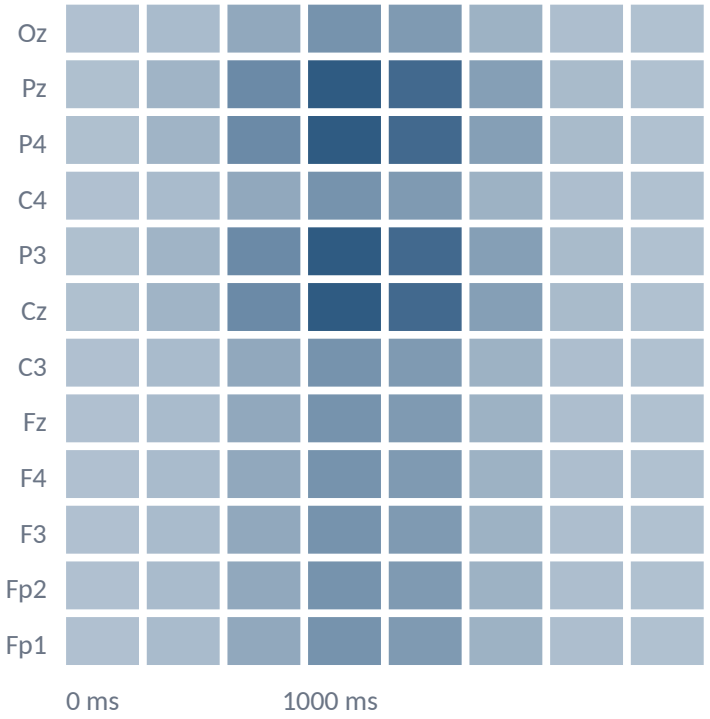
**One epoch = 12 × 800 = 9,600 values, but only 30 participants: a high-dimensional signal with very few people**

Sources: [25], [37], [46] · thumbnails: real L12 probe data — B: one single epoch; C: mean Pz power over 148 probe epochs (not the same trial) · Dryad

# From a 12 × 800 Epoch to 96 Features

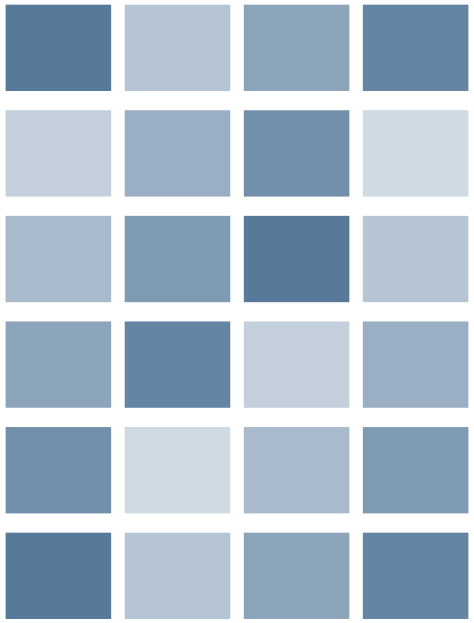
*Why 96 features: LDA and SVM take a fixed-length vector per trial; 96 interpretable window means replace the 6,000 raw samples.*

**0–1,000 ms: 12 channels × 8 windows of 125 ms**




average each cell

**96 features: $x_1$ … $x_{96}$**



(24 of 96 shown)

**Step 0 · select the segment**

Released epoch: −300 to +1,298 ms (800 samples). Keep 0–1,000 ms: 500 samples at 500 Hz.

**$\Sigma w_i x_i + b$**
**96 weights**

**score per trial**

**One cell = a window mean**

$x = (v_1 + \ldots + v_n) / n$ · 125 ms = 62.5 samples at 500 Hz (bins alternate 62/63); exactly 16 at 128 Hz, as in the results

**Kept: coarse shape and scalp pattern · lost: timing within 125 ms and frequency content**

Sources: [25], [37] · illustrative windows, not an optimised feature set · the same 96 features feed LDA and SVM in the results slides

# Feature Engineering in EEG-CIT

*Why its own slide: in small EEG datasets, feature choice often matters as much as the classifier.*

### Handcrafted physiological features

- P300 amplitude, latency, area, window means
- P3 combined with frontal EEG asymmetry [31]
- P300-based feature extraction for classification [25]
- Grounded in prior knowledge of the component

### Data-driven feature selection

- Many candidate features from channels, time windows or coefficients
- Features ranked by a criterion, e.g. F-score [37]
- Top-ranked subset passed to the classifier
- Grounded in the training data

| | | |
|---|---|---|
| **Interpretability** | Handcrafted: higher | Data-driven: lower |
| **Flexibility** | Handcrafted: lower | Data-driven: higher |
| **Overfitting risk in small samples** | Handcrafted: grows with features tried | Data-driven: high unless nested |

**Rule**

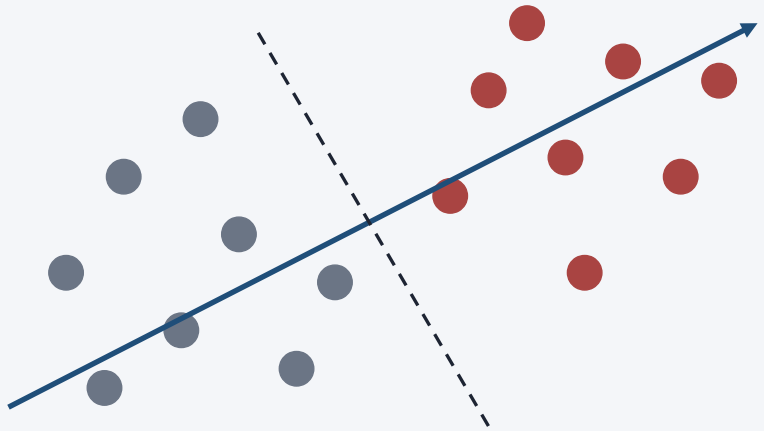
Fit selection inside each training fold

Sources: [25], [31], [37]

# Classical Classifiers: LDA vs SVM

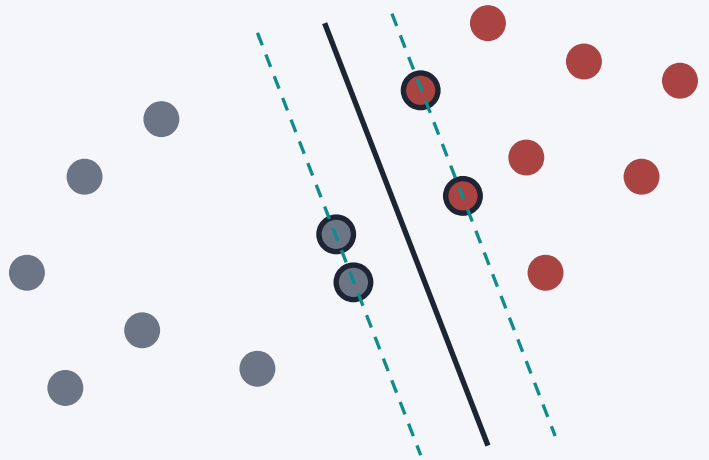
*Why explain, not just name: the classifier's geometry determines what kind of separation it can find.*

## LDA — maximize class separation along a projection

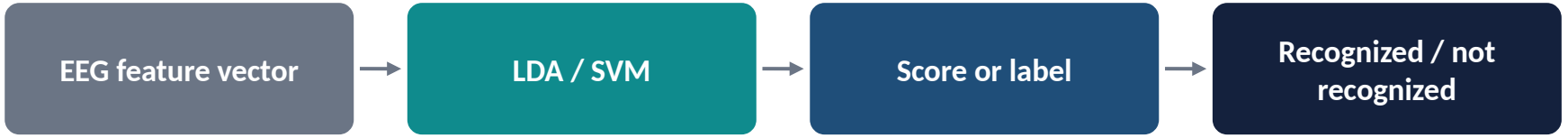


*Arrow: projection direction w · dashed: threshold. Maximizes between-class / within-class variance.*

## SVM — maximize the margin around a boundary



*Solid: boundary · dashed: margin · ringed: support vectors. Kernels allow curved boundaries.*

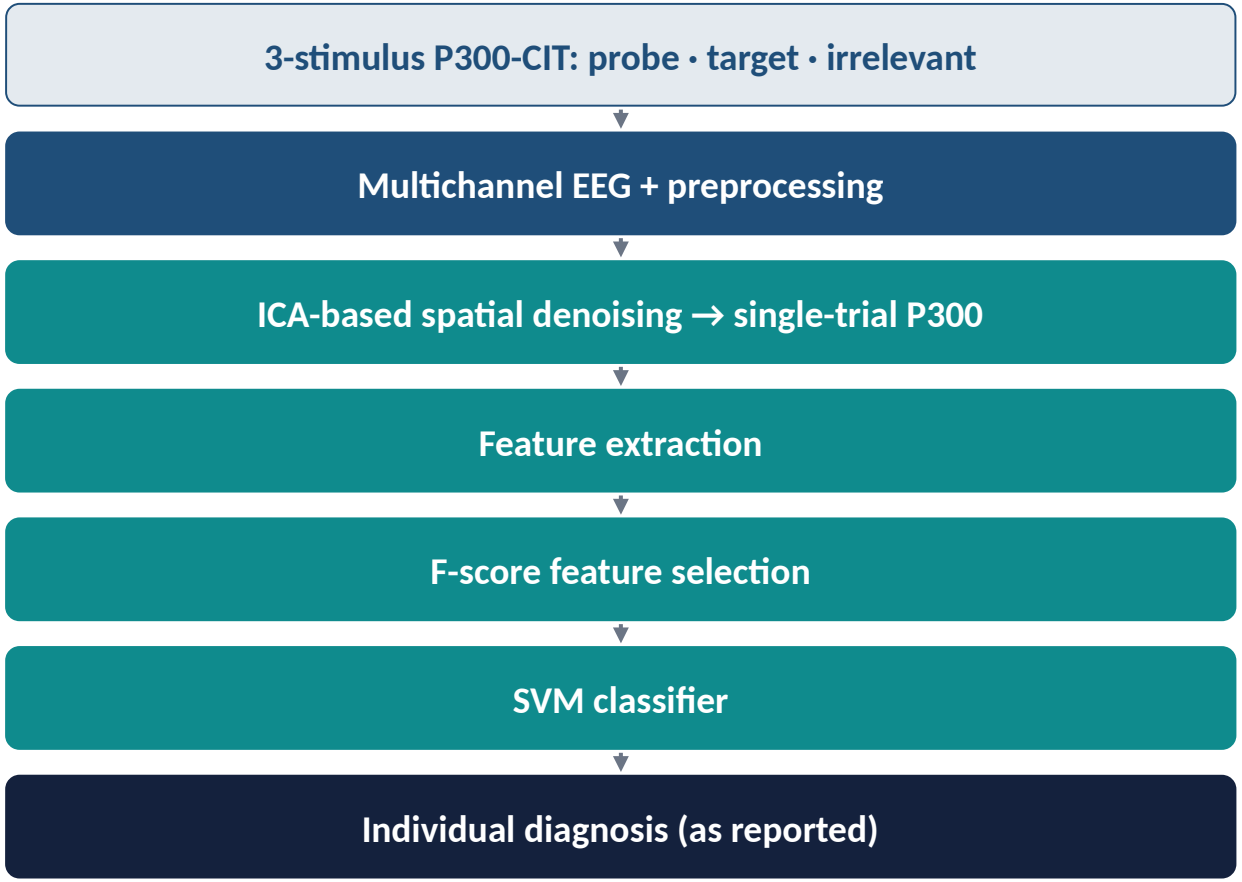



*Schematic; general ML background. In the reviewed literature: LDA [46], SVM [37]*

Sources: [37], [46]

# An End-to-End Pipeline: Gao et al. [37]

*Why one case in depth: a complete pipeline shows where each design decision enters — and where validity can fail.*




### Why it counts as P300-CIT

Probe, target and irrelevant stimuli; guilty and innocent groups; P300 measured and enhanced.

### Computational contribution

Single-trial P300 enhancement before classification.

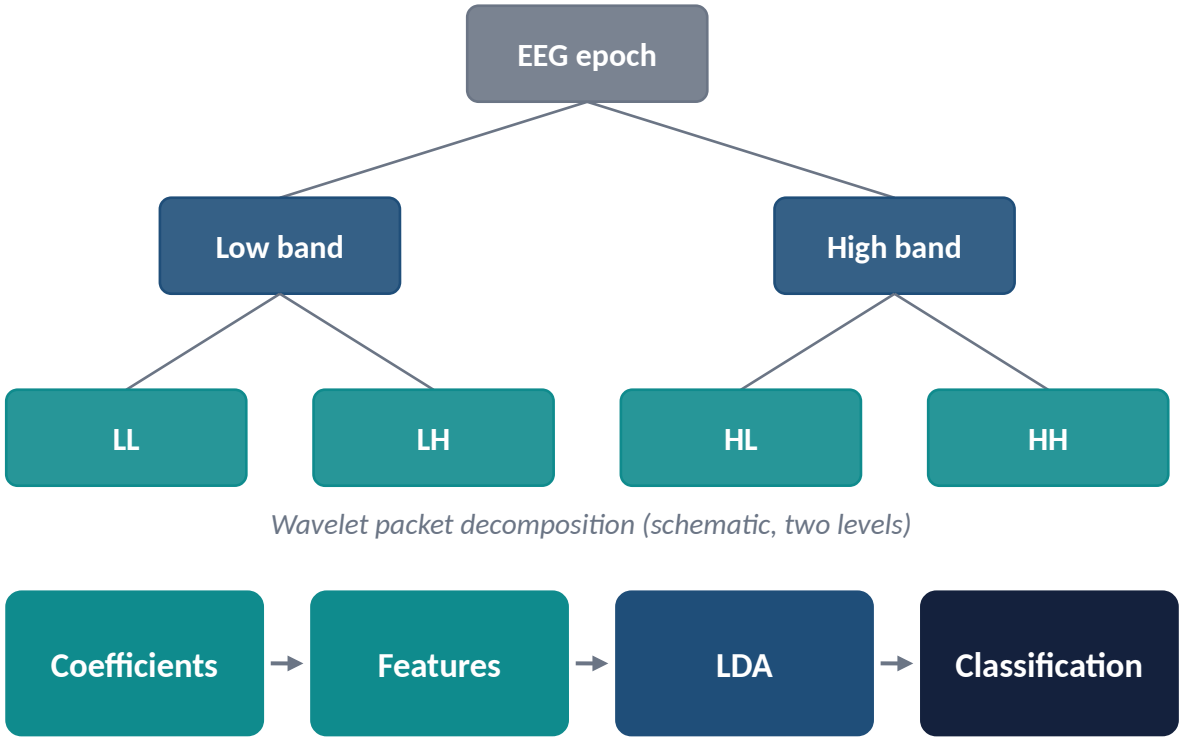
### Caution

Subject-wise CV (1 guilty + 1 innocent per fold); trials are averaged in groups of five first. Nesting of F-score selection and the 90% threshold is not fully reported: a leakage risk, not a shown flaw.

Sources: [37]

# Another Route: Wavelet-Based EEG Features

*Why include it: it shows what a time-frequency representation adds beyond a single P300 amplitude value.*




*Wavelet packet decomposition (schematic, two levels)*

## What a wavelet representation adds

- Energy localized in both time and frequency
- Can describe oscillatory activity and its timing
- Retains information that one peak-amplitude value discards

## Status in this seminar

Related EEG/deception study [46] — not established here as classical P300-CIT. Shown as a methodological example; no performance reported.

Sources: [46]

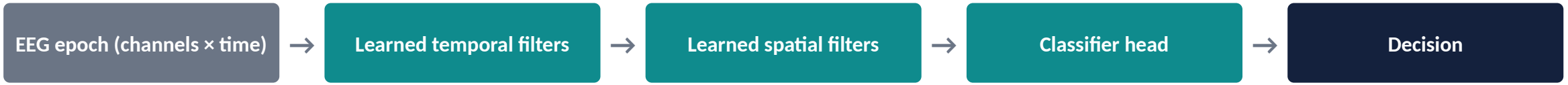
# From Handcrafted Features to Representation Learning

*Why include it: modern EEG analysis increasingly learns the representation instead of designing it.*

### Handcrafted route (as in Gao et al. [37])




### Representation learning




### Candidate architectures

- Compact EEG CNNs
- CNN + temporal modelling
- Attention / transformer models
- Self-supervised pretraining

### What changes

- Features are learned from data, not designed
- Needs more data and strict subject-independent validation
- Interpretation requires explainable AI

### Caution

- Promising in EEG generally; CIT-specific validation remains limited
- With N = 30, shortcuts (participant identity, artifacts, stimulus differences) are a real risk

General EEG / machine-learning background — not drawn from the reviewed CIT literature · illustrative

# One Epoch, One Tensor: What the Network Receives

*Why first: before any filter, it must be clear which numbers enter the network and how they are arranged.*

**Released epoch**
**12 × 800**
−300 to +1,298 ms · 500 Hz

→

**Select 0–1,000 ms**
**12 × 500**
500 samples at 500 Hz

→

**Resample to 128 Hz**
**12 × 128**
one trial: the network input

→

**Mini-batch**
**64 × 12 × 128**
64 epochs per training step

| View | Shape | Meaning |
|---|---|---|
| **One trial** | 12 × 128 | 12 simultaneous time series (rows = electrodes), 128 samples each, ≈ 7.8 ms apart |
| **Mini-batch** | 64 × 12 × 128 | 64 epochs processed together in one training step |
| **Inside EEGNet** | 64 × 1 × 12 × 128 | a singleton axis for the 2-D convolution layers — not an extra electrode |

**Not a flattened vector: unlike the 96 LDA features, channels stay rows and time stays columns**

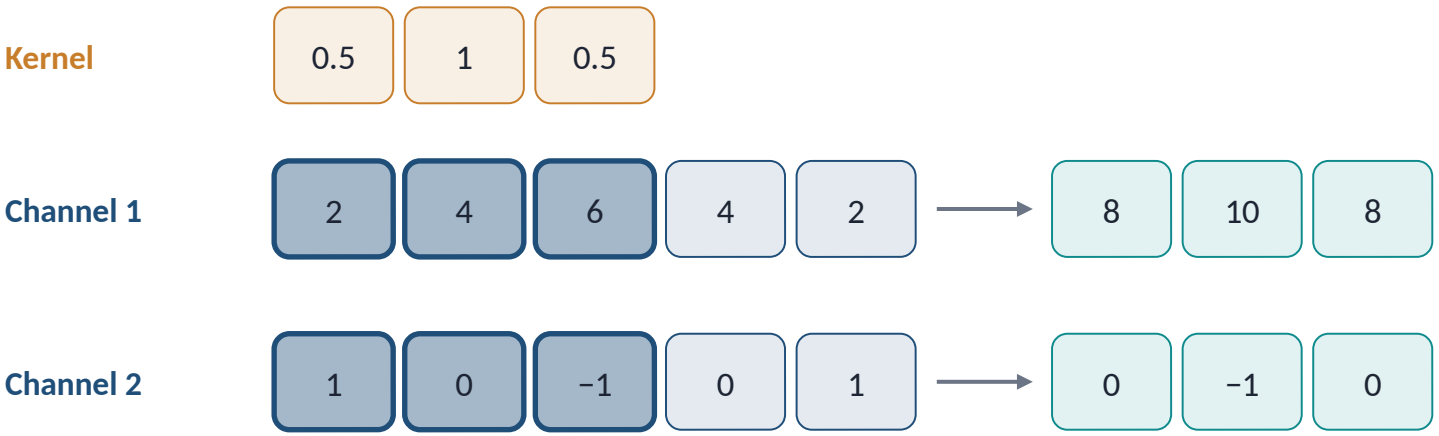
**The network receives 12 × 128 numbers per trial: the same epoch, cropped and resampled — not a second recording**

Sources: [S1], [37] · shapes of the evaluated EEGNet and CNN-LSTM

# What a 1D Convolution Does to EEG

*Why a toy example: convolution is a sliding weighted sum, and five samples are enough to see it.*

### 1 EEGNet-style: one temporal kernel applied to each channel




Channel 1, first output: 2·0.5 + 4·1 + 6·0.5 = 8; the window then slides one sample

**Generic multichannel Conv1D:** a filter may hold a distinct set of weights for each channel and combine them into one feature map.

### 2 Spatial weights mix the channels

0.8 × channel 1 (8, 10, 8)
+ 0.3 × channel 2 (0, −1, 0)

**= virtual channel**




**Two kinds of weight**

- Temporal: one shape, reused everywhere
- Spatial: one weight per channel

**Temporal weights learn what the response looks like; spatial weights learn where it appears**

Sources: General deep-learning methodology · EEGNet-style illustration [S1] · illustrative fixed weights, not learned values

# More Filters, More Feature Maps

*Why more than one: one waveform template cannot capture every relevant EEG pattern, so networks learn several.*

| Filter | Temporal kernel | Channel 1 map | Channel 2 map | Spatial mix | Feature map |
|---|---|---|---|---|---|
| **A** | [0.5, 1, 0.5] | [8, 10, 8] | [0, −1, 0] | 0.8·Ch1 + 0.3·Ch2 | [6.4, 7.7, 6.4] |
| **B** | [−1, 0, 1] | [4, 0, −4] | [−2, 0, 2] | 0.4·Ch1 − 0.2·Ch2 | [2, 0, −2] |

| Operation | Weights per output filter (C = 12 channels, K = 3) |
|---|---|
| **EEGNet-style temporal filter** | K = 3 weights, reused along every channel; the spatial step adds 12 channel weights per map |
| **Generic multichannel Conv1D** | C × K = 12 × 3 = 36 weights (+ bias): all channels and 3 time points in one sum |
| **More filters (either design)** | each filter has its own learned weights; filters do not share one set |

**Evaluated EEGNet: 8 temporal filters × 64 samples = 512 weights · 16 spatial filters × 12 electrodes = 192 weights**

**Two filters give two feature maps: learned patterns, not new electrodes**

Sources: General deep-learning methodology · EEGNet-style illustration [S1] · illustrative fixed weights

# From Feature Maps to a Score

*Why follow the numbers: the feature maps still have to be compressed and combined before a single score exists.*

| Step | Operation | Output |
|---|---|---|
| **1 Feature maps** | A = [6.4, 7.7, 6.4] · B = [2, 0, −2] | 2 maps × 3 |
| **2 Nonlinearity** | ReLU(x) = max(0, x) | A same · B = [2, 0, 0] |
| **3 Pooling** | global max over time | [7.7, 2.0] |
| **4 Dense layer** | z = 0.4·7.7 − 0.2·2.0 − 2 | z = 0.68 |
| **5 Score** | sigmoid(z) = 1 / (1 + $e^{-z}$) | ≈ 0.66 |

| Stage | Size |
|---|---|
| **Input** | 2 × 5 |
| **1 temporal filter, K = 3** | 2 × 3 |
| **Spatial mix** | 1 × 3 |
| **2 filters** | 2 × 3 |
| **Global pool** | 2 |
| **Dense → score** | 1 |

**The evaluated EEGNet differs in detail**

ELU instead of ReLU · average pooling · a separable convolution · a dense layer over 64 numbers — same logic, different choices

**An uncalibrated score for the trained label (target-like) — not a statement about deception or guilt**

Sources: General deep-learning methodology · illustrative generic CNN head, not the evaluated EEGNet

# Deep Learning for EEG: Three Architecture Families

*Why these three: they are established EEG-decoding architectures — not established P300-CIT solutions.*

## EEGNet [S1], step by step

*Original released epoch: 12 × 800 (500 Hz) · evaluated network input: 12 × 128 (0–1 s, resampled to 128 Hz)*




### DeepConvNet / ShallowConvNet [S2]

- End-to-end decoding from raw EEG
- Visualized the spectral and spatial features it learned
- Evaluated on imagined or executed movement

### EEGNet [S1]

- Compact CNN: depthwise + separable convolutions
- Four paradigms, including P300 visual-evoked potentials
- Within- and cross-subject comparisons

### EEG Conformer [S3]

- Convolution for local temporal / spatial features
- Self-attention for global dependencies
- Evaluated on motor imagery and emotion recognition

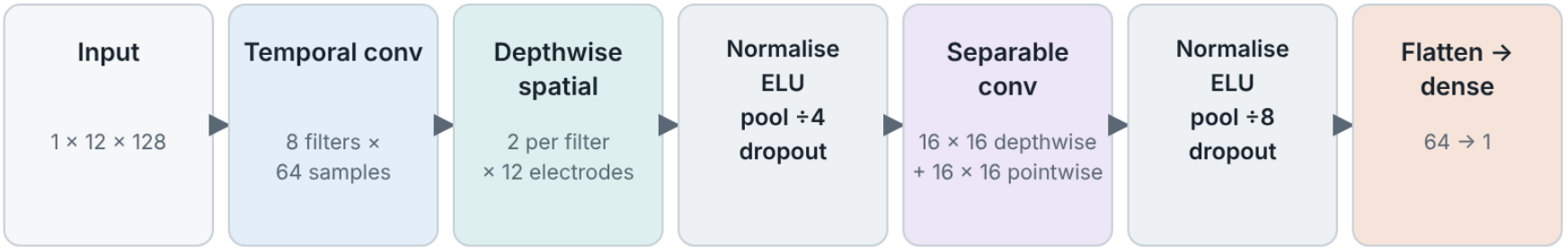
**Deep learning does not remove preprocessing or experimental design — it changes how the representation is learned**

Supplementary references (outside the main reference list): [S1] Lawhern et al. 2018 · [S2] Schirrmeister et al. 2017 · [S3] Song et al. 2023 · EEGNet-type models are evaluated in the results slides

# EEGNet Layer by Layer, With Tensor Shapes

*Why trace the shapes: each block is a temporal filter, a spatial filter or a weighted sum seen on the previous slide.*

**Tensor shape after each block (filters × electrodes × time):**




1. temporal filters: waveform / frequency shape
2. spatial filters: electrode weights per filter
3. separable conv: mixes the 16 maps cheaply
4. one weighted sum: 64 numbers → score

**Total 1,361 learned numbers · illustrative input: 12 channels × 1 s at 128 Hz**

**Illustrative compact network: 12 channels × 128 samples (1 s at 128 Hz), distinct from the original 12 × 800-sample released epoch**

Sources: [S1] Lawhern et al. 2018 · this configuration is trained on the 0–1 s, 128-Hz epochs in the results slides

# CNN vs EEGNet vs CNN-LSTM: What Changes?

*Why compare: the three families differ in what they assume about EEG, not just in size.*

| Model | Core operation | What it assumes about EEG | Learned numbers | Evaluated here? |
|---|---|---|---|---|
| **Generic CNN (e.g. ShallowConvNet [S2])** | Temporal + spatial convolution, full filter weights | Useful information is a local pattern in time and across electrodes | Design-dependent | No |
| **EEGNet [S1]** | Temporal filters → depthwise spatial filters → separable convolution | A few filters suffice; each temporal filter has its own spatial pattern | 1,361 (shown config.) | Yes (results) |
| **CNN-LSTM** | EEGNet-style front end → LSTM over 32 time steps | Models the sequence of CNN-derived feature vectors: how informative EEG patterns evolve across time within the epoch | 6,929 (shown config.) | Yes (results) |

### For a P300

A fixed scalp pattern at a fairly fixed latency: convolution captures it; recurrence may add little.

### Status

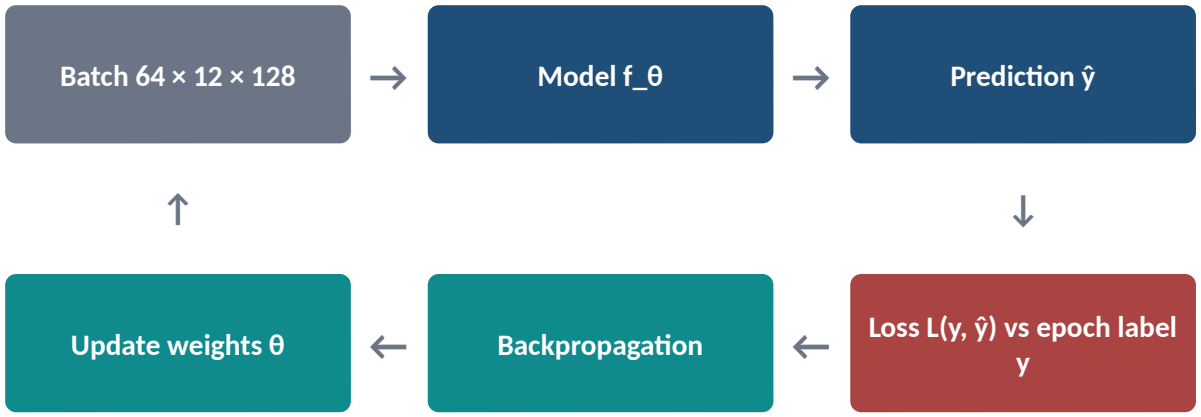
EEGNet and CNN-LSTM, as configured here, are trained and evaluated in the results slides; the generic CNN is not.

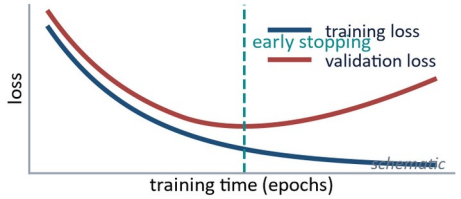
Sources: [S1], [S2] · own configurations for a 12 × 128 epoch (0–1 s at 128 Hz)

# How a Network Learns — and How It Is Used

*Why separate them: the model learns from labelled people, then is applied, frozen, to a new person.*

## Training: learn from labelled participants

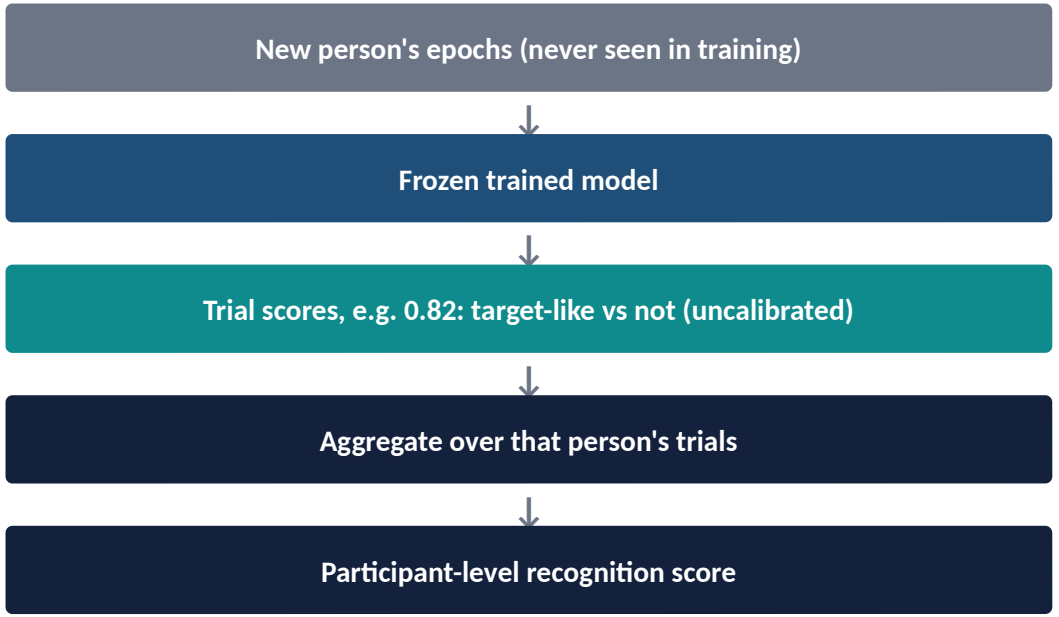



Overfitting: training loss keeps falling while validation loss rises

Mitigations, not guarantees: early stopping, regularization, compact models

## Inference: apply the frozen model to a new person




**A trial score describes a target-like brain response — not a calibrated probability, and never a probability of guilt**

*Training label y: target (1) vs irrelevant (0) for each epoch · knowledgeable/control labels evaluate person-level decisions; they are never the epoch training target*

General machine-learning methodology · batch shape as in the evaluated models (64 epochs, 12 channels, 128 samples)

# Training, Validation and Testing

*Why first: how the data are split decides what a performance number means.*

**Training set**
- Learns the model weights

**Validation set**
- Hyperparameters, early stopping, model and threshold selection

**Test set**
- Used once, for the final unbiased estimate

**Correct: split by participant**




S01–S20 train · S21–S24 validation · S25–S30 test *(balance knowledgeable and control participants in each set)*

**Wrong: split one participant's epochs at random**




**× the same person on both sides → the model can learn the person, not recognition**

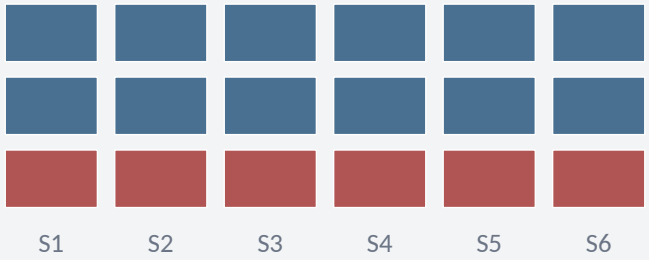
**With N = 30: repeated stratified participant-wise CV or leave-one-pair-out (1 knowledgeable + 1 control held out), with preprocessing, feature selection, hyperparameters and thresholds fitted inside each training fold**

# The Generalization Problem

*Why decisive: a real examinee is always someone the model has never seen.*

### SUBJECT-DEPENDENT

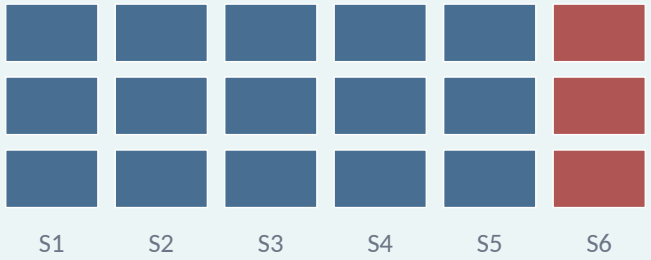
Same people in training and test




Weakest test

### SUBJECT-INDEPENDENT

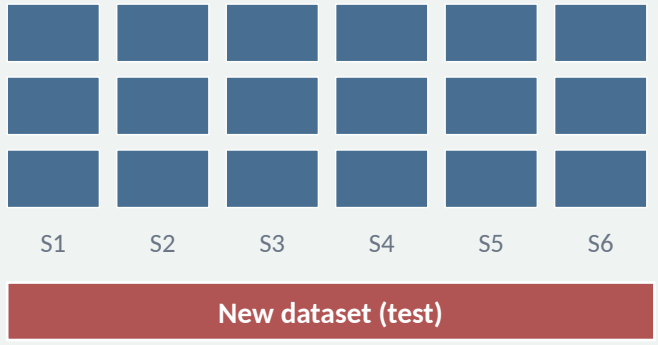
Held-out people (e.g. leave-one-subject-out)




Stronger test

### EXTERNAL / CROSS-DATASET

Different lab, sample or protocol




Strongest test

Train | Test

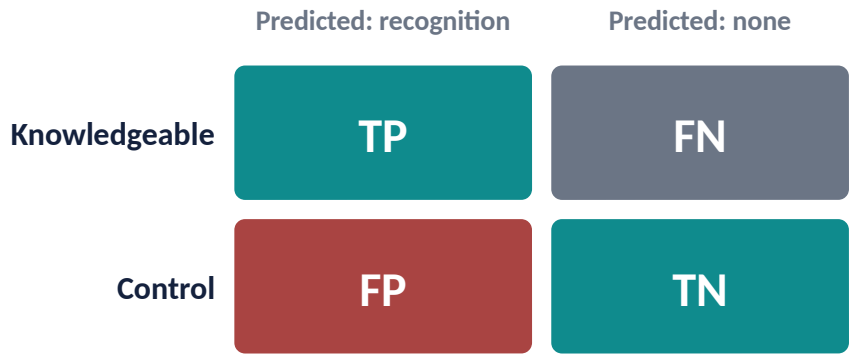
Columns = subjects; rows = trial blocks. The reviewed literature does not show that CIT research has solved generalization.

Sources: [37], [46], [49]

# How Should an EEG-CIT Classifier Be Evaluated?

*Why not accuracy alone: false positives and false negatives have very different consequences.*

### Participant-level confusion matrix




### Imbalance, uncertainty and stakes

- In the Gao dataset: 1 probe per 4 irrelevants, so trial accuracy misleads
- Report confidence intervals and class-aware metrics
- Specificity matters: false recognition is costly

### Threshold-based

- Sensitivity = TP / (TP + FN): knowledgeable participants detected
- Specificity = TN / (TN + FP): controls correctly not flagged
- Precision = TP / (TP + FP);  F1 = 2 · precision · recall / (precision + recall)
- Balanced accuracy = (sensitivity + specificity) / 2

### Threshold-free and probabilistic

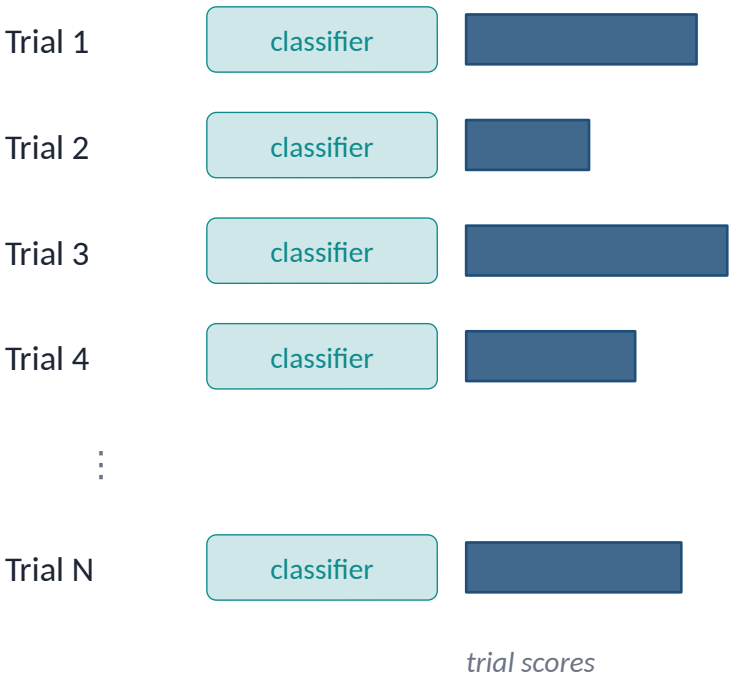
- ROC-AUC: discrimination across all thresholds
- Calibration / Brier score: do probabilities match observed frequencies?

**Report participant-level metrics when the inference concerns participants · with 15 + 15 people, one error moves sensitivity or specificity by 6.7 points**

General ML methodology · Gao et al. report sensitivity, specificity and balanced accuracy [37]

# From Trial Classification to Person-Level Decision

*Why decisive: an investigation asks about a person, but most classifiers score trials.*







### What is classified?

- a trial
- a stimulus type
- a probe response
- the person

### Report the unit

Trial-level accuracy ≠ person-level accuracy. Statistical route: within-person bootstrap on averaged ERPs [21], [24], [47].

Sources: [21], [24], [37], [47]

# From Model Output to Application

*Why end here: the endpoint of the pipeline is an inference about recognition, not about guilt.*

EEG model output → Recognition score (calibrated only if checked) → Participant-level evidence → Research or application context

## Possible research and application contexts

- Concealed-information research
- Recognition-memory assessment
- Forensic research under controlled conditions
- BCI and cognitive-state decoding methods
- Human-factors and security research
- Multimodal behavioural research

**Model output supports an inference about recognition. It does not independently determine deception, criminal responsibility or guilt.**

Sources: [11], [14], [53]

# Analysis Setup: Fixed Before Any Result

*Why show the setup first: a result is only as credible as the split, the settings and the decision rule behind it.*

### Data

- 30 people: 15 knowledgeable, 15 control
- 23,388 epochs after rejecting > ±100 µV
- 0–1 s · 12 EEG channels · 500 → 128 Hz

### Validation

- Leave-one-pair-out: 15 folds, 1 knowledgeable + 1 control held out
- Standardisation, weights, early stopping fitted on training people only

### Training task

- Target vs irrelevant epochs
- Probe epochs never used in training

### Methods

- Classical Pz test (no ML): mean 300–800 ms
- LDA, linear SVM: 96 window means
- EEGNet, CNN-LSTM: 12 × 128 epochs [S1]

### Decision rule (all methods)

- Knowledgeable if probe scores > irrelevant scores
- One-sided permutation test, $p < 0.05$

### Metrics

- Trial AUC (target vs irrelevant)
- Person AUC, sensitivity, specificity, balanced accuracy, 95% CIs

**Plan fixed before results · the only later change: Pz read from the true row (channel order from the raw .cnt header)**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · [S1]

# Results: Metrics, Uncertainty and Errors

*Why this table: it gives the exact numbers, how wide the uncertainty is, and which kind of error each method makes.*

| Method | Trial AUC | Person AUC | Person AUC perm. p[1] | Detected TP / FN | Cleared TN / FP | Sensitivity [95% CI] | Specificity [95% CI] | Balanced accuracy |
|---|---|---|---|---|---|---|---|---|
| **Classical Pz test (no ML)** | 0.62 | 0.87 | < 0.001 | 12 / 3 | 13 / 2 | 0.80 [0.55, 0.93] | 0.87 [0.62, 0.96] | 0.83 |
| **LDA** | 0.90 | 0.94 | < 0.001 | 14 / 1 | 13 / 2 | 0.93 [0.70, 0.99] | 0.87 [0.62, 0.96] | 0.90 |
| **SVM** | 0.90 | 0.94 | < 0.001 | 14 / 1 | 12 / 3 | 0.93 [0.70, 0.99] | 0.80 [0.55, 0.93] | 0.87 |
| **EEGNet** | 0.91 | 0.92 | < 0.001 | 14 / 1 | 13 / 2 | 0.93 [0.70, 0.99] | 0.87 [0.62, 0.96] | 0.90 |
| **CNN-LSTM** | 0.91 | 0.94 | < 0.001 | 14 / 1 | 14 / 1 | 0.93 [0.70, 0.99] | 0.93 [0.70, 0.99] | 0.93 |

[1] *Group labels of the 30 people shuffled 10,000 times; null: person AUC = 0.5. Separate from each person's decision: probe/irrelevant labels of that person's epoch scores shuffled 2,000 times, one-sided, $p < 0.05$.*

### Trial vs person

- Trial AUC: target vs irrelevant epochs
- Person AUC: knowledgeable vs control people

### Uncertainty

- 95% Wilson CIs, 15 people per group
- One person = 6.7 points of sensitivity or specificity; CIs overlap

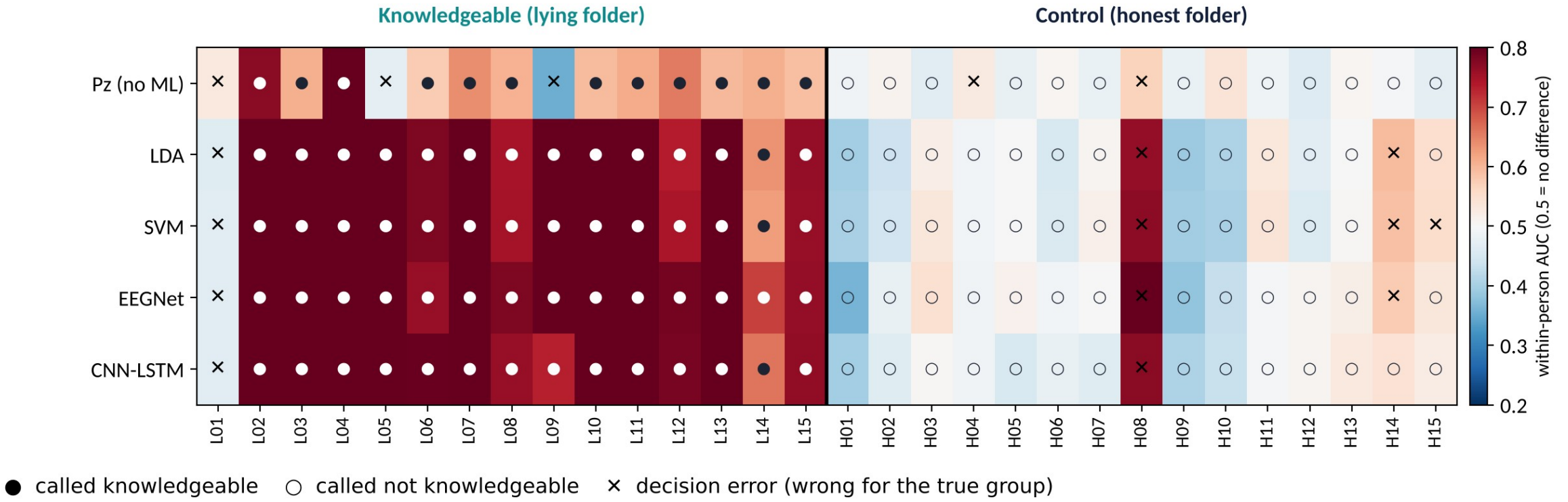
### Errors

- All five: L01 missed, H08 flagged
- Pz test also: L05, L09 missed; H04 flagged
- H14: LDA, SVM, EEGNet · H15: SVM

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons

# Every Person, Every Method

*Why this view: a single accuracy hides who is misclassified; the grid shows whether errors belong to the methods or to particular people.*

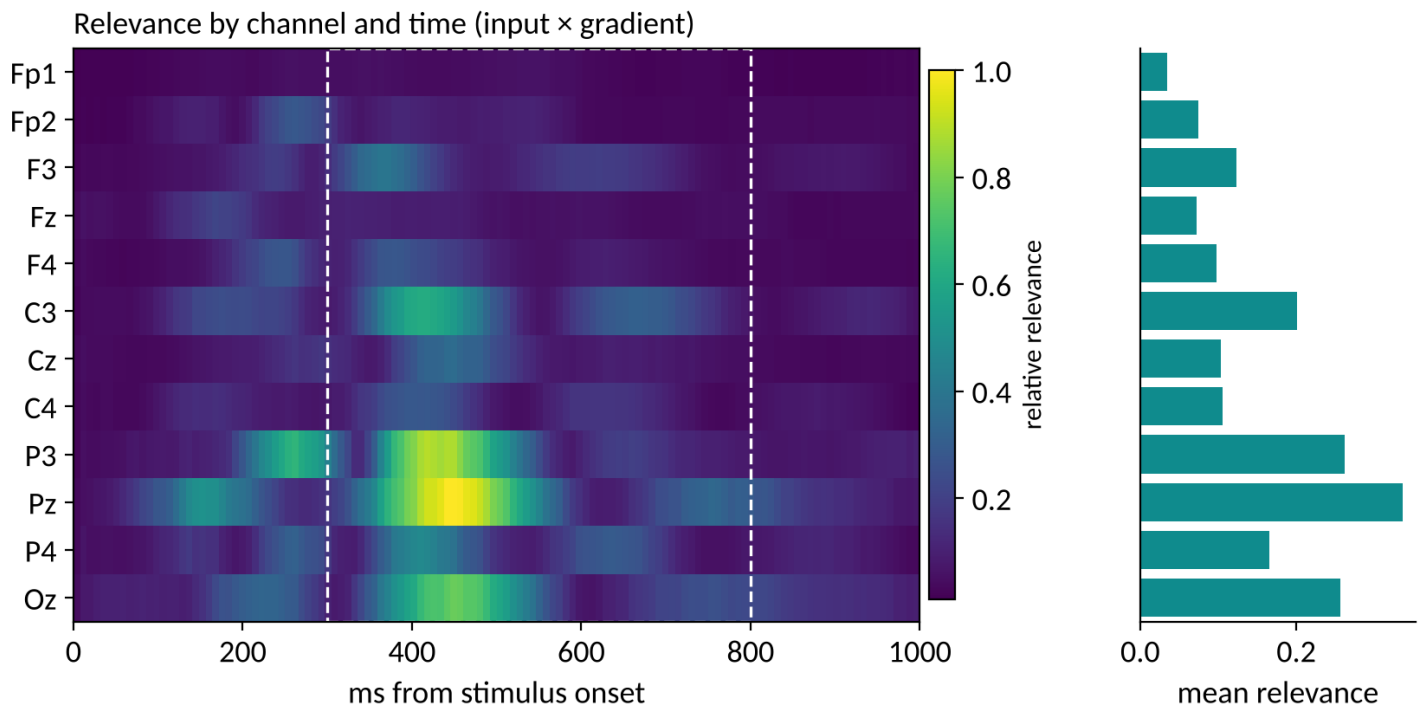



**All five methods miss L01 and flag H08 — some errors follow the person, not the algorithm**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons

# EEGNet Relevance on Held-Out Target Epochs

*Why check: a model that scores well for the wrong reason, such as eye activity or noise, would fail on new data.*




Channel names from the raw .cnt header (one file checked; same export assumed). Rows ordered front to back. Dashed box = 300–800 ms.

### How to read

- Colour: output sensitivity to each channel and time
- Bars: mean relevance per channel

### What it shows

- Largest at Pz, then P3, Oz, C3
- Peak ≈ 414 ms; 65% in 300–800 ms

### Scope

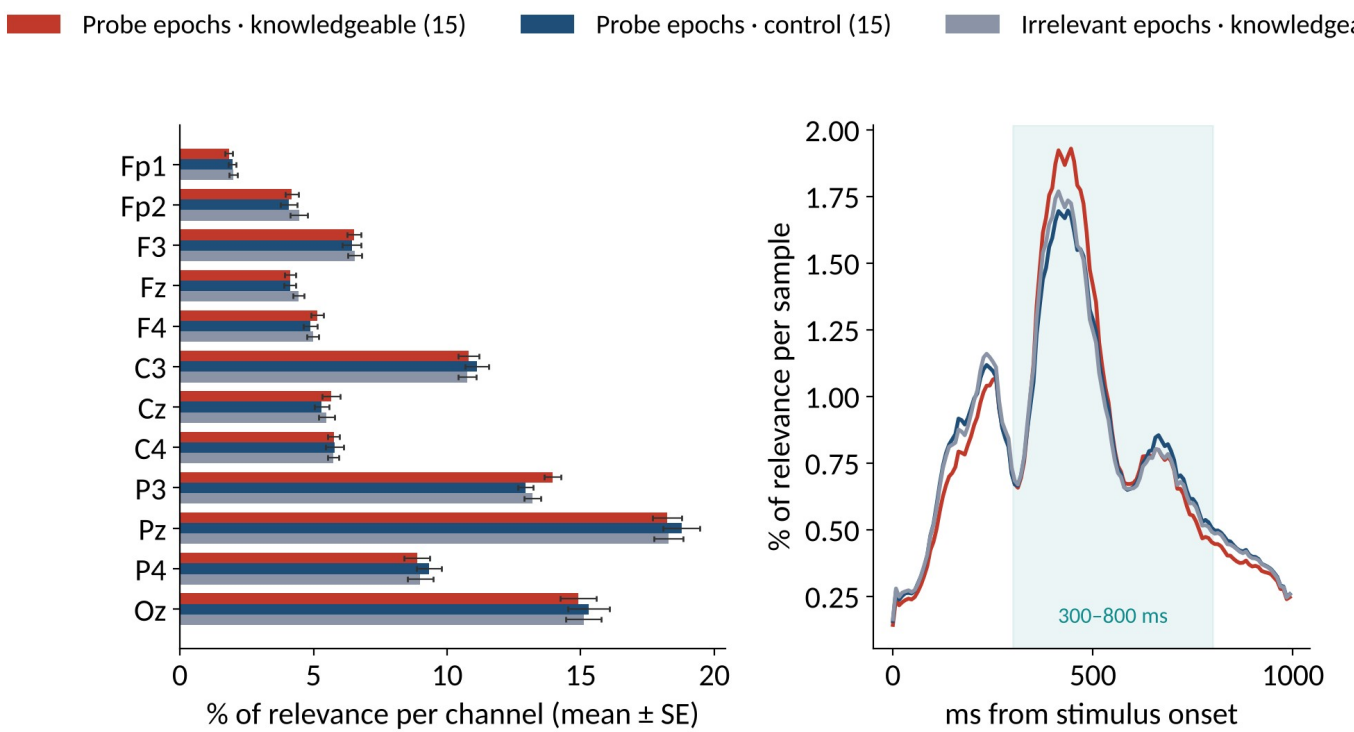
Target-vs-irrelevant predictions only; not a cortical source

**Target predictions depend most on parietal inputs in the P300 window; Fp1/Fp2 contribute least**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · input × gradient, held-out target epochs

# Does the Same Pattern Hold for Probe Epochs?

*Why test it: people are judged on their probe scores, so the explanation has to be checked on probe epochs, not only on targets.*




Held-out people only · each person normalised to 100% · channel names from the raw .cnt header (one file checked)

## What it shows

- Same pattern for probe and irrelevant epochs, in both groups
- Pz largest; parietal-occipital ≈ 56%; Fp1 + Fp2 ≈ 6%

## Per person

Top channel posterior: 15/15 per group · 63–66% in 300–800 ms

## Limits

Shows which inputs influence scores under this method — not a cortical source, unique mechanism or causal dependence

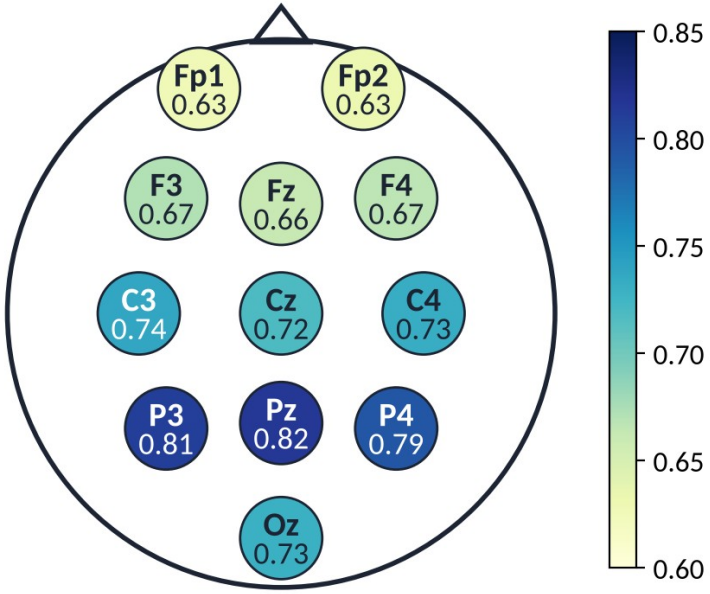
**Probe scores rely on the same posterior, P300-window inputs as target scores, in both groups**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · input × gradient, each fold's model on its own two held-out people

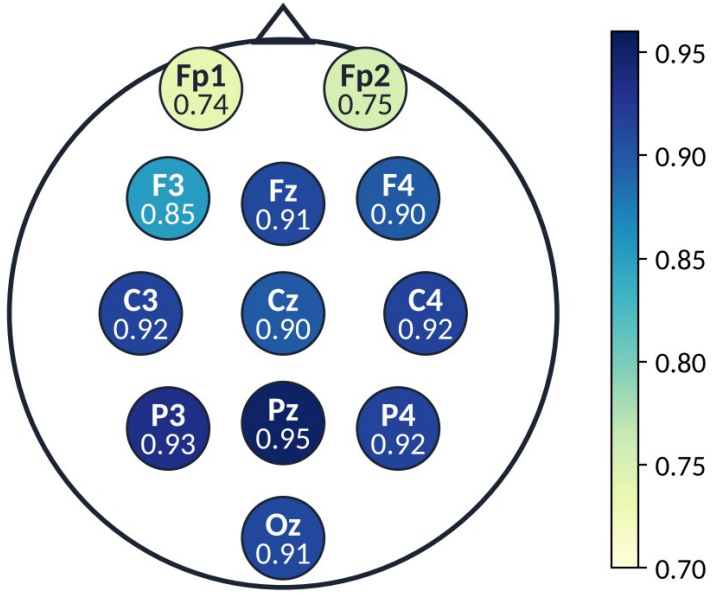
# Which Electrodes Carry the Information?

*Why test it: a relevance map shows what one network uses; one model per electrode measures what each site carries on its own.*

**Trial AUC (target vs irrelevant epochs)**




**Person AUC (knowledgeable vs control)**




*nose at top · single-channel LDA, 8 time-window features per channel · channel order from the raw .cnt header*

### How it works

- Same LDA, folds and decision rule
- Input: one electrode, 8 window means (0–1 s)

### What it shows

- Trial AUC highest at Pz, P3, P4 (0.79–0.82)
- Person AUC: Pz 0.95; ≥ 0.85 everywhere except Fp1, Fp2

**Predictive information peaks over parietal electrodes, where the P300 is largest; frontopolar sites carry least**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · same folds and decision rule as the main results

# Single Electrodes: Person-Level Metrics

*Why every metric: one error in a 15-person group moves sensitivity or specificity by 6.7 points; precision and F1 depend on the whole matrix.*

| Electrode | Region | Trial AUC | Person AUC | Detected | Cleared | Accuracy | Precision | Sensitivity | Specificity | F1 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Fp1** | Frontopolar | 0.63 | 0.74 | 10/15 | 11/15 | 0.70 | 0.71 | 0.67 | 0.73 | 0.69 |
| **Fp2** | Frontopolar | 0.63 | 0.75 | 9/15 | 11/15 | 0.67 | 0.69 | 0.60 | 0.73 | 0.64 |
| **F3** | Frontal | 0.67 | 0.85 | 11/15 | 14/15 | 0.83 | 0.92 | 0.73 | 0.93 | 0.81 |
| **Fz** | Frontal | 0.66 | 0.91 | 12/15 | 13/15 | 0.83 | 0.86 | 0.80 | 0.87 | 0.83 |
| **F4** | Frontal | 0.67 | 0.90 | 11/15 | 13/15 | 0.80 | 0.85 | 0.73 | 0.87 | 0.79 |
| **C3** | Central | 0.74 | 0.92 | 11/15 | 13/15 | 0.80 | 0.85 | 0.73 | 0.87 | 0.79 |
| **Cz** | Central | 0.72 | 0.90 | 12/15 | 13/15 | 0.83 | 0.86 | 0.80 | 0.87 | 0.83 |
| **C4** | Central | 0.73 | 0.92 | 13/15 | 12/15 | 0.83 | 0.81 | 0.87 | 0.80 | 0.84 |
| **P3** | Parietal | 0.81 | 0.93 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |
| **Pz** | Parietal | 0.82 | 0.95 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |
| **P4** | Parietal | 0.79 | 0.92 | 14/15 | 11/15 | 0.83 | 0.78 | 0.93 | 0.73 | 0.85 |
| **Oz** | Occipital | 0.73 | 0.91 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |
| **All 12 channels** | Reference | 0.90 | 0.94 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |

*Shaded: Pz, highest single-electrode trial and person AUC · 15 knowledgeable + 15 control people · decision: permutation test $p < 0.05$ · channel names from the raw .cnt header (one file checked)*

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · same folds and decision rule as the main results

# Electrode Regions: Where Is the Effect?

*Why regions: single electrodes are noisy; grouping neighbours tests whether a scalp region, not one site, carries the effect.*

| Region | Electrodes | Trial AUC | Person AUC | Detected | Cleared | Accuracy | Precision | Sensitivity | Specificity | F1 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Frontal** | Fp1 Fp2 F3 Fz F4 | 0.75 | 0.92 | 12/15 | 12/15 | 0.80 | 0.80 | 0.80 | 0.80 | 0.80 |
| **Frontal without Fp1/Fp2** | F3 Fz F4 | 0.69 | 0.93 | 12/15 | 13/15 | 0.83 | 0.86 | 0.80 | 0.87 | 0.83 |
| **Central** | C3 Cz C4 | 0.76 | 0.92 | 13/15 | 13/15 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 |
| **Parietal-occipital** | P3 Pz P4 Oz | 0.87 | 0.94 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |
| **All 12 without Fp1/Fp2** | 10 channels | 0.90 | 0.94 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |
| **All 12 channels** | Reference | 0.90 | 0.94 | 14/15 | 13/15 | 0.90 | 0.88 | 0.93 | 0.87 | 0.90 |

### Reading

- Parietal-occipital vs all 12: identical decisions, person by person
- Trial AUC 0.87 vs 0.90 — not equivalence
- Without Fp1/Fp2: same observed metrics

### Cautions

- 18 sets, 30 people: 1–2-decision gaps may be sampling variability
- Ocular artifacts not ruled out by this alone

### Meaning

- Posterior, P300-compatible scalp pattern
- Scalp data do not localise generators

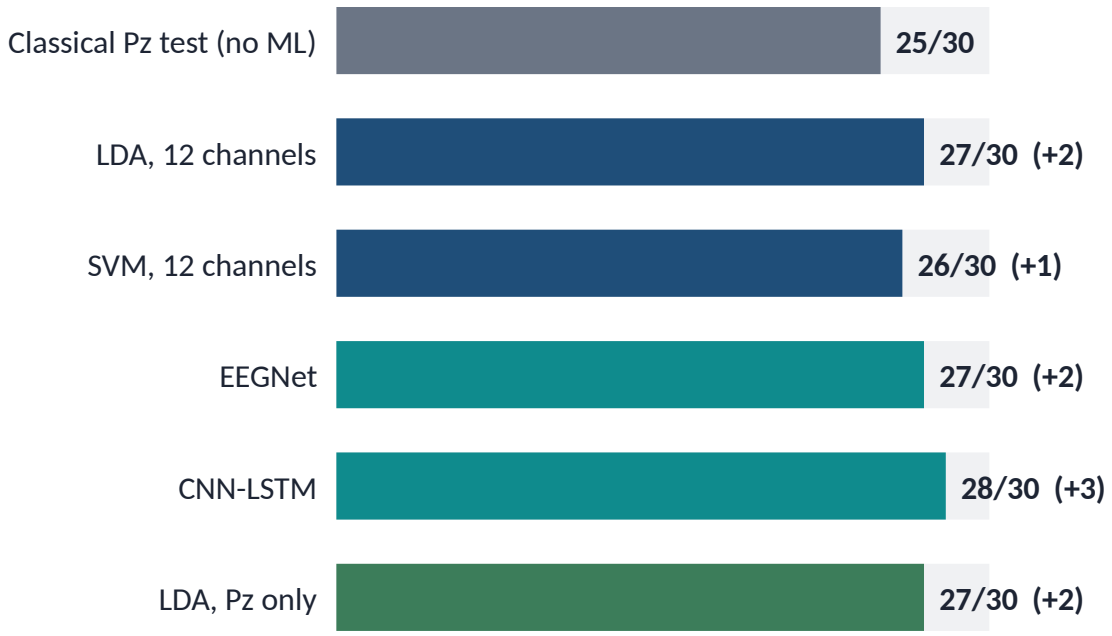
**Predictive information is concentrated over parietal electrodes, where the P300 is largest**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · same folds and decision rule as the main results

# Interpreting the Results (1): How Much Does ML Add?

*Why ask: the classical Pz test is the baseline that every model has to beat before its complexity is justified.*

**Correct person-level decisions out of 30**




*Knowledgeable detected + controls cleared · bars start at 0*

**Trial level: large gain**

Trial AUC 0.62 (Pz test) → 0.90–0.91 (trained models)

**Person level: small, uncertain gain**

vs Pz test: SVM +1 · LDA, EEGNet +2 · CNN-LSTM +3

**Similar, not proven equal**

Deep networks ≈ LDA here; overlapping CIs do not prove equivalence

**Gains of one to three people over the classical test are small and uncertain with 30 people**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons

# Interpreting the Results (2): Scalp Pattern and Errors

*Why both: a credible result needs a plausible scalp pattern and an honest account of the people it fails on.*

### A posterior scalp pattern

- Relevance, single electrodes and regions all point to parietal electrodes
- Peak ≈ 414 ms, inside the P300 window
- Scalp data do not localise neural generators

**Compatible with a recognition-enhanced P300 (Section II)**

### Errors follow people

- L01 missed by all five: probe scores not target-like
- H08 flagged by all five: target-like probe scores in a control
- L09: missed by the Pz test, found by all four models

**A different classifier does not repair these people**

### What the score means

- Models learn targets; probes are never trained on
- Flag = probe scores resemble target scores
- Not deception; not proof of recognition in one person

**Evidence compatible with recognition, not proof of it**

**The models find target-like probe scores over parietal electrodes — compatible with recognition, not proof**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons

# What These Results Show — and What They Do Not

*Why end the section here: the boundary of the claim decides how the limitations in Section V should be read.*

### The results show

- Within one dataset, held-out knowledgeable people: 14 of 15 detected
- Best methods flag 1–2 of 15 controls
- Predictive information over parietal electrodes, P300 window
- LDA and deep networks: similar observed performance

**Claim level: recognition-compatible responses of people, within one lab dataset**

### They do not show

- Deception or guilt: a control (H08) can look recognised [11], [52]
- Transfer to other labs, stimuli or people (1 dataset, 30 people)
- Robustness to countermeasures, delay or real base rates
- Neural generators: scalp patterns do not localise sources
- A replication of Gao et al.'s accuracy: their pipeline differs [37]
- Channel order beyond the one raw file checked

**Each gap is a question for Section V or VI**

**Recognition evidence under laboratory conditions; Section V asks what limits its value**

Sources: Own analysis · data: Gao et al. public Dryad release [37], doi:10.5061/dryad.2qc64 · plan fixed before results; no tuning on test persons · [11], [52]

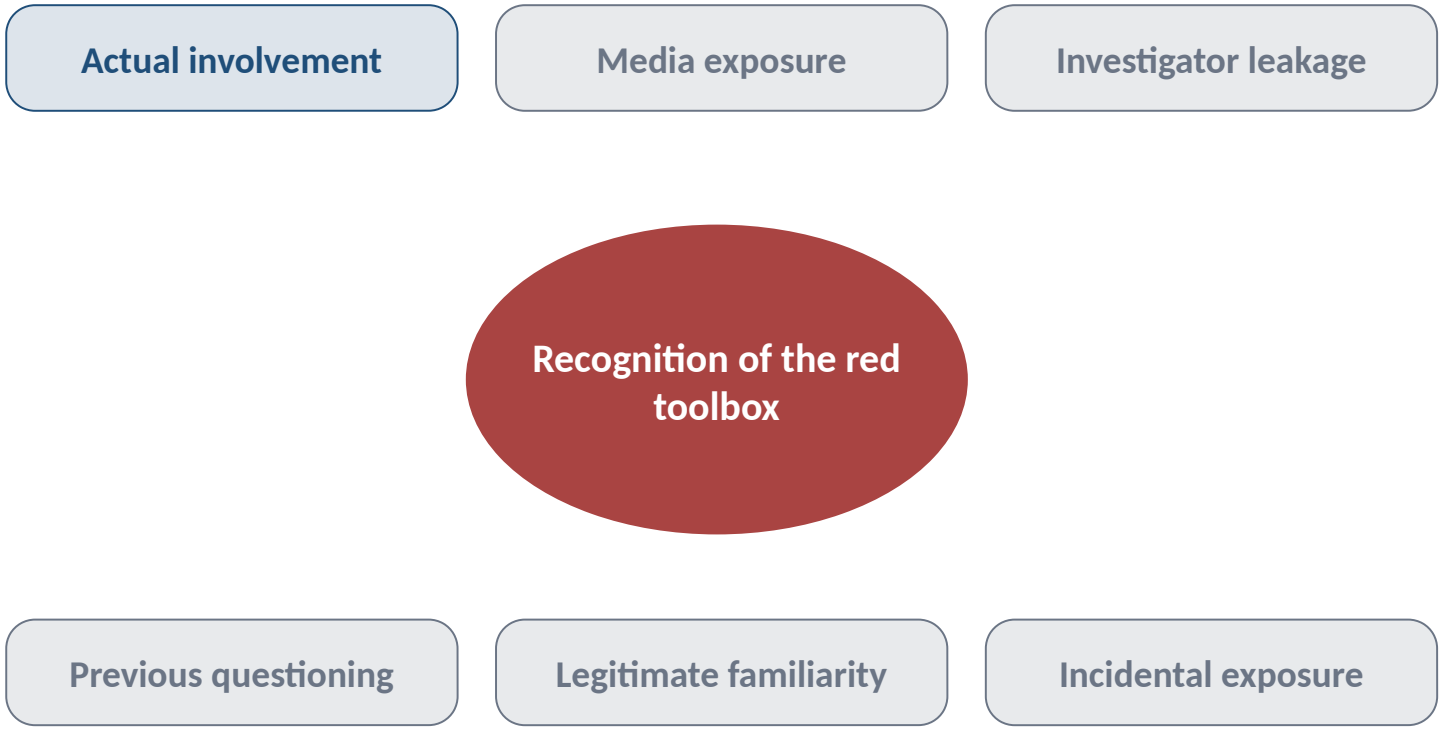
# Limitations: What Prevents Real-World Use?

Recognition vs guilt · memory · countermeasures and variability · methodological limits · a synthesis · ethics and law.

I · What & why | II · Methods | III · Experiment & data | IV · Computation | **V · Limitations** | VI · Future

# Recognition ≠ Guilt, Revisited

*Why return here: the opening distinction is the limitation that no method in this seminar removes.*




## What safeguards do

- Careful item selection [44], [51]
- Target stimuli and memory manipulations [11], [52]

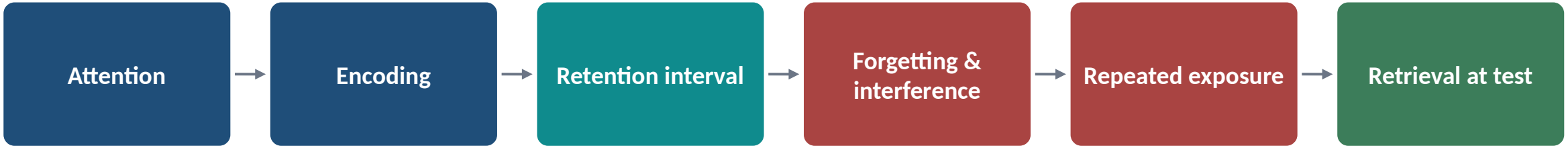
## What they cannot do

Turn recognition into guilt. That inference remains legal and investigative [14].

Sources: [11], [14], [20], [36], [44], [51], [52]

# Memory Is Not Perfect

*Why it matters: the test can only detect what was encoded and is still accessible.*




### Encoding

- Peripheral details may never be stored
- Arousal and alcohol can alter encoding [6]

### Retention

- Delay can reduce memory for crime details [15]
- Interference changes P300-CIT outcomes [52]

### Implication

Failure to detect recognition does not demonstrate absence of prior knowledge.

Sources: [6], [15], [20], [36], [52]

# Countermeasures and Human Variability

*Why it matters: individuals are not interchangeable signal generators.*

**Deliberate countermeasures**

[8], [18]

**Motivation and strategy**

**Attention and fatigue**

**Example of a cost**

Pretest screening improved diagnosticity but left about one third of participants undiagnosed in one autonomic study [38].

**Individual ERP variability**

**Physiological reactivity**

[38]

**Traits, e.g. psychopathic traits**

[19]

**Individuals are not interchangeable signal generators**

Sources: [8], [18], [19], [38], [57]

# Methodological Limitations in the Literature

*Why list them: they explain why reported performance can exceed validated performance.*

- Small samples
- Laboratory paradigms
- Heterogeneous protocols
- Preprocessing variation
- Decision-rule variation
- Potential data leakage
- Subject-dependent evaluation
- Limited external validation
- Publication bias risk
- Limited real-world datasets

## Fair reading

- Not every study has every weakness
- Some studies state limitations explicitly, e.g. not preregistered [58]
- Synthesis shows moderators, not uniform quality [34], [57]

## Consequence

Reported performance ≠ credibly validated performance.

Sources: [21], [34], [37], [46], [57], [58]

# What Prevents Forensic Use Today?

*Why a matrix: each threat changes the signal or the decision in a specific way, and every mitigation leaves some uncertainty.*

| Threat | Effect on signal or decision | Mitigation | Residual uncertainty |
|---|---|---|---|
| **Informed innocents, leaked details [11], [52]** | Recognition without involvement → false positive | Nonpublic, detailed items [51]; targets, memory manipulations [11], [52] | Leakage never fully excluded |
| **Memory failure [15], [52]** | Weak or absent probe response → false negative | Short retention; salient details | Negative ≠ no knowledge |
| **Countermeasures [8], [18]** | Irrelevants made salient → smaller difference | Protocols such as the CTP [24] | Shown only for tested strategies |
| **Individual variability [19], [38]** | One rule performs unevenly across people (e.g. L09) | Within-person tests; person-level validation | Some people stay undetected |
| **Artifacts and shortcuts** | Eye or muscle activity mimics a probe effect | Artifact handling; eye channels; relevance checks | Stimulus-locked artifacts remain |
| **Small samples, leakage [34], [57]** | Optimistic accuracy | Person-wise splits; nested selection; external data | Few external replications |
| **Mock-crime realism [15], [40]** | Lab effects may not transfer to real cases | Self-initiated acts, real memories [36], [42] | Little operational evidence |

**Together these explain why EEG-based GKT is not yet a reliable real-world forensic tool**

Sources: [8], [11], [15], [18], [19], [24], [34], [36], [38], [40], [42], [51], [52], [57]

# Ethical and Legal Questions

*Why include them: a test about concealed knowledge in people raises questions that accuracy alone cannot answer.*

### Legal admissibility

- Use of GKT evidence in court is contested [14]
- Known error rates are needed for the protocol and population used

### Consequences of error

- A false recognition inference can harm an innocent person
- Leaked details make false positives more likely [11], [52]

### Consent and mental privacy

- The test probes information a person may not wish to reveal
- Voluntariness and data protection need explicit safeguards

### Scope of the claim

- Results support recognition, not deception or guilt [14], [53]
- Model scores need calibrated uncertainty, not verdicts

**Scientific validity is necessary, not sufficient: real-world use also needs legal, ethical and procedural safeguards**

Sources: [11], [14], [52], [53] · consent and privacy points are general ethical considerations, not drawn from the reviewed literature

# Future Directions: What Is Needed for Translation?

Better experiments · better computational science · AI research opportunities · a ladder toward responsible translation.

# Better Experimental Science

*Why first: no analysis method can recover validity that the experiment did not build in.*

| | | | | | |
|---|---|---|---|---|---|
| 1 | **Larger samples** | Stable individual-level estimates | 2 | **Multi-site studies** | Independence from one laboratory |
| 3 | **Standardized protocols** | Comparable results | 4 | **Transparent reporting** | Reproducible pipelines |
| 5 | **Preregistration** | Less analytic flexibility | 6 | **Realistic scenarios** | Self-initiated acts, real memories [36], [42] |
| 7 | **Careful probe construction** | Encoding-matched items [44] | 8 | **Informed-innocent controls** | Leakage tested, not assumed [11], [52] |
| 9 | **Countermeasure conditions** | Robustness under strategy | 10 | **Longer retention intervals** | Memory under realistic delay [15] |

Sources: [11], [15], [36], [40], [42], [44], [52], [58]

# Better Computational Science: A Research Roadmap

*Why staged: each stage depends on the one before; better models cannot compensate for weaker validation.*

**1** Stage 1
**Better representations**

- Single-trial ERP features
- Time-frequency features
- Multichannel learned representations

→

**2** Stage 2
**Better validation**

- Subject-independent evaluation
- Stratified participant-wise or leave-one-pair-out validation
- External datasets

→

**3** Stage 3
**Better adaptation**

- Transfer learning
- Domain adaptation
- Few-shot personalization

→

**4** Stage 4
**Better trust**

- Explainable AI
- Uncertainty estimation
- Physiological validation

Reproducible pipelines throughout

*"The objective should be generalization, not merely maximizing within-dataset accuracy." Proposed directions — not completed work.*

# Modern AI: What Could Come Next

*Why frame as gaps: the reviewed literature contains no strong CIT-specific evidence for these methods.*

## Representation learning

- Deep learning on EEG
- Temporal / convolutional networks
- Learned features
- Multimodal fusion

## Generalization

- Transfer learning
- Domain adaptation
- Few-shot personalization
- Cross-dataset validation

## Trustworthiness

- Explainable AI
- Uncertainty estimation
- Physiological plausibility
- Calibrated decisions

**Research opportunities / evidence gaps — not established EEG-CIT capabilities**

# A Research Design for Testing Transfer

*Why sketch it: it turns the generalization gap into a testable question — without claiming an answer.*




## Prerequisite: audit each dataset

- Paradigm: is it really a CIT?
- Labels: what was concealed?
- Markers, timing, channels, reference, units
- Deception-labelled ≠ CIT [46]

## What would count as success

- Person-level AUC and balanced accuracy on B (95% CI), only if A and B have comparable knowledgeable/control ground truth
- Otherwise, trial-level recognition results reported separately
- Threshold fixed on A; Brier checked on B
- Features consistent with physiology

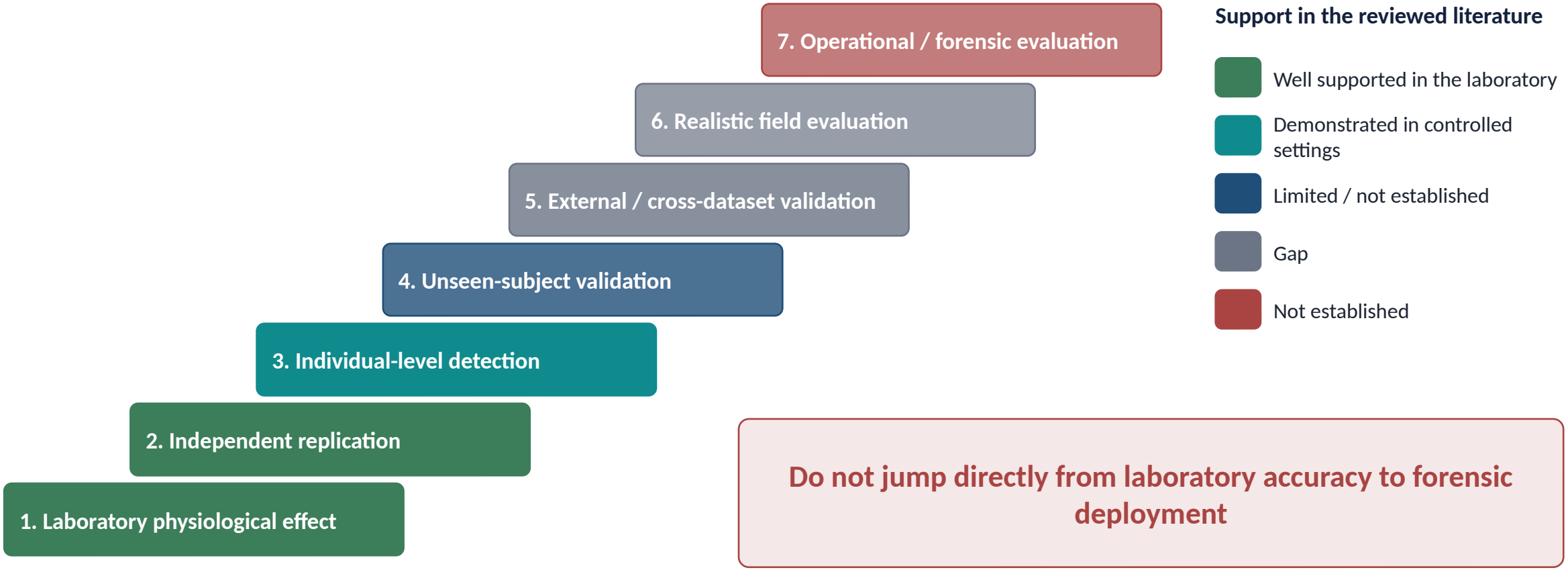
**CONCEPT ONLY**

A design for future work — not performed, not evidence.

Sources: [46]

# A Translation Ladder for EEG-Based CIT

*Why a ladder: each rung is a distinct claim that requires its own evidence.*




**Do not jump directly from laboratory accuracy to forensic deployment**

Sources: [14], [34], [36], [40], [57]

CONCLUSION

# Five Take-Home Messages

*Why five: each message corresponds to one stage of the scientific story.*

1. GKT/CIT is fundamentally a recognition-based paradigm.
2. Concealed information can influence autonomic, behavioral, ocular and neural responses.
3. P300 is an important neural marker for concealed-information research — but it is not a lie signal.
4. Signal processing and AI expand analytical capability but do not remove experimental and interpretive limitations.
5. Real-world translation requires rigorous individual-level, subject-independent, externally validated and ecologically meaningful evaluation.

Sources: [1], [3], [34], [47], [57]

CONCLUSION · CENTRAL MESSAGE

# EEG-Based GKT Does Not Read Lies

*Why end on one sentence: every result in this seminar fits inside this boundary.*

> ***"EEG-based GKT does not read lies."***
>
> It measures neural responses that may be consistent with recognition of concealed information.

**RECOGNITION ≠ GUILT**

# Where Should EEG-Based GKT Go Next?

*“Can concealed-information detection become scientifically robust enough for responsible real-world use?”*




## Thank You

Questions & Discussion

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# References (Supplementary)

*Added for the deep-learning slides, plus the dataset used for the real-data slides.*

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